- Again, the tangential acceleration, a and the angular acceleration, α can be related as follows

$$a = \frac{v-u}{t} = \frac{r\omega - r\omega o}{t} = r\left(\frac{\omega - \omega_o}{t}\right), \text{ but } \frac{\omega - \omega_o}{t} = \alpha$$

$$\boxed{a = r\alpha}$$

Conceptual Examples

1. When a wheel of radius R rotates about a fixed axis, do all points on the wheel have the same angular speed? do they all have the same linear speed?

   Explanation: Yes, all points on the wheel have the same angular speed. But, not all points on the wheel have the same linear speed. At $r = 0$, $v = \omega r = 0$, at $r = \frac{R}{2}$, $v = \omega r = \frac{\omega R}{2}$ and at $r = R$, $v = \omega r = \omega R$.

2. A wheel rotates counter clockwise in the xy palne. What is the direction of the angular velocity?

   Explanation: Using the right hand rule, the angular velocity is directed out of the page, i.e., towards the positive z – axis.

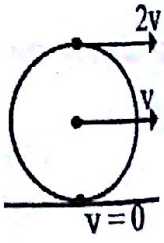
3. An object at rest begins to rotate with a constant angular acceleration. If it rotates an angle θ in time t, through what angle did it rotate in time $t/2$?

   Hint: $\theta = \omega_o t + \frac{1}{2}\alpha t^2 \Rightarrow \theta \sim t^2$. Therefore in time $t/2$, it rotated through $\theta/4$.

4. A car with tires of radius 32 cm moves at 60 km/h.

   A. What is the angular speed of the tires?

   B. What is the linear speed of the top of the tires?



Explanation: A. $v = r\omega \Rightarrow \omega = \frac{v}{r} = \frac{16.7}{0.32} = 52$ rad/s

B. At the top, $v = 2(60) = 120$ km/h

Illustrative Examples

5. A wheel starts from rest and rotates with constant angular acceleration to an angular speed of 12 rad/sec in 3 sec. Find:
   a) The angular acceleration of the wheel.
   b) The angle through which it rotates in this time.

Solution: a) $\alpha = \dfrac{\Delta\omega}{\Delta t} = \dfrac{\omega - \omega_o}{t} = \dfrac{12-0}{3} = 4 \text{ rad/sec}^2$

b) $\theta = \omega_o t + \dfrac{1}{2}\alpha t^2 = \dfrac{1}{2}(4)(3)^2 = 18 \text{ rad}$

6. A disk initially rotating at 120 rad/sec, slows down at 4 rad/s^2.
   a) How much time did it take the disk to stop?
   b) Through what angle did it turn before coming to rest.

Solution: a) $\omega = \omega_o - \alpha t \Rightarrow \omega = 0$ and $t = \dfrac{\omega}{\alpha} = 30 \text{sec}$

b) $\theta = \left(\dfrac{\omega + \omega_o}{2}\right)t = \left(\dfrac{0+120}{2}\right)30 = 1800 \text{rad}$

7. A flywheel is rotating at 180 rpm and after 20 sec it is running at 120 rpm. If the retardation is uniform.
   a) How many revolutions will it make, and
   b) What time will elapse before it stops?

Solution: a) $\omega_o = 180 \text{ rpm} = 2\pi\,(180)/60 = 6\pi \text{ rad/sec}$

$\omega = 120 \text{ rpm} = 2\pi\,(120)/60 = 4\pi \text{ rad/sec}$

$$\alpha = \frac{\omega - \omega}{t} = \frac{4\pi - 6\pi}{20} = -0\pi \text{rad/sec}^2$$

And $\omega^2 = \omega_o^2 + 2\alpha\theta \Rightarrow \theta = 180\pi \text{ rad} = 90 \text{ rev}$

$$\omega = \omega_o + \alpha t \Rightarrow t = \frac{-6\pi}{-0.1\pi} = 60 \text{sec} = 1 \text{min}$$

8. A shaft is uniformly accelerated from 10 rev/sec to 18 rev/sec in 4 sec. The shaft continues to accelerate at this rate for the next 8 sec. There after the shaft rotates with a uniform angular speed. Find the total time taken to complete 400 rev.

Solution: $\omega_o = 10 \text{ rev/s} = 2\pi(10) \text{ rad/s} = 20\,\pi \text{ rad/s}$

$\omega = 18 \text{ rev/s} = 2\pi\,(18) \text{ rad/s} = 36\,\pi \text{ rad/s}$

First of all, at $t = 4$ sec, $\alpha = \dfrac{36\pi - 20\pi}{4} = 4\pi \text{ rad/sec}^2$

$$\theta_1 = \omega_o t + \frac{1}{2}\alpha t^2 = 20\pi(4) + \frac{1}{2}(4\pi)(4)^2 = 56\text{rev}$$

Next, for t = 8 sec, $\omega = \omega_o + \alpha t = 36\pi + 4\pi(8) = 68\pi$ rad/sec

$$\theta_2 = \omega_o t + \frac{1}{2}\alpha t^2 = 36\pi(8) + \frac{1}{2}(4\pi)(8)^2 = 208\text{rev}$$

Finally, we have, $400 - (56 + 208) = 136$ rev

$t = 4 + 8 + 4 = 16$ sec

9. A car is moving at 72 km/h, if the wheels are 75 cm in diameter, find the angular velocity of the wheel about its axis. If the car comes to rest in 20m, under a uniform retardation, find the angular retardation of the wheels.

Solution: v = 72 km/h = 20 m/sec, r = 0.375m,

$$\omega = \frac{v}{r} = 53.3\text{rad/sec} \text{ and } v^2 = u^2 + 2as \Rightarrow a = -10\ \text{m/sec}^2$$

$$a = \alpha r,\ \alpha = \frac{a}{r} = \frac{-10}{0.375} = -26.7\text{rad/sec}^2$$

10. If an airplane propeller rotates at 2000 rev/min while the airplane flies at a speed of 480 km/h relative to the ground, what is the speed of a point on the tip of the propeller at radius 1.5m, as seen by;

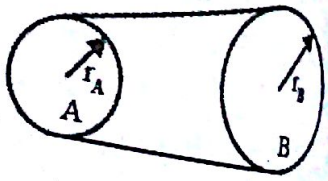
a) the pilot, and

b) an observer on the ground

Solution: a) $v = \omega r = 2000\ \frac{(2\times 3.14)}{60} \times 1.5 = 314\text{m/s}$

b) $v = \sqrt{\left(\frac{480}{3.6}\right)^2 + (314)^2} = 341\,\text{m/s}$

11. Wheel A of radius 10 cm is coupled by a belt to wheel B of radius 25 cm. Wheel A increases its angular speed from rest at a uniform rate of 1.6 rad/sec^2. Find the time for wheel B to reach a rotational speed of 100 rev/min assuming the belt does not slip.



Solution: If belt does not slip, $v_A = v_B$

$(\alpha)(t)(r_A) = (\omega)(r_B)$

$$t = \frac{\omega r_B}{\alpha r_A} = \frac{100(2\times 3.14)\times 0.25}{60\times 1.6\times 0.1} = 16\,\text{sec}$$

12. Two gears wheels which are meshed together have radii of 0.5 and 0.15 cm. Through how many revolutions does the smaller turn when the larger turns through 3 rev.

Solution: $s_1 = s_2 \Rightarrow (0.5)(3\text{ rev}) = (0.15)(\theta) \Rightarrow \theta = 10$ rev

13. The angle of rotation of a reel of a tape recorder of diameter 1.2cm is given by $\theta = 1.2t + 0.1t^2$. Compute the linear speed and acceleration of a point on the rim of the reel at

a) $t = 0$ b) $t = 30$ sec

Solution: $\omega = \frac{d\theta}{dt} = 1.2 + 0.2t$

$$\alpha = \frac{d\omega}{dt} = 1.2\text{rad/s}^2$$

a) $t = 0 \Rightarrow \omega = 1.2 + 0.2t = 1.2$ rad/s and $\alpha = 1.2$ rad/s^2
$v = r\omega = 0.72$ cm/s and $a = r\alpha = 0.72$ cm/s^2

b) $t = 30 \Rightarrow \omega = 1.2 + 0.2(30) = 7.2$ rad/s and $\alpha = 1.2$ rad/s^2
$v = r\omega = 4.32$ cm/s and $a = 0.72$ cm/s^2

14. A body is rotated with a uniform speed in a circle of radius 1m. What is the centripetal acceleration in m/s^2 if it completes 12 rotations in 3 sec?

A. 36π B. $64\pi^2$ C. $72\pi^2$ D. $144\pi^2$

Solution: $\omega = \frac{\theta}{\Delta t} = \frac{12(2\,p)\,\text{rad}}{3} = 8\,\pi\text{rad/sec}$

$$a_c = \frac{v^2}{r} = \omega^2 r = (8\pi)^2(1) = 64\pi^2 \text{m/s}^2$$ **Answer: B**

15. The angular position of a swinging door is described by $\theta = 5 + 10t + 2t^2$. What is the angular speed and acceleration of the door at $t = 3$ sec?

A. 22 rad/s, 4 rad/s^2 C. 22 rad/s, 12 rad/s^2
B. -17 rad/s, 4 rad/s^2 D. 12 rad/s, 3 rad/s^2

Solution: $\omega = \frac{\Delta\theta}{\Delta t}$, as, $\Delta t \to 0 \Rightarrow \omega = 10 + 4t = 22$ rad/s

$$\alpha = \frac{\Delta\theta}{\Delta t}, \text{ as } \Delta t \to 0 \Rightarrow \alpha = 4 \text{ rad/s}^2$$ **Answer: A**

16. A wheel starts from rest and rotates with constant angular acceleration to reach an angular speed of 12 rad/s in 3 sec. What is the magnitude of the angular acceleration of the wheel in *rad*/s² and the angle in *rad* through which it rotates in this time interval respectively?

A. 4, 18 B. 3, 18 C. 18, 5 D. 13, 4

Solution: $\alpha = \dfrac{\omega - \omega_o}{\Delta t} = 4 \text{ rad/s}^2$

$$\theta = \omega_o t + \frac{1}{2}\alpha t^2 = 18 \text{ rad}$$

Answer: A

17. A sphere initially revolving at 5π rad/s slows down uniformly and stops in 10 sec. How many revolutions does the sphere make during this time?

A. 6.25 B. 25 C. 15 D. 12.5

Solution: $\alpha = \dfrac{\omega - \omega_o}{\Delta t} = -0.5 \text{ rad/s}^2$

$$\theta = \omega_o t + \frac{1}{2}\alpha t^2 = 25\pi \text{rad} = 12.5 \text{ rev}$$

Answer: D

18. A disc initially at rest rotates with constant angular acceleration through an angle of 22.5 rad in 3 sec. What is the angular speed of the disk after 4 second?

A. 10 rad/s B. 20 rad/s C. 15 rad/s D. 25 rad/s

Solution: $\theta = \omega_o t + \dfrac{1}{2}\alpha t^2 \Rightarrow \alpha = 2(22.5)/3^2 = 5 \text{ rad/s}^2$

$$\omega = \omega_o + \alpha t = 5(4) = 20 \text{ rad/s}$$

Answer: B

19. A wheel initially at rest is rotated with constant angular acceleration of 100 rad/s in 20 sec. If the radius of the wheel is 0.3m. What is the tangential acceleration of a point at the rim of the wheel?

A. 1.5 m/s² B. 2 m/s² C. 2.5 m/s² D. 3 m/s²

Solution: $\alpha = \dfrac{\omega - \omega_o}{\Delta t} = 5 \text{ rad/s}^2$, $a = r\alpha = 1.5 \text{ m/s}^2$

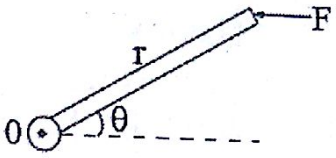
Answer: A

6.2 Rotational Dynamics

The quantities such as torque, rotational kinetic energy, rotational work and power and angular momentum are the focus of rotational dynamics at this level.

Torque (τ, tau): Is the twisting or turning effect of a force.

☞ The magnitude of torque or moment as it is usually called is obtained by multiplying the force with the perpendicular distance from the axis of rotation to the point of application of the force.



That is, suppose a force is applied to a lever of length r as shown. If the pivot point or axis of rotation is at 0, then the torque is obtained by,

Torque = (Force)(perpendicular distance)

$\tau = (F)(r \sin\theta)$

Torque is a vector quantity and its unit is Newton meter (N.m).

☞ More generally, torque is defined as the cross product between the position vector $(\vec{r})$ and the force vector $(\vec{F})$.

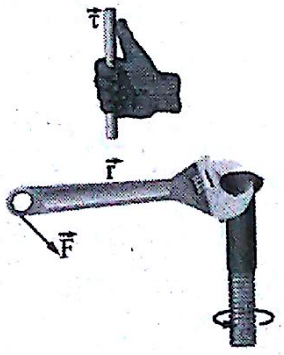
$\vec{\tau} = \vec{r} \times \vec{F}$ ⟵ Vector form of torque

And, we can have

$\tau = Fr \sin\theta$ ⟵ Magnitude of torque

What is more, since torque is a vector quantity, its direction is found to be determined using the right hand rule.

Right hand rule "If you curl your right hand fingers from vector $\vec{r}$ to vector $\vec{F}$, then your outstretched thumb will point in the direction of the torque, $\vec{\tau}$."



☞ **Important points to remember**

- The perpendicular distance from the axis of rotation or pivot point to the point of application of the force is usually called *lever arm*.

- By convention, on counter clockwise torque is taken as positive and the clockwise torque as negative.

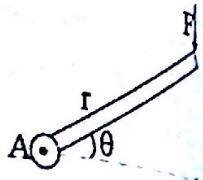
Conceptual Examples

20. Both torque and work are products of force and distance. How are they different?

Explanation: First of all, $W = F.s = Fs\cos\theta$ . . . scalar

$\tau = r \times F = Fr\sin\theta$ . . . vector moreover, The unit of work is Nm (Joule), but torque has a unit N.m.

21. What is the magnitude of the torque about point A for the given arrangement?



Explanation: $\tau = (F)(r_1)$

$\tau = Fr\cos\theta$

Illustrative Examples

22. Force $F = (2i - 3k)N$ acts on a body with a position vector $r = (0.5j - 2k)m$. What torque acts on the body?

a) About the origin

b) About a point with coordinate (2, 0, -3)m?

Solution:

a) $\tau = r \times F = \begin{vmatrix} i & j & k \\ 0 & 0.5 & -2 \\ 2 & 0 & -3 \end{vmatrix}$

+ve direction

$= (-1.5i - 4j - k)$ N. m

b) $\tau = r \times F \begin{vmatrix} i & j & k \\ -2 & 0.5 & 1 \\ 2 & 0 & -3 \end{vmatrix} = (-1.5i - 4j - k)$ N. m

$r = (-2i, 0.5j, k)$ m

23. Force $F = (-8i + 6j)N$ acts on a particle with position vector $r = (3i + 4j)$ m what are;

a) The torque on the particle.

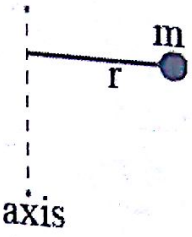
b) The angle between the position and the force vectors.

Solution: a) $\tau = r \times F = \begin{vmatrix} i & j & k \\ 3 & 4 & 0 \\ -8 & 6 & 0 \end{vmatrix} = 50\hat{k}$ N.m

$$|r \times F| = Fr \sin\theta \Rightarrow \theta = \sin^{-1}\left(\frac{50}{10\times 5}\right) = 90°$$

Moment of Inertia: Is the rotational mass that measures the resistance of a body to change in its rotational motion.

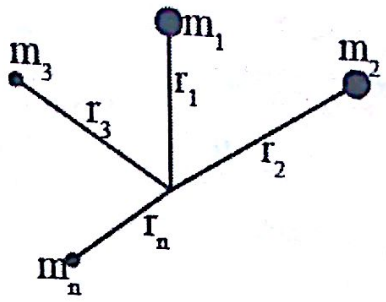
- Moment of inertia depends on the mass of the body and the distribution of that mass about the axis of rotation. For example, the moment of inertia of a point mass is the product of the mass and the square of the distance of the mass from the axis of rotation.



$$I = mr^2$$

The unit of moment of inertia is kilogram meter square (kgm^2).

- If we think of a number of point masses at different distances from the axis of rotation, the total moment of inertia of the system is the sum of the moment of inertia of the individual point masses. That is, if the masses, $m_1, m_2 \ldots, m_n$ are at distances $r_1, r_2, \ldots, r_n$ respectively from the axis of rotation, then the total moment of inertia is obtained by



$$I = I_1 + I_2 + \ldots + I_n$$

$$I = m_1r_1^2 + m_2r_2^2 + \ldots + m_nr_n^2$$

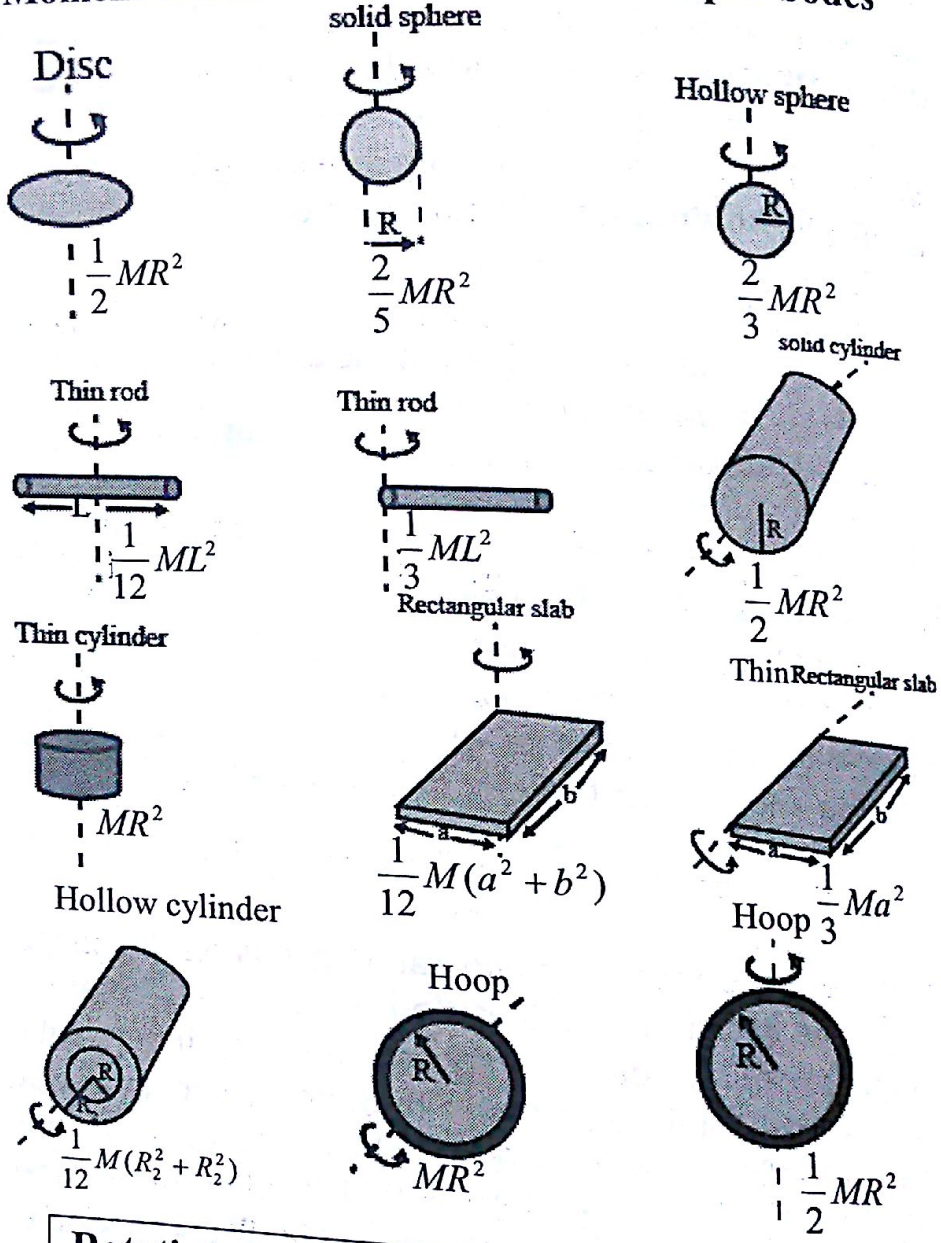
What is more, the moment of inertia, in general, depends on the shape, mass distribution and the location or orientation of the axis if rotation. As a result, the moment of inertia of a sphere and a cylinder, solid sphere and hollow sphere, solid sphere axis through center and solid sphere axis tangent to surface can be mentioned as examples. So you can see that, in simple terms, a sphere and a cylinder of the same mass and the same radius can have different value of moment of inertia due to their difference in shape.

Just to mention some, the moment of inertia of:

- A point mass m at distance r from axis, $I = mr^2$.
- A disc of mass M and radius R axis center, $I = \frac{1}{2}MR^2$.
- A sphere of mass M and radius R axis center, $I = \frac{2}{5}MR^2$.

- A thin rod of mass M and length L axis, center, $I = \frac{1}{12}ML^2$.
- A hoop of mass M and radius R, axis, center, $I = MR^2$.

Moment of inertia of some different shaped bodes



Rotational Kinetic Energy: Is the energy possessed by a body due to its rotational motion, since a rotating body consists of mass in motion.

Therefore, the rotational kinetic energy of a body of moment of inertia I and angular speed ω is given by

$$k_E = \frac{1}{2}I\omega^2$$

- The greater the moment of inertia, the greater the kinetic energy of the rotating body.

Rotational Work Done: Is the product of the torque and angular displacement.

- Pedaling a bicycle, a rotating motor shaft driving a power tool, a car engine propelling the vehicle, all involve rotational work done.

The rotational work done by a constant torque τ in turning a body through an angular displacement θ is given by

$$W = \tau\alpha\theta$$

From, the rotational analog of Newton's second law, we can have, $\tau = I\alpha$.

Therefore, $W = \tau\,\alpha\,\theta$

- Again from the work energy theorem, the work done is equal to the change in rotational kinetic energy

$$W = \Delta k_E r = \frac{1}{2}I(\omega^2 - \omega_o^2)$$

$$\tau\theta = I\,\alpha\,\theta = \frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_o^2$$

- It turns out that the net torque causes the rotational velocity of a body to change.

Rotational Power: Is the rate at which rotational work is done.

$$P = \frac{W}{t} = \frac{\tau\theta}{t} = \tau\omega$$

Conceptual Examples

24. Tons of dust and small particles rain down into the earth from space every day. As a result, does the Earth's moment of inertia increase, decrease or stay the same? Explain

Explanation: In principle, the earth's moment of inertia increases because both its mass and radius increase.

25. Two forces produce the same torque. Does it follow that they have the same magnitude? Explain

Explanation: No. Torque depends both on the force and the distance at which the force is applied from the axis of rotation. A

small force can produce the same torque as a large force if it is applied farther from the axis of rotation.

26. A skater pulls in her arms, decreasing her moment of inertia by half and doubling her angular speed. What will happen to her angular kinetic energy?

Explanation: $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{I}{2}\right)(2\omega)^2 = 2kE_1$. The kinetic energy become twice the original kinetic energy.

27. If the polar ice caps were to melt and release their water into the oceans, will the earth's moment of inertia increase or decrease? Explain

Explanation: It would increase. This is because mass that had been near the axis of rotation would now be distributed uniformly. With more of the mass at greater distance from the axis of rotation, the moment of inertia as well as the length of the day would increase.

28. A solid sphere, a hollow sphere, a solid cylinder and a hollow cylinder are released from an inclined plane at the same time. Which one will reach the bottom first?

Explanation: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}cMR^2\frac{v^2}{R^2}$

$\Rightarrow v = \sqrt{\frac{2gh}{1+C}}, C = \frac{2}{5}$ for solid sphere, etc. small C, large V

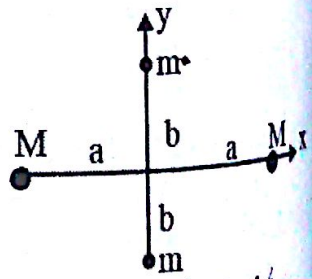
Illustrative Examples

29. Four point masses are fastened to the massless rods as shown. Find the moment of inertia and rotational energy.
   a) If the axis of rotation is the y axis.
   b) If the axis of rotation is the origin (z axis).

Solution: a)

a) $I_y = \sum m_i r_i^2 = Ma^2 + Ma^2 = 2Ma^2$

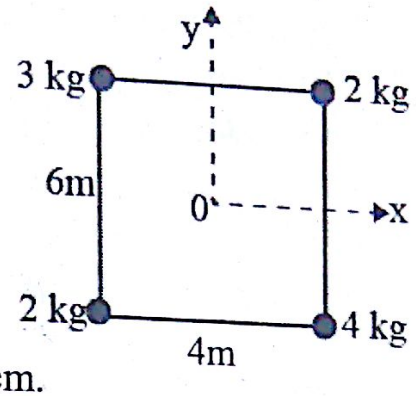
$kE_r = \frac{1}{2}I_y\omega^2 = Ma^2\,\omega^2$



b) $I_E = \sum m_i r_i^2 = Ma^2 + Ma^2 + mb^2 + mb^2 = 2Ma^2 + 2mb^2$

$$kE_r = \frac{1}{2}I_z\omega^2 = (Ma^2 + mb^2)\omega^2$$

30. The four particles are connected by rigid rods of negligible mass. The origin is at the center of the rectangle. If the system rotates about the z axis with an angular speed of 6 rad/sec. Calculate:



a) The moment of inertia of the system.
b) The rotational energy of the system.

Solution: $r = \left(\sqrt{6^2 + 4^2}\right)/2 = 3.6m$

a) $I_z = \sum_{i=1}^{n} m_i r_i^2 = (2+3+2+4)(3.6)^2 = 143\ kgm^2$

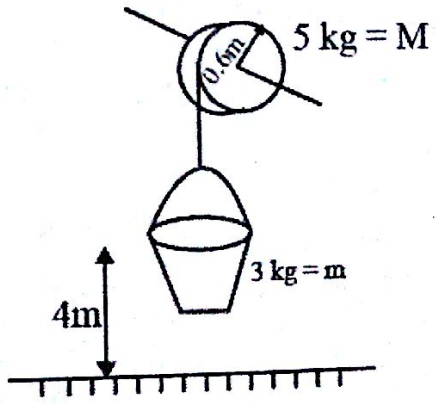
b) $k_{Er} = \frac{1}{2}I\omega^2 = \frac{143}{2}(6)^2 = 2574J$

31. A constant torque of 25 Nm is applied to a grind stone whose moment of inertia is 0.13 $kg.m^2$, neglecting friction, find the angular speed after the grindstone has made 15 revolutions.

Solution: $W = \Delta k_{Er} \Rightarrow \tau\theta = \frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_o^2$

$$\Rightarrow \omega^2 = \frac{2\tau\theta}{I} = \frac{2(25)(15)(2\times 3.14)}{0.13} = 190\ rad/sec$$

32. Determine the angular speed of the spool after the 3 kg bucket has fallen 4m starting from rest.



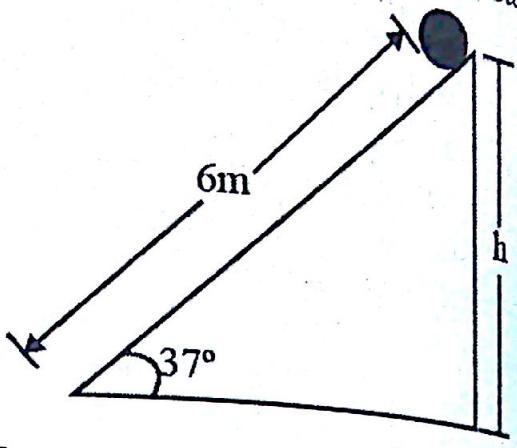
Solution: $k_{Er} = G_{PE}$

$$\frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2}\left(\frac{1}{2}(Mr^2)\omega^2\right) = mgh$$

$$\omega^2 = \frac{4mgh}{Mr^2} = \frac{4(3)(10)(4)}{5(0.6)^2} = 256\ sec = 16\ rad/s$$

33. A 240N sphere 0.2m in radius rolls without slippling 6m down a ramp that is inclined at 37° with the horizontal. What is the angular speed of the sphere at the bottom of the slop if its starts from rest?



Solution: $M_{E_i} = M_{E_f}$

$mgh_i + k_{E_{ri}} + k_{E_{ti}}$

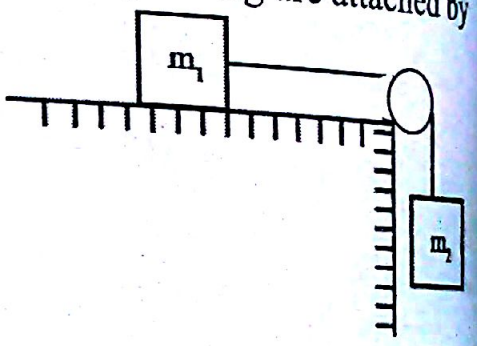
$= mgh_f + k_{E_{rf}} + k_{E_{tf}}$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2, v = r\omega$$

$$mgh = \frac{1}{2}mR^2\omega^2 + \frac{1}{2}I\omega^2, I = \frac{2}{5}mR^2$$

$$\omega^2 = \frac{2gh}{\frac{7}{5}R^2} = \frac{2(10)(6)(\sin 37)}{1.4(0.2)^2} \Rightarrow \omega = 36\text{rad/sec}$$

34. Two blocks with masses $m_1 = 5$ kg and $m_2 = 7$ kg are attached by a string over a pulley of mass $M = 2$ kg and radius 0.05m. The horizontal surface has coefficient of friction 0.35. What is the speed of the system when the 7 kg block has dropped 2m?



Solution: $\Delta P_E = W_f + \Delta kE_t + \Delta kE_r$

$$m_2gh = \mu_k m_1 gh + \frac{1}{2}m_1v^2 + \frac{1}{2}m_2v^2 + \frac{1}{2}I\omega^2$$

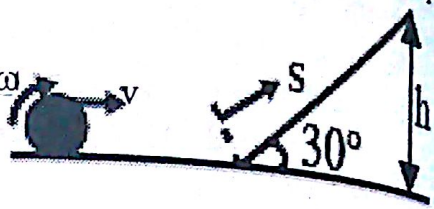
$$m_2gh - \mu_k m_1 gh = \frac{1}{2}m_1v^2 + \frac{1}{2}m_2v^2 + \frac{1}{2}Mv^2, \frac{I}{r^2} = M$$

$$\Rightarrow v = \sqrt{\frac{2gh(m_2 - \mu_k m_1)}{m_1 + m_2 + M}} = 3.9\text{m/s}$$

Solving for a, $a = 0.38\ m/s^2$

b) From equation (*), $T_1 = 7.96N$, and from equation (**), $T_2 = 9.9N$

41. A solid disk is rolling without slipping on a level surface at a constant speed of 4 m/sec. How far it roll up a 30° inclined plane before it stops?



Solution: $M_{Ei} = M_{Ef}$

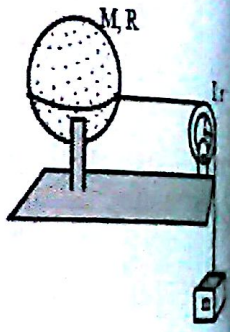
$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 = mgh$$

$$\frac{1}{2}v^2 + \frac{1}{4}v^2 = gh \Rightarrow h = \frac{3v^2}{4g} = \frac{3(4)^2}{4(10)} = 1.2m$$

$$s = \frac{h}{\sin\theta} = \frac{1.2}{\sin 30} = 2.4\,m$$

42. A spherical shell of mass M and radius R rotates about a vertical axis. A massless cord passes around the shell, over a pulley of rotational inertia I and radius r, and is attached to a small object of mass m. If there is no friction and the cord doesn't slip, what is the speed of the object after it has fallen a distance h from rest?



Solution: $M_{Ei} = M_{Ef}$

$$mgh = \frac{1}{2}I_s\omega^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$2mgh = \frac{2}{5}MR^2\left(\frac{v}{R}\right)^2 + I\left(\frac{v}{r}\right)^2 + mv^2 \Rightarrow v = \sqrt{\frac{10mgh}{(2M + 5m + 5I/r^2)}}$$

43. The pulley shown has radius 0.16m and moment of inertia 0.4 kgm^2. The rope does not slip on the pulley rim. Calculate the speed of the 4 kg block just before it strikes the ground.

Solution: $M_{Ei} = M_{Ef}$

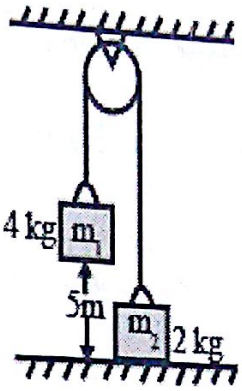
$$m_1gh = \frac{1}{2}m_1v^2 + \frac{1}{2}I\omega^2 + m_2gh$$

$$m_1gh = \frac{1}{2}m_1v^2 + \frac{1}{2}I\left(\frac{v}{r}\right)^2 + m_2gh$$

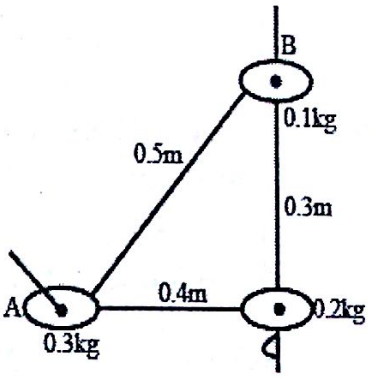
$$4(10)(5)$$

$$= \frac{1}{2}(4)v^2 + \frac{1}{2}(0.48)\frac{v^2}{(0.16)^2} + 2(10)(5)$$

$$\Rightarrow v = 3 \text{ m/s}$$



44. What is the moment of inertia of the system shown?
   a) About an axis through the center of disk A.
   b) About an axis through the center of disks B & C.

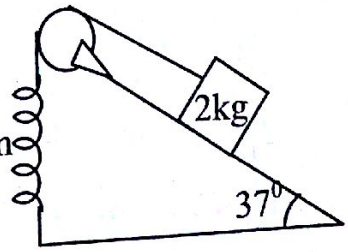


Solution:

a) $I = \sum m_i r_i^2 = (0.1)(0.5)^2 + (0.2)(0.4)^2 = 0.057 \text{ kgm}^2$

b) $I = \sum m_i r_i^2 = (0.3)(0.4)^2 = 0.048 \text{ kgm}^2$

45. The system shown is released from rest with the spring in the upstretched position. If friction is neglected and the pulley has radius 30 cm and moment of inertia 0.5 kgm^2.
   a) How far will the mass slide down the incline?
   b) What will be the speed of the mass when the mass has slid 1m down the incline?



Solution: a) $M_{Ei} = M_{Ef}$

$$mgx\sin\theta = \frac{1}{2}kx^2$$

$$\Rightarrow (2)(10)(x)(0.6) = \frac{1}{2}(20)(x)^2 \Rightarrow x = 1.2\text{m}$$

b) $M_{Ei} = M_{Ef}$

$$mg\,x\sin\theta = \frac{1}{2}kx^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2, \omega = \frac{v}{r}$$

$$(2)(10)(0.6)=\frac{1}{2}(10)(1)^2+\frac{1}{2}(0.5)\left(\frac{v}{0.3}\right)^2+\frac{1}{2}(2)v^2 \Rightarrow v=0.73\ m/s$$

46. A cord 3m long is coiled around the axle of a wheel. The cord is pulled with a force of 40N. When the cord leaves the axle, the wheel is rotating at 2 rev/s. Determine the moment of inertia of the wheel and axle. Neglect friction.

Solution: $W=\Delta kE \Rightarrow Fs=\frac{1}{2}I\omega^2 \Rightarrow I=\frac{2Fs}{\omega}$

$$I=\frac{2(40)(3)}{[2(2\times 3.14)]^2}=1.52 kgm^2$$

47. A 32 kg wheel with radius 1.2m, is rotating at 280 rev/min. If you want to stop it in 15 sec.
   a) How much work must be done to stop it?
   b) What is the required power?

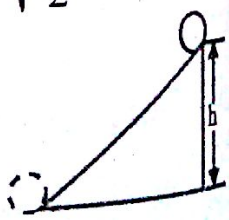
Solution: a) $W=\Delta kE=\frac{1}{2}I\omega^2=\frac{1}{2}(mR^2)\omega^2$, $\omega=29.3$ rad/sec

$$W=\frac{1}{2}(32)(1.2)^2(29.3)^2=19.8\ kJ$$

$$b)\ P=\frac{W}{t}=\frac{19800}{15}=1.32kW$$

48. A hoop of mass M and radius R is at rest at the top of an inclined as shown. The hoop rolls down the plane without slipping. If the hoop reaches bottom, then what is its angular momentum about its center of mass?

A. $MR\sqrt{gh}$ B. $MR\sqrt{2gh}$ C. $MR\sqrt{\frac{gh}{2}}$ D. Mgh



Solution: $M_{Ei}=M_{Ef}$

$$Mgh=\frac{1}{2}Mv^2+\frac{1}{2}I\omega^2, I=MR^2, \omega=\frac{v}{R}$$

$$\Rightarrow v=\sqrt{gh}, \omega=\frac{v}{R}=\frac{\sqrt{gh}}{R}$$

$$L = I\omega = MR^2\left(\frac{\sqrt{gh}}{R}\right) = MR\sqrt{gh}$$

Answer: A

49. Which of the following objects has the largest moment of inertia assuming they all have the same mass and radius?

A. A solid disk
B. A solid sphere
C. A circular loop
D. A solid Cylinder

Solution: $I_{disc} = \frac{1}{2}MR^2$, $I_{sphere} = \frac{2}{5}MR^2$

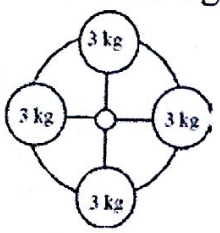
$$I_{loop} = MR^2, I_{cylinder} = \frac{2}{5}MR^2$$

Answer: C

50. The radius of the wheel shown below is 0.75m. What is its moment of inertia?

A. 2.25 kgm^2 B. 3.38 kgm^2 C. 6.75 kgm^2 D. 9.00 kgm^2

Solution:



$$I = \sum_{i=1}^{n} m_i r_i = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2$$

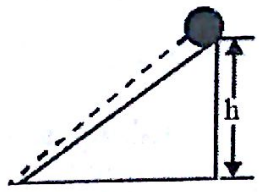
$$I = 4mr^2 = 4(3)(0.75)^2 = 6.75\ kgm^2$$

Answer: C

51. A uniform ball of radius r and mass m starts from rest and rolls down a frictionless incline of height h. When the ball reaches the ground, how fast is it going?

A. $\sqrt{\frac{10}{7}gh}$ B. $\sqrt{\frac{9}{10}gh}$ C. $\sqrt{\frac{10}{9}gh}$ D. $\sqrt{\frac{7}{10}gh}$

Solution: $M_{E_i} = M_{E_f}$



$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2, I = \frac{2}{5}mr^2, \omega = \frac{v}{r}$$

$$\Rightarrow v = \sqrt{\frac{10}{7}gh}$$

Answer: A

52. A 50 Nm torque acts on a wheel of moment of inertia 200 kgm^2. If the wheel starts from rest, how long will it take the wheel to reach an angular speed of 1 rad/s?

A. 4 sec B. 2 sec C. 1 sec D. 0.25 sec

Solution: $\tau = I\alpha \Rightarrow \alpha = 0.25\ rad/s^2$

$$\Delta t = \frac{\omega - \omega_o}{\alpha} = \frac{1}{0.25} = 4\,\text{sec}$$

Answer: A

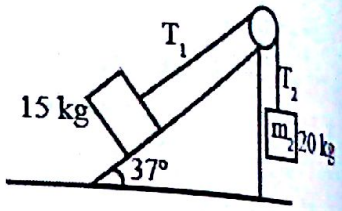
53. As shown in the figure, two blocks are connected by a string of negligible mass passing over a pulley to radius r = 0.2 and moment of inertia I. The block on the friction less incline is moving with a constant acceleration a = $2m/s^2$ up along the incline. What is the moment of inertia of the pulley?

A. 0.8 kgm^2 B. 1.25 kgm^2 C. 4 kgm^2 D. 40 kgm^2

Solution: $T_1 = m_1 g \sin\theta = m_1 a \Rightarrow T_1 = 120N$

$$m_2 g - T_2 = m_2 a \Rightarrow T_2 = 160N$$

$$(T_2 - T_1)r = I\alpha = I\frac{\alpha}{r}$$

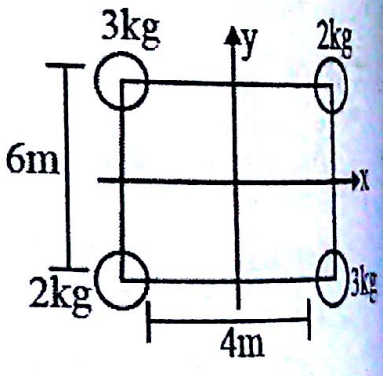


$$I = \frac{(T_2 - T_1)r^2}{a} = \frac{(160-120)}{2}(0.2) = 0.8\text{kgm}^2$$

Answer: A

54. The particles shown are connected by rectangular rigid rods of negligible mass. The origin is at the center of the rectangle. The system rotates in the xy plane about the z axis with an angular speed of 60 rad/s. What is the rotational kinetic energy of the system? (UEE: 2007)

A. 2574 J
B. 2340J
C. 11287J
D. 1170J



Solution: $k_E = \frac{1}{2}I\omega^2$

Taking moment about the origin,

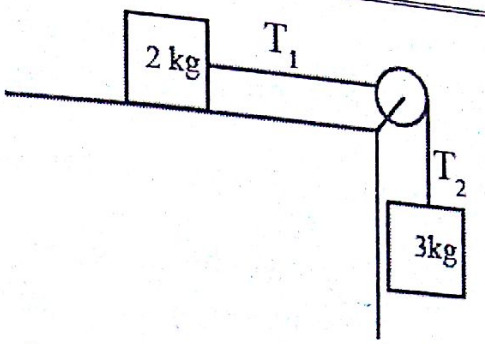
$r = \sqrt{3^2 + 2^2} = \sqrt{13}$

$I = 2(2)(13) + 2(3)(13) = 130\ kgm^2$

$$k_E = \frac{1}{2}(30)(6)^2 = 2340J$$

Answer: B

55. Two blocks are connected by a string of negligible mass passing over a pulley of radius r = 0.2m and moment of inertia I. The block on the horizontal frictionless surface is moving with a constant acceleration a = 2 m/s^2. If the pulley axis is frictionless, the string is inextensible and does not slip on the pulley, what should be the moment of inertia of the pulley? (UEE: 2007)



A. 0.4 kgm^2 C. 4 kgm^2
B. 0.8 kgm^2 D. 8 kgm^2

Solution: $T_1 = 4 \Rightarrow T_1 = 4$

$30 - T_2 = 6 \Rightarrow T_2 = 24$

$$(T_2 - T_1)r = I\alpha = I\frac{a}{r} \Rightarrow I = \frac{(T_2 - T_1)}{a}r^2$$

$$I = \frac{(24-4)(0.2)^2}{2} = 0.4\text{kgm}^2$$

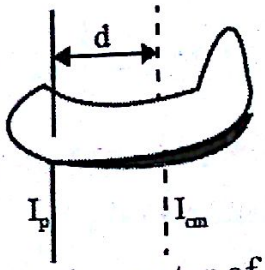
Answer: A

Parallel Axis Theorem

A body, in fact has infinitely many moment of inertia. This is so because, there are infinitely many axis about which it might rotate. Parallel axis theorem enables us to find the moment of inertia of a body about any axis of rotation.

Parallel axis theorem states that; "When the moment of inertia of a body about its center is known, then its moment of inertia about any pivot point is determined by:"

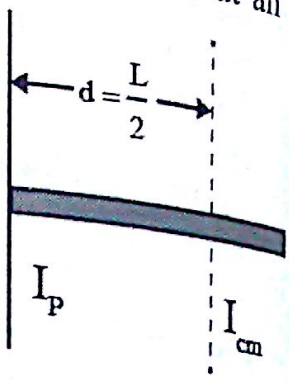
$$I_P = I_{cm} + Md^2$$



Where: I_{cm}; moment of inertia about the center of mass
M: mass of the body
d: distance between the two axes
I_P = moment of inertia about any axis that is parallel to the original central axis.

Illustrative Examples

56. The moment of inertia of a uniform rod of mass M and length L about an axis passing through its center is given by $I = \frac{1}{12}ML^2$. What would be the moment of inertia of the rod about an axis passing through its end point?



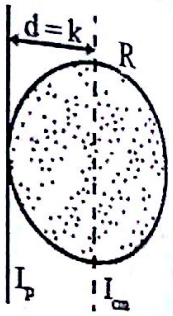
Solution:

$$I_P = I_{cm} + Md^2$$

$$I_P = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2$$

$$I_P = \frac{1}{3}ML^2$$

57. The moment of inertia of a uniform solid sphere of mass M and radius R about an axis passing through its center is given by $I = \frac{2}{5}MR^2$. What is the moment of inertia of the sphere about an axis passing through its edge?



Solution: $I_P = I_{cm} + Md^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$

58. The moment of inertia of a solid cylinder of mass M and radius R about an axis passing through its center is givne by $I = \frac{1}{2}MR^2$. What would be the moment of inertia of the cylinder about an axis passing through its edge?

Solution:

$$I_{P'} = I_{cm} + Md^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

$$I_{P'} = I_{cm} + Md^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

59. The moment of inertia of an object of mass 4 kg is 20 kgm^2. Use the parallel axis theorem to find the moment of inertia of the object about an axis that has displacement $\begin{bmatrix}3\\4\end{bmatrix}_m$.

Solution: $d = \sqrt{3^2 + 4^2} = 5$

$I_P = I_{cm} + Md^2 = 20 + 4(5)^2 = 120\ kgm^2$

Angular Momentum

When an object of mass m moves with a speed v in a straight line, we say that it has a linear momentum P = mv. When the same object moves with an angular speed ω along a circle of radius r, we say that it has angular momentum, L. Therefore,

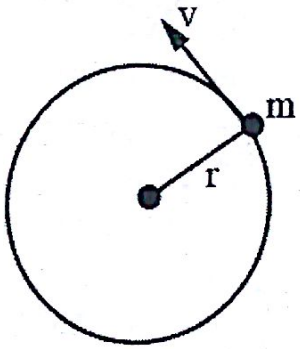
$$\boxed{L = I\omega}$$

The unit of angular momentum is kgm^2/s.

- If we consider a point mass m moving in a circle of radius r, then the angular momentum can be obtained as,

  $L = I\omega$, but $I = mr^2$ and $v = r\omega$

  $\Rightarrow \boxed{L = rmv = rp}$, $mv = P$



Which leads us to another possible definition that, angular momentum equals the vector product of the position vector, $\vec{r}$ and the linear momentum, $\vec{P}$,

$$\boxed{\vec{L} = \vec{r} \times \vec{P} = \vec{r} \times m\vec{v}}$$

- Angular momentum is a vector quantity whose direction is in the direction of the angular velocity.
  - The net external torque exerted on the body during a given time interval is given by:

    $\vec{\tau} = \dfrac{\Delta\vec{L}}{\Delta t}$, from this we can have;

    $\boxed{\tau\Delta t = \Delta L}$ ←Angular impulse

Conservation of Angular Momentum

Angular momentum is said to be conserved when its value doesn't change with time.

Law of conservation of Angular momentum states that "In the absence of external net torque, the total angular momentum of a system remains constant."

That is, if $\sum\tau = 0$ then

$$\boxed{\sum L_i = \sum L_f} \text{ Or } \boxed{I_i\omega_i = I_f\omega_f}$$

☞ When the angular momentum is conserved, then the angular impulse becomes zero.

- An ice skater, spins and pulls her arms inwards to speed up, a diver springs into air and folds his body to speed his rotation, a hurricane draws circulating air at ground level and rise so high. In all these cases angular momentum is at work.

Conceptual Examples

60. Can a particle moving in a straight line have angular momentum?
Explanation: A particle has angular momentum about the origin as long as its line of motion does not pass through the origin. Its angular momentum is zero only if the line of motion passes through the origin, in this case $\vec{r}$ is parallel to $\vec{v}$ and $L = r \times P = r \times mv = rmv \sin\theta$, $\theta = 0 \Rightarrow L = 0$.

61. A particle moves in a straight line and you are told that the net torque acting on it is zero about some unspecified origin. Does this necessarily mean that the net force on the particle is zero? Can you say that its velocity is constant?
Explanation: The net force is not necessarily zero. If the line of action of the net force passes through the origin, the net torque about an axis through the origin is zero while the net force is non-zero. Because the net force is none zero, you cannot conclude that its velocity is constant.

62. A student sits on a rotating chair holding a pair of weight. The student is set in rotating motion with his arms outstretched. What will happen to the angular speed of the system as he pulled inward his arms?
Explanation: Increased! Because, the weights are now closer to the axis of rotation, reducing the moment of inertia since $\sum\tau = 0$. $I_i\omega_i = I_f\omega_f$, but $I_f < I_i \Rightarrow \omega_f > \omega_i$.

63. If global warming continues, it's likely that some ice from the polar ice caps of the Earth will melt and the water will be distributed to the equator. If this occurs, what will happen to the length of the day? Will it increase or decrease?

Explanation: Assuming $\sum \tau = 0$, $L_i = L_f$

$I_i\omega_f = I_f\omega_f$, but $I_f > I_i$

$$\Rightarrow \omega_f < \omega_i \Rightarrow \frac{2\pi}{T_f} < \frac{2\pi}{T_i} \Rightarrow T_i < T_f$$

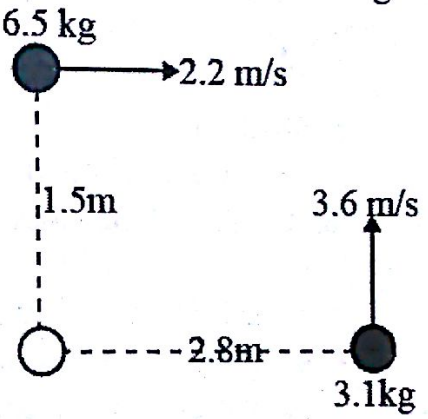
$\Rightarrow$ Length of the day increases!

64. Experienced cooks can tell whether an egg is raw or boiled by rolling it down a slope. How is this possible?

Explanation: The raw is like a hollow sphere and hence it will have larger rotational energy, thus it will reach the bottom after the boiled.

Illustrative Examples

65. Two bodies moving as shown below, what is their total angular momentum about point o?



Solution: $L = L_{\text{counter clockwise}} + L_{\text{clock wise}}$

$= (2.8)(3.6)(3.1) - (1.5)(2.2)(6.5) = 9.8\ \text{kgm}^2/\text{s}$

66. At a certain instant, the position of a 0.25 kg body is r = 2i – 2k, its velocity is v = -5i + 5k and a net force F = 4j is acting on it.

a) What is the angular momentum of the body about the origin?

b) What torque acts on it?

Solution:

a) $$L = \vec{r} \times \vec{v}m = \begin{vmatrix} i & j & k \\ 2 & 0 & -2 \\ -1.25 & 0 & 1.25 \end{vmatrix} = 0$$

b) $$\tau = \vec{r} \times \vec{F} = \begin{vmatrix} i & j & k \\ 2 & 0 & -2 \\ 0 & 4 & 0 \end{vmatrix} = (8i + 8k)N.m$$

67. What is the magnitude of the angular momentum about the Earth's center, of an 84 kg person on the equator due to the rotation of the earth?

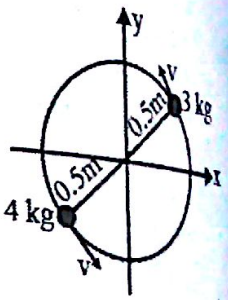
Solution: $L = rv\,m\sin\theta$, $v = \dfrac{2\pi r}{T}$, $T = 24hr$

$$L = (6400000)(465)(84),\quad v = \frac{2(3.14)(6400000)}{(24)(60)(60)} = 465m/s$$

$$L = 2.5 \times 10^{11}\ kgm^2/s$$

$$\text{Or, } L = I\omega = mr^2\left(\frac{2\pi}{(24)(60)(60)}\right) = 2.5 \times 10^{11} kgm^2/s$$

68. A light rigid rod 1m in length rotates about an axis perpendicular to its length and through its center. Two particles of masses 4 kg and 3 kg are connected to the ends of the rod. What is the angular momentum of the system if the speed of each particle is 5 m/s?



Solution: $L = L_1 + L_2 = (3)(0.5)(5) + (4)(0.5)(5) = 17.5\ kgm^2/s$

69. A horizontal platform in the shape of a circular disk rotates in a horizontal plane about a frictionless vertical axle. The platform has mass M = 100 kg and radius R = 2m. A student whose mass m = 60 kg walks slowly on the platform towards the center. If the angular speed of the system is 2 rad/s when the student is at the edge. Calculate the angular speed when the student has reached a point 0.5m from the center.

Solution: $I_i = I_P + I_s = \dfrac{1}{2}MR^2 + mR^2 = 440 kgm^2$

$$I_f = I_p + I_s = \frac{1}{2}MR^2 + mR^2 = 215 kgm^2$$

$$L_i = L_f \Rightarrow I_i\omega_i = I_f\omega_f \Rightarrow \omega_f = \frac{I_i}{I_f}\omega_i = 4.1\ rad/s$$

70. A merry – go – round of radius R = 2m has a moment of inertia I = 250 kgm^2 and is rotating at 10 rev/min. A 25 kg child jumps onto the edge of the merry – go – round. What is the new angular speed of the merry – go – round?

Solution: $I_i = 250\ kgm^2$

$$I_f = 250 + mr^2 = 250 + (25)(2)^2 = 350\ kgm^2$$

$$L_i = L_f \Rightarrow I_i\omega_i = I_f\omega_f$$

$$\omega_f = \frac{I_i}{I_f}\omega_i = \frac{250}{350}(10) = 7.1 rev/min$$

71. An artificial satellite is placed into an elliptical orbit about the earth. If its speed is 8450 m/s at a distance of 8.37×10^6m from the center of the earth, what will be its speed at a distance of 25.1×10^6m from the earth's center?

Solution: $I_1\omega_1 = I_2\omega_2 \Rightarrow mr_1^2\left(\frac{v_1}{r_1}\right) = mr_2^2\left(\frac{v_2}{r_2}\right) \Rightarrow v_2 = \frac{r_1 v_1}{r_2} = 2820 m/s$

72. For a classroom demonstration, a student sits on a piano stool holding masses on each hand. Initially, the student holds his arms out stretched and spins about the axis of the stool with an angular speed of 3.74 rad/s. The moment of inertia in this case is 5.33 kgm^2. While still spinning, the student pulls his arms into his chest, reducing the moment of inertia to 1.6 kgm^2. What is the student's angular speed now?

Solution: $L_I = L_f \Rightarrow I_i\omega_i = I_f\omega_f \Rightarrow \omega_f = 12.5$ rad/s

73. The angular momentum of a particle is:

A. independent of the specific origin of coordinates
B. zero when its position and momentum vectors are parallel
C. zero when its position and momentum vectors are perpendicular
D. not zero for none zero position and momentum vectors

Solution: $L = \vec{r} \times \vec{P} = 0$ when $L = \vec{r}\ //\ \vec{P}$. **Answer: B**

74. A 1.5 kg particle moves in the xy lane with velocity v = (4.2i – 3.6j) m/s. What is the angular momentum when its position vector is r = (1.5i + 2.2j)m?

A. $-22\ \hat{k}\ kgm^2/s$ C. $-14.6\hat{k}\ kgm^2/s$

B. $14.6\hat{k}\ kgm^2/s$ D. $22\ \hat{k}\ kgm^2/s$

Solution: $L = m(\vec{r} \times \vec{v}) = -22\hat{k} kgm^2/s$ **Answer: A**

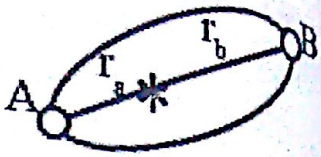
5. A planet of mass m is circling a star in an elliptical orbit as shown below. If its velocity at point A is 60,000 m/s and $r_b = 8$ ra, what is its velocity at point B?

A. 7500 m/s B. 8571 m/s C. 60,000 m/s D. 488,000 m/s

Solution: $L_a = L_b$

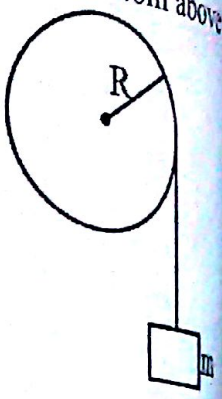
$mv_a r_a = mv_b r_b$

$$v_B = \frac{r_a}{r_b} v_A = \frac{r_a}{8r_a} v_A \Rightarrow v_B = 7500 m/s$$



76. An object with a mass of m = 3kg is attached to the free end of a light string wrapped around a pulley of radius R = 0.5m and moment of inertia I = 0.75 kgm^2. The pulley is free to rotate in a vertical plane about a fixed frictionless axis passing through its center. The suspended object is released from rest at 3.6m above the floor. What is the angular momentum of the pulley at the time when the object strikes the floor (UEE: 2007)

Answer: A



A. 4.5 kgm^2/s C. 12.3 kgm^2/s

B. 9 kgm^2/s D. 18 kgm^2/s

Solution: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$$mgh = \frac{1}{2}m\omega^2 r^2 + \frac{1}{2}I\omega^2 \Rightarrow \omega = 12 \text{ rad/s}$$

$$L = I\omega = (0.75)(12) = 9 \text{ kgm}^2/\text{s}$$

Answer: B

Center of Mass

There is one point in any system of bodies that has special significance. The center of mass, cm. One of the reasons the center of mass is so special is the fact that, in many ways, a system behaves as if all of its mass were concentrated there. As a result, a system can be balanced at its center of mass.

> **Center of Mass**: Is a point at which all of the mass of the body is considered to be concentrated.

For a system of n particles in the xy plane with masses $m_1, m_2 \ldots$ and co-ordinates from the arbitrary origin $(x_1, y_1), (x_2, y_2), \ldots (x_n, y_n)$ the x and y co-ordinate of the center of mass is given by. x coordinate of the center of mass

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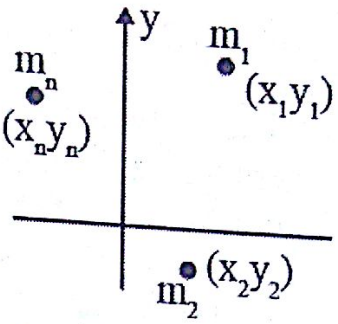
$$x_{cm} = \frac{m_1x_1 + m_2x_2 + ... + m_nx_n}{m_1 + m_2 + ... + m_n}$$

y coordinate of the center of mass

$$y_{cm} = \frac{m_1y_1 + m_2y_2 + ... + m_ny_n}{m_1 + m_2 + ... + m_n}$$



The position of the center of mass the system is then Cm = (x_{cm}, y_{cm}).

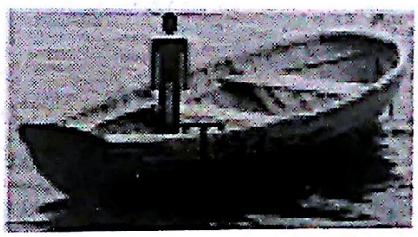
☞ In simple terms, the center of mass is a point that represents the average location for the total mass of a system.

Conceptual Example

77. A projectile is fired into the air and suddenly explodes into several fragments. What can be said about the motion of the center of mass of the fragments, after the explosion?

Explanation: The center of mass of the system will follow parabolic path as the unexploded projectile. And the center of mass of the system of fragments will land at the same place as the unexploded projectile would land.

78. A boy stands at one end of a small boat that is stationary relative to the shore. He then walks to the opposite end of the boat. Does the boat move? Explain

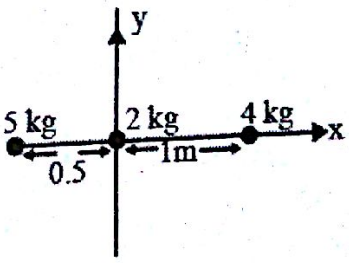


Explanation: Yes, the boat moves towards the shore. The center of mass of the boat and boy system remains fixed relative to the shore. As the boy moves away from the shore, the boat must move towards the shore such that the center of mass of the system remains fixed in position.

Illustrative Examples

79. Find the center of mass of the system of particles shown.

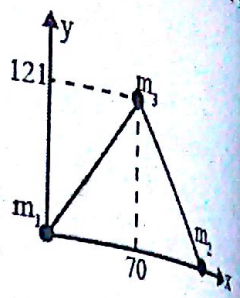
Solution: $x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$



$$= \frac{(5)(-0.5) + (2)(0) + (4)(1)}{5 + 2 + 4}$$

$$m = x_{cm} = 0.134m$$

80. The particles of mass $m_1 = 1.2$ kg, $m_2 = 2.5$kg and $m_3 = 3.4$ kg are located at the corners of an equilateral triangle of side s = 14[illegible] cm. Where is the center of mass?



Solution: $x_{cm} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$

$$= \frac{(1.2)(0) + (2.5)(140) + (3.4)(70)}{7.1}$$

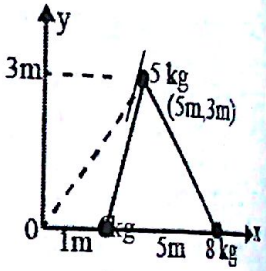
$$= 83 \text{ cm}$$

$$y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$= \frac{(1.2)(0) + (2.5)(0) + (3.4)(121)}{7.1} = 58 \text{ cm}$$

$$m = (x_{cm}, y_{cm}) = (83, 58)$$

81. Locate the center of mass of the three particle system.



Solution: $r_{cm}(m_1 + m_2 + m_3)$

$= m_1(x_1, y_1) + m_2(x_2, y_2) + m_3(x_3, y_3)$

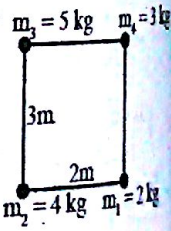
$r_{cm}(6 + 8 + 5)$

$= 6(1i) + 8(5i) + 5(5i + 3j)$

$19\, r_{cm} = 71i + 15j$

$\Rightarrow r_{cm} = (3.7, 0.8)m$

82. Locate the center of mass of particles $m_1 = 2$kg, $m_2 = 4$ kg, $m_3 = 5$ kg and $m_4 = 3$ kg each on the vertices of a 2m x 3[illegible] rectangle and m_1 is at the origin.



Solution: $x_{cm} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$

$$= \frac{2(0) + 4(-2) + 5(-2) + 3(0)}{14} = -1.28m$$

$$y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4}{m_1 + m_2 + m_3 + m_4}$$

$$= \frac{2(0) + 4(0) + 5(3) + 3(3)}{14} = 1.71m = (-1.28, 1.71)m$$

83. Find the position of the center of mass of the system shown below.

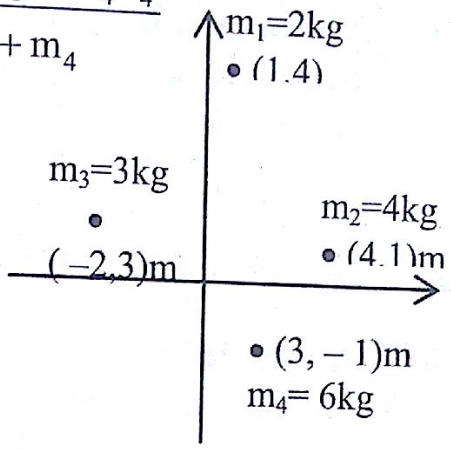
Solution: $x_{cm} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$

$= \dfrac{2(1) + 4(4) + 3(-2) + 6(3)}{2 + 4 + 3 + 6} = 2m$

$y_{cm} = \dfrac{m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4}{m_1 + m_2 + m_3 + m_4}$

$= \dfrac{2(4) + 4(1) + 3(3) + 6(-1)}{2 + 4 + 3 + 6} = 1m$

c. $m = (x_{cm}, y_{cm}) = (2, 1)m$



Additional Examples

Circular Motion

84. A particle moves along a circle of radius $\dfrac{20}{\pi}$m. If the velocity of the particle is 80 m/s at the end of the second revolution after motion has begun, the tangential acceleration is:

A. 40 m/s^2 B. 640πm/s^2 C. 160Πm/s^2 D. 40πm/s^2

Solution: $\omega^2 = \omega_o^2 + 2\alpha\theta$, $\omega_o = 0$, $\omega = \dfrac{v}{r}$ and $\alpha = \dfrac{a}{r}$

$\omega^2 = 2\alpha\theta$

$\dfrac{v^2}{r^2} = 2\dfrac{a}{r}\theta$ or $a = \dfrac{v^2}{2r\theta} = \dfrac{80^2}{2\left(\dfrac{20}{\pi}\right)4\pi} = 40m/s^2$

Answer: A

85. A one kg stone attached to the end of a 60 cm chain is revolving at the rate of 3 rev/s. If after 30 second, it is making only one revolution per second, find the mean torque

A. 0.45Nm B. 0.25Nm C. 0.35Nm D. 0.15Nm

Solution: $\tau = I\alpha = mr^2\dfrac{\Delta\omega}{\Delta t} = (1)\left(\dfrac{60}{100}\right)^2\dfrac{2\pi(3) - 2\pi(1)}{30}$

$\tau = 0.15$ Nm **Answer: D**

86. A particle is acted up on a force of constant magnitude which is always perpendicular to the velocity of the particle. It follows that

A. Its kinetic energy is constant
B. Its acceleration is constant
C. Its velocity is constant
D. It moves in a straight line

Answer: A

87. Which of the following statement is FALSE for a particle moving in a circle with a constant angular speed?
A. The acceleration vector points to the center of the circle.
B. The acceleration vector is tangent to the circle.
C. The velocity vector is tangent to the circle.
D. The velocity and acceleration vectors are perpendicular to each other.

Answer: B

88. A particle moves in a circle of radius 0.25m at two revolution per second. The acceleration of the particle in m/s^2 is?
A. π^2 B. $8\pi^2$ C. $4\pi^2$ D. $2\pi^2$

Solution: $a = \omega^2 r = 4\pi^2 v^2 r = 4\pi^2(2)^2(0.25) = 4\pi^2 m/s^2$

Answer: C

89. In uniform circular motion, the factor that remains constant is?
A. Linear velocity
B. Centripetal force
C. Acceleration
D. Speed

Answer: D

90. A ball tied to a string is swung in a vertical circle. Which of the following remains constant?
A. Tension in the string
B. Speed of the ball
C. Centripetal force
D. Earth's pull on the ball

Answer: D

91. When the road is dry and the coefficient of friction is μ, the maximum speed of a car in a circular path is 10 m/s. If the road becomes wet and $\mu' = 0.5\mu$. What is the maximum speed permitted?

A. 5 m/s B. 10 m/s C. $10\sqrt{2}$m/s D. $5\sqrt{2}$m/s

Solution: $v = \sqrt{\mu gr}$ and $v' = \sqrt{\mu' gr}$

Taking ratio, $\frac{v'}{v} = \sqrt{\frac{\mu'}{\mu}} \Rightarrow v' = 5\sqrt{2}$m/s

Answer: D

92. When a particle moves in a circle with a uniform speed
A. Its velocity and acceleration are both constant.
B. Its velocity is constant but the acceleration changes.
C. Its acceleration is constant but the velocity changes.
D. Its velocity and acceleration both change.

Answer: D

93. A stone is tied to a string and is rotated in a circle with the other end of the string as the center. If the string breaks, the stone will move

A. Towards the center C. Along a tangent
B. Away from the center D. Will stop **Answer: C**

94. A particle is projected so as to just move along a vertical circle. The ratio of the tensions in the string when the particle is at the lowest and highest point on the circle is?

A. 1 B. finite C. zero D. infinite

Hint: As per the condition in the problem, tension at the highest point is zero. $\frac{T_L}{T_H} = \infty$.

Answer: D

95. A coin placed on a rotating turn table just slips if it is placed at a distance of 4cm from the center. If the angular velocity of the turn table is doubled, it will just slip at a distance of

A. 1cm B. 2cm C. 4cm D. 8cm

Solution: $f = mr^2\omega^2$, when ω doubled, r is reduced by a factor of 4.

Answer: A

96. A mass of 1 kg is connected to a string of 1m and whirled in a circle in 1 second. The tension in the string is

A. π^2N B. $2\pi^2$N C. $3\pi^2$N D. $4\pi^2$N

Solution: $T = mr\omega^2 = (1)\,(1)4\pi^2(1)^2 = 4\pi^2$N **Answer: D**

97. A particle moves with uniform speed in a circular path. The angle between instantaneous velocity and acceleration is

A. 0° B. 180° C. 90° D. 45°

Explanation: Speed is towards the tangent and acceleration is towards the center. **Answer: C**

98. The radius of the path of a body in uniform circular motion is doubled. The centripetal force needed, if its speed remains the same is

A. Half as great as before C. Twice as great
B. Same as before D. Four times as great

Hint: $F = \frac{mv^2}{r} \Rightarrow F \sim \frac{1}{r}$ **Answer: A**

99. If the frequency of the rotating platform is f and the distance of a boy from the center is r, what is the area swept out per second by the line connecting the boy to the center?

A. πrf B. $2\pi rf$ C. $\pi r^2 f$ D. $2\pi r^2 f$

Hint: Area per second = (Area swept in one rotation). (Number of rotations per time.)

100. A fiter air craft is moving in a vertical circle of radius r. The minimum velocity at the height point is? Answer: C

A. $\sqrt{gr}$ B. $\sqrt{0.05gr}$ C. $\sqrt{2gr}$ D. $\sqrt{3gr}$

Hint: $\frac{mv^2}{r} = mg \Rightarrow v = \sqrt{gr}$

101. A body crosses the top most point of a vertical circle with critical speed. What will be its acceleration when the string is horizontal? Answer: A

A. g B. 2g C. 3g D. 6g

Hint: $v^2 - gr = 2gr \Rightarrow v^2 = 3gr$ or

$$\frac{v^2}{r} = 3g = a$$

102. When a bucket containing water is rotated fast in a vertical circle of radius R, the water in the bucket doesn't spill provided. Answer: C

A. The bucket is whirled with a maximum speed of $\sqrt{2gR}$.

B. The bucket is whirled with a minimum speed of $\sqrt{\frac{1}{2}gR}$.

C. The bucket is having a rpm of $\sqrt{900g/\pi^2R}$.

D. The bucket is having a rpm of $\sqrt{3600g/\pi^2R}$.

Solution: At the highest point, $mg = mR\omega^2 \Rightarrow \omega = \sqrt{\frac{g}{R}} = 2\pi f$

$$f = \frac{1}{2\pi}\sqrt{g/R} = \sqrt{\frac{g}{4\pi^2R}}, \quad rpm = 60\,f = 60\sqrt{g/4\pi^2R}$$

$$rpm = \sqrt{\frac{900g}{\pi^2R}}$$

103. The string of a pendulum is horizontal. The mass of the bob attached to it is m. Now the string is released. The tension in the string in the lowest position is? Answer: C

A. mg B. 2mg C. 3mg D. 4mg

Solution: $T - mg = \frac{mv^2}{r} \Rightarrow T = mg + \left(\frac{m}{r}\right)(2gr) = 3mg$

Answer: C

104. A spinning wheel initially has an angular velocity of 20 rad/s north. Twenty second later, its angular velocity is 20 rad/s south. If the angular acceleration is constant, what is the angular displacement over 20 second?

A. 0 B. 10 rad C. 20 rad D. 60 rad

Explanation: The average angular velocity is zero. The only effect of angular acceleration over the 20 second is to reverse axis of rotation.

Answer: A

105. Through how many revolutions must the 60 cm diameter wheel of a car turn as the car travels 2.5km?

A. 2500 B. 2000 C. 1000 D. 1325

Solution: $\theta = \frac{s}{r}$, No. of revolutions $\frac{\theta}{2\pi} = \frac{s}{2\pi r} = \frac{(2500)(100)(7)}{(30)(2)(22)}$

$= 1325$

Answer: D

106. The circular motion of a particle with constant speed is

A. Periodic and simple harmonic.
B. Neither periodic nor simple harmonic.
C. Periodic but not simple harmonic.
D. Simple harmonic but not periodic.

Explanation: The motion is repeating after regular intervals of time. So it is periodic. But, the requirements of SHM are not met. So, motion is periodic only.

Answer: C

107. The moment of inertia of a body does not depend on.

A. Mass of the body. C. Angular velocity of the body.
B. Distribution of the mass. D. Axis of rotation of the body.

Answer: C

Moment of Inertia

108. The radius of gyration of a body about an axis at a distance of 6 cm, from its center of mass is 10 cm. The radius of gyration about a parallel axis through its center of mass is

A. 2cm B. 4cm C. 8cm D. 12cm

Solution: $I = I_g + Md^2$ or $Mk^2 = Mk_g^2 + Md^2$

$k^2 = k_g^2 + d^2 \Rightarrow 10^2 = k_g^2 + 6^2 \Rightarrow k_g = 8cm$

Answer: C

109. Two discs A and B have same mass and same thickness but A is made of aluminum and B of lead. About the central axis, the moment of inertia of

A. Both the discs are same
B. Disc A is larger
C. Disc B is larger
D. The discs can't be same

Explanation: In the case of aluminum, density is less. The distribution of mass is farther from the axis. **Answer: B**

110. A closed tube, partly filled with a liquid and set horizontal is rotated about a vertical axis passing through its center. In the process, the moment of inertia of the system about its axis would

A. Increases always
B. Decrease always
C. Remain constant
D. Increase if tube is less than half filled

Explanation: Centrifugal force drives liquid outward thereby increasing moment of inertia. **Answer: A**

111. The moment of inertia of a solid cube is minimum about an axis

A. Passing through its center & perpendicular to one of the faces.
B. Passing through a corner and parallel to one edge.
C. Passing through a corner and parallel to diagonal of one face.
D. Coinciding with the diagonal of the cube.

Hint: Distribution of mass is nearest in cade d. **Answer: D**

112. If I_1, I_2 and I_3 are moment of inertia of solid sphere, hollow sphere and a spherical shell of same mass and radius, which of the following statements holds good?

A. $I_1 > I_2 > I_3$
B. $I_3 > I_2 > I_1$
C. $I_2 > I_1 > I_3$
D. $I_2 > I_3 > I_1$

Solution: $I_1 = \frac{2}{5}MR^2, I_2 = \frac{2}{5}M\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3}, I_3 = \frac{2}{3}MR^2$

$\therefore I_3 > I_2 > I_1$

Answer: B

113. A uniform rod is $\sqrt{3}$ m long and weighs 1 kg. Its radius of gyration about an axis through its one end and perpendicular to its length is

A. 0.25m B. 0.5m C. 0.75m D. 1m

Solution: $Mk^2 = \frac{m\ell^2}{3} \Rightarrow k = \frac{\ell}{\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{3}} = 1\text{m}$

Answer: D

114. For the same total mass, which of the following will have the largest moment of inertia about an axis passing through the center of gravity and perpendicular to the plane of the body?

A. A ring of radius ℓ C. a square lamina of side 2ℓ

B. A disc of radius ℓ D. four rods forming square of side 2ℓ

Hint: Distribution of mass is farthest in case d. **Answer: D**

115. Which of the following has, the highest moment of inertia when each of them has the same mass and the same radius?

A. A hollow sphere about one of its diameter.
B. A solid sphere about one of its diameters.
C. A disc about its central axis perpendicular to the plane of the disc.
D. A ring about its central axis perpendicular to the plane of the ring.

Hint: Distribution of mass is farthest in case d. **Answer: D**

116. Two discs, have same mass and thickness. Their materials are of densities ρ_1 and ρ_2. The ratio of their moment of inertia about central axis will be?

A. $\rho_1 : \rho_2$ B. $\rho_1\rho_2 : 1$ C. $1 : \rho_1\rho_2 : 1$ D. $\rho_2 : \rho_1$

Solution: $M = \pi R^2 t \rho$, m and t same

$R^2\rho = \text{constant or } R^2 = \frac{1}{\rho}$ again, $I = \frac{1}{2}MR^2$ or $I \sim R^2$ or $I \sim \frac{1}{\rho}$

Answer: D

117. Two solid spheres are made of the same material. The ratio of their diameter is 1:2. The ratio of their moments of inertia about their respective diameter is

A. 1: 1 B. 1: 4 C. 1: 16 D. 1: 32

Solution: $I = \frac{2}{5}MR^2 = \frac{2}{5}\left(\frac{4}{3}\pi R^3\rho\right)R^2$ or $I \sim R^5$

$\therefore \frac{I_1}{I_2} = \frac{1^5}{2^5} = \frac{1}{32}$ **Answer: D**

118. A circular disc X of radius R is made from an iron plate of thickness t, and another disc Y of radius 4R is made from an iron plate of thickness 0.25t. Then, the relation between moment of inertias is

A. $I_y = 64I_x$ C. $I_y = 16I_x$

B. $I_y = 32I_x$ D. $I_y = I_x$

Solution: $I = \frac{1}{2}MR^2 = \frac{1}{2}(\pi R^2 t\,\rho)R^2 \Rightarrow I \sim tR^2$

$$\frac{I_x}{I_y} = \frac{tR^4}{\frac{t}{4}(4R)^4} \text{ or } I_y = 64\,I_x$$

Answer: A

119. Two circular loops A and B of radii r_A and r_B are made from the same uniform wire. The ratio of their moments of inertia about axes passing through their centers and perpendicular to their planes is $I_B/I_A = 8$. Then, r_B/r_A is,

A. 2 B. 4 C. 6 D. 8

Solution: Mass of A, $M_A = 2\pi r_A m$, m is mass per length and $M_B = 2\pi r_B m$

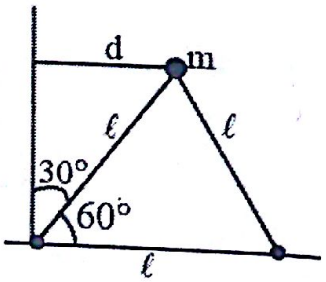
$$\frac{I_B}{I_A} = \frac{M_B r_B^2}{M_A r_A^2} \Rightarrow \frac{r_B}{r_A} = \sqrt[3]{\frac{I_B}{I_A}} \Rightarrow \sqrt[3]{8} = 2$$

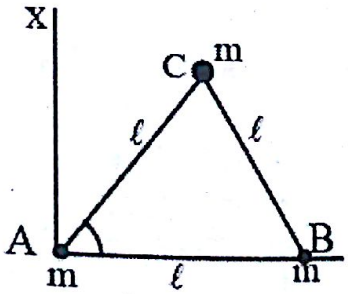
Answer: A

120. What is the moment of inertia of the system about a line AX perpendicular to AB and in the plane of ABC

A. $2m\ell^2$ B. $\frac{5}{4}m\ell^2$ C. $\frac{3}{2}m\ell^2$ D. $\frac{3}{4}m\ell^2$

Solution:





$$\sin 30° = \frac{d}{\ell} \text{ or } \frac{1}{2} = \frac{d}{\ell} \text{ or } d = \frac{\ell}{2}$$

$$\therefore I = m\ell^2 + \frac{m\ell^2}{4} = \frac{5m\ell^2}{4}$$

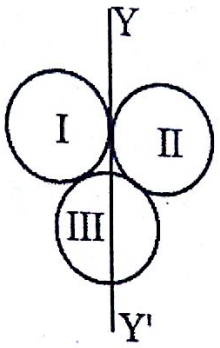
Answer: B

121. What is the moment of inertia of the three identical rings each of mass M and radius R above the YY' axis?

A. $\frac{1}{2}MR^2$ B. MR^2 C. $\frac{5}{2}MR^2$ D. $\frac{7}{2}MR^2$

Solution: Moment of inertia of ring I & II about

$$YY' = 2\left(\frac{3}{2}MR^2\right) = 3MR^2$$



Moment of inertia of ring III about

$$YY' = \frac{1}{2}MR^2$$

Total moment of inertia

$$= 3MR^2 + \frac{1}{2}MR^2 = \frac{7}{2}MR^2$$

Answer: D

122. The diameter of a flywheel increased by 1%. The percentage increase in moment of inertia about its axis passing through the center will be

A. 2% B. 2.5% C. 1.5% D. 3%

Solution: $I = \frac{1}{2}MR^2, I \sim R^2$ percentage increase in moment of inertia $= 2 \times 1\% = 2\%$. **Answer: A**

123. A copper ring of mass M and radius R is suspended from a horizontal nail in a vertical wall. The moment of inertia of the ring about the nail is

A. MR^2 B. $2MR^2$ C. $4MR^2$ D. $8MR^2$

Solution: $I = MR^2 + MR^2 = 2MR^2$ **Answer: B**

124. The spokes are used in a cycle wheel because it

A. Increases the strength of the wheel.
B. Increases the moment of inertia of the wheel.
C. Gives a better shape to the wheel.
D. Has become conventional

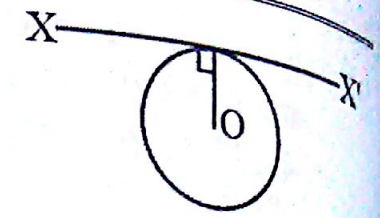
Answer: B

125. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with center at 0. The moment of inertia of the loop about XX' is

A. $\frac{\rho L^3}{8\pi^2}$ B. $\frac{\rho L^3}{16\pi^2}$ C. $\frac{5\rho L^3}{16\pi^2}$ D. $\frac{3\rho L^3}{8\pi^2}$

Solution: $I = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

$= \frac{3}{2}(L\rho)\left(\frac{L}{2\pi}\right)^2 = \frac{3\rho L^3}{8\pi^2}$



Answer: D

126. Two discs A and B having same mass and thickness have densities 7 g/cm^3 at 9 g/cm^3 respectively. The ratio of the moment of inertia of A and B about an axis perpendicular to plane and passing through center is

A. $\frac{7}{9}$ B. $\frac{9}{7}$ C. 63 D. $\frac{1}{63}$

Solution: $m_A = \pi r_A^2\, t\, d_A$, $m_B = \pi r_B^2 t\, d_B$

but, $m_A = m_B \Rightarrow r_A^2 d_A = r_B^2 d_B$

again, $\frac{I_A}{I_B} = \frac{r_A^2}{r_B^2} = \frac{d_B}{d_A} = \frac{9}{7}$

Answer: B

127. The moment of inertia of a uniform semicircular disc of mas M and radius r about a line perpendicular to the plane of the disc through the center is

A. $\frac{1}{2}Mr^2$ B. Mr^2 C. $\frac{2}{5}Mr^2$ D. $\frac{1}{4}Mr^2$

Solution: Mass of disc = 2M. and

Moment of inertia of disc $= \frac{1}{2}(2M)R^2 = MR^2$

Moment of inertia of semi – disc $= \frac{1}{2}MR^2$

Answer: A

Rotational Mechanics

128. A solid sphere is rotating in free space. If the radius of the sphere is increased keeping mass same, which one of the following will not be affected?

A. Angular velocity C. Moment of Inertia
B. Angular momentum D. Rotational kinetic energy

Hint: No external force and no external torque in free space.

Answer: B

129. In an orbital motion, the angular momentum vector is

A. Along the radius vector.

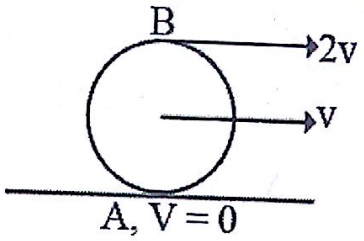
B. Parallel to the linear momentum.
C. In the orbital plane.
D. Perpendicular to the orbital path.

Answer: D

130. The direction of the angular velocity vector is along
A. The tangent to the circular path.
B. The inward radius.
C. The outward radius.
D. The axis of rotation.

Answer: D

131. A disc rolls over a horizontal floor without slipping with a linear speed of 5 m/s. The linear speed of a particle on its 4im, wrt the floor, when it is in its highest position is?



A. 10 m/s C. 2.5cm/s
B. 5cm/s D. 0

Explanation: Velocity is twice since the highest point is at a double distance.

Answer: A

132. A fan of moment of inertia 0.6 kgm^2 is turned up to a working speed of 0.5 rps. The angular momentum of the fan is
A. $0.6\pi\ kgm^2/s$ C. $3kg\ m^2/s$
B. $6\ kg\ m^2/s$ D. $\pi/6\ kgm^2/s$

Solution: $L = I\omega = (0.6)(2\pi)(0.5) = 0.6\pi\ kgm^2/s$

Answer: A

133. A sphere rolling on a horizontal surface gradually slows down and stops. The frictional force has the tendency to
A. Increase the linear velocity.
B. Increase the angular velocity.
C. Increase the linear momentum.
D. Decrease the angular velocity.

Answer: B

134. A rigid body of moment of inertia I has an angular acceleration α. If the power supplied to the body is P, its instantaneous angular velocity is
A. $P I\alpha$ B. $P/I\alpha$ C. PI/α D. $P\alpha/I$

Solution: $P = \tau\omega = I\,\alpha\,\omega \Rightarrow \omega = P/I\alpha$

Answer: B

135. A solid sphere of mass 1 kg rolls on a table with linear speed 1 m/s. Its total kinetic energy is?
A. 1J B. 0.5J C. 0.7J D. 1.4J

Solution: $k_E = k_{Er} + k_{Et} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$$= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{5}mR^2\right)\left(\frac{v}{R}\right)^2 = \frac{7}{10}mV^2$$

$k_E = 0.7\ J$

Answer: C

136. The angular momentum of a 10g particle moving with velocity 5i m/s and having position vector 10i + 6jm about the origin is?

A. $-0.1\,\hat{k}$ Js B. $-0.2\,\hat{k}$ Js C. $-0.3\,\hat{k}$ Js D. $-0.4\,\hat{k}$ Js

Solution: $\vec{L} = \vec{r} \times \vec{P} = m(\vec{r} \times \vec{v}) = -0.3\,\hat{k}$Js

Answer: C

137. A wheel whose moment of inertia is 12 kgm^2 has an initial angular velocity of 40 rad/s. A constant torque of 20Nm acts on the wheel. The time in which the wheel is accelerated to 100 rad/s is

A. 72s B. 16s C. 8s D. 36s

Solution: $\tau = I\alpha$, and $\alpha = \frac{\omega - \omega_o}{t} \Rightarrow t = 36s$

Answer: D

138. A meter stick is held vertically with one end on the floor and is then allowed to fall. The speed of the other end when it hits the floor ($g = 9.8\ m/s^2$)

A. 3.2 m/s B. 5.4 m/s C. 7.6m/s D. 9.2m/s

Solution: $\frac{1}{2}I\omega^2 = \frac{1}{2}mg\ell \Rightarrow \frac{m\ell^2}{3}\left(\frac{v^2}{\ell^2}\right) = mg\ell \Rightarrow v = 5.4m/s$

Answer: B

139. If the Earth shrinks to half the present radius, without any change in mass, then the duration of day and night becomes?

A. 24hr B. 12hr C. 6hr D. 3hr

Solution: $\frac{2}{5}MR^2\left(\frac{2\pi}{T}\right) = \text{constant}$ or $\frac{T}{R^2} = \text{constant}$

$$T'/R^2/4 = T'/R^2 \text{ or } \frac{4T'}{R^2} = \frac{24}{R^2} \Rightarrow T' = 6hr$$

Answer: C

140. A constant torque of 400Nm turns a wheel of moment of inertia 100 kgm^2 about an axis through its center. The angular velocity gained in 4 second is?

A. 12 rad/s B. 16 rad/s C. 20 rad/s D. 24 rad/s

Solution: $\tau = I\alpha \Rightarrow \alpha = 4\ rad/s^2$ and $\omega = \alpha t = 16$ rad/s

Answer: B

141. A particle undergoes uniform circular motion. About which point on the plane of the circle, will the angular momentum of the particle remain conserved?

A. Center of the circle
B. On the circumference
C. Inside the circle.
D. Outside the circle

Answer: A

142. The motion of planets in the solar system is an example of the conservation of

A. Mass
B. Linear momentum
C. Angular momentum
D. Energy

Answer: C

143. A fly wheel is a uniform disc of mass 72 kg and radius 50 cm, when it is rotating at the rate of 70 rpm, its kinetic energy is?

A. 142j B. 242J C. 342J D. 300J

Solution: $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 = \frac{1}{4}mr^2\omega^2$

$$= \frac{1}{4}(72)\left(\frac{50}{100}\right)^2\left(\frac{2\pi \times 70}{60}\right)^2 = 242J$$

Answer: B

144. A wheel having moment of inertia 2 kgm^2 about its vertical axis rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in one minute would be?

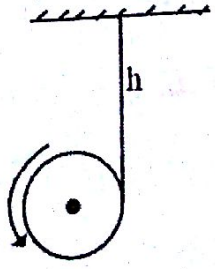
A. $\pi/2$ Nm B. $\pi/15$ Nm C. $\pi/18$ Nm D. $\frac{2\pi}{15}$ Nm

Solution: $\tau = I\alpha = \frac{I\omega}{t} = \frac{I2\pi f}{t} = \frac{2(2\pi)(60)}{(60)(60)} = \frac{\pi}{15}$ Nm

Answer: B

145. A disc like reel with massless thread unrolls itself while falling vertically downward. The acceleration of its fall is?

A. Zero B. g C. $\frac{2}{3}g$ D. $g/2$



Solution: $P_E = k_E$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\frac{v^2}{r^2}$$

$$\Rightarrow v^2 = \frac{4}{3}gh$$

Comparing with $v^2 = 2ah$, we get

$$a = \frac{2}{3}g$$

Answer: C

146. A flywheel of mass 25kg has a radius of 0.2m. What force should be applied tangentially to the rim of the flywheel so that it acquires an angular acceleration of 2 rad/s^2?

A. 1N B. 3N C. 5N D. 8N

Solution: $\tau = FR = I\alpha$ or $F = \frac{I\alpha}{R} = \frac{1}{2}\frac{MR^2}{R}\alpha$

$$F = \frac{1}{2}MR\alpha = \frac{1}{2}(25)(0.2)(2) = 5N$$

Answer: C

147. A rod of length 1m and mass $\frac{1}{2}$kg rotates at angular speed 6 rad/s about one of its ends. The kinetic energy of the rod is?

A. 1J B. 2J C. 3J D. 4J

Solution: $\frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{3}m\ell^2\right)\omega^2 = \frac{1}{6}\left(\frac{1}{2}\right)(1)(6)^2 = 3J$ **Answer: C**

148. A wheel of moment of inertia 30 kgm^2 is rotating at 10 rev/min. The work done in increasing its speed to 5 times its initial value will be

A. $10\pi^2$J B. $20\pi^2$J C. $30\pi^2$J D. $40\pi^2$J

Solution: $W = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_o^2 = \frac{1}{2}I4\pi^2(v^2 - u^2)$

$$= \frac{1}{2}(30)(4\pi^2)\left(\frac{25}{36} - \frac{1}{36}\right) = 40\pi^2 J$$

Answer: D

149. If the angular momentum of a body increases by 50%, its kinetic energy of rotation increases by

A. 50% B. 125% C. 200% D. 100%

Solution: L is increased by a factor of $\frac{3}{2}$, k_E is increased by a factor of $\frac{9}{4}$. % tage increase $= \frac{\frac{9}{4} - 1}{1} \times 100 = 125$

Answer: B

150. A drum of radius R and mass M rolls down without slipping along an inclined plane of angle θ. The frictional force:
A. Decreases the rotational motion.
B. Converts translational energy to rotational energy.
C. Decreases the rotational and translational motion.
D. Dissipates, energy as heat.

Hint: The frictional force provides the torque required for rotational motion.

Answer: B

151. A flywheel at rest is to reach an angular velocity of 36 rad/s in 6 second, with a constant angular acceleration. The total angle turned through during this interval is
A. 216rad B. 144rad C. 108rad D. 72 rad

Solution: $\theta = \left(\frac{\omega + \omega_o}{2}\right)t = \left(\frac{0+36}{2}\right)6 = 108 \text{ rad}$ **Answer: C**

152. A body rolls down an inclined plane. If its kinetic energy of rotational motion is 40% of its kinetic energy of translation, then the body is
A. Ring
B. Solid sphere
C. Solid disc
D. Cylinder

Solution: $\frac{40}{100} = \frac{\frac{1}{2}I\omega^2}{\frac{1}{2}mv^2} = \frac{\frac{Iv^2}{R^2}}{mv^2} \Rightarrow I = \frac{2}{5}mR^2 \rightarrow$ solid sphere

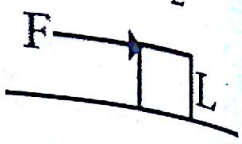
Answer: B

153. A body is acted upon by a constant torque. In 4 sec, its angular momentum changes from L to 4L. What is the magnitude of the torque?
A. $\frac{3}{4}L$ B. 3L C. $\frac{1}{4}L$ D. 12L

Solution: $\tau = \frac{\Delta L}{t} = \frac{4L - L}{4} = \frac{3}{4}L$ **Answer: A**

154. A cubical block of side L rests on a rough horizontal surface. A force F is applied on the block as shown. What is the minimum force required to topple the block?
A. mg B. mg/2 C. mg/4 D. $mg(1-\mu)$

Solution: The torque FL is clockwise and the torque $\frac{mgL}{2}$ of the block is anticlockwise.



Applying condition for rotational equilibrium $-FL + mg\frac{L}{2} = 0 \Rightarrow F = \frac{mg}{2}$

Answer: B

155. The torque of a force $(3i + 2j + 3k)N$ acts at a point with position vector of $(2i - 3j + 4k)m$ is

A. 6j B. $13\hat{k}$ C. $-17i$ D. $-17i + 6j + 13\hat{k}$

Solution: $\tau = \vec{r} \times \vec{F}$

Answer: D

156. When a sphere rolls without slipping, the ratio of kinetic energy of translation to its total kinetic energy is

A. 1: 7 B. 1: 2 C. 1: 1 D. 5: 7

Hint: $kE_t = \frac{1}{2}mv^2$ and $kE_{total} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

Answer: D

157. The angular velocity of a body is $\omega = 2i + 3j + 4\hat{k}$. A torque $\vec{\tau} = i + 2j + 3k$ acts on it. The rotational power is

A. 10W B. 20W C. 30W D. 40W

Hint: $P = \vec{\tau}.\vec{\omega}$

Answer: B

158. A loop and a disc have the same mass and roll without slipping with the same linear velocity v. If the total kinetic energy of the loop is 8J, the kinetic energy of the disc must be

A. 8J B. 6J C. 16J D. 4J

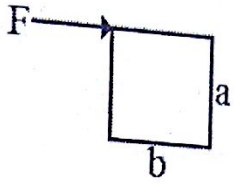
Solution: For loop, $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}(mr^2)\frac{v^2}{r^2} = mv^2 = 8$

For disc, $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\frac{v^2}{r^2} = \frac{3}{4}mv^2$

$\Rightarrow \frac{3}{4}(8) = 6J$

Answer: B

159. A person supports a book by his finger. If the book has a weight of W, then the person is producing a torque on the book of



A. $W\frac{a}{2}$ anticlockwise

B. $W\frac{b}{2}$ anticlockwise

C. Wa anticlockwise

D. Wa clockwise

Solution: Torque due to weight of book $\frac{\omega b}{2}$ clockwise

This is balanced by $\frac{\omega b}{2}$ anticlockwise **Answer: B**

160. A solid cylinder of mass 3kg is placed on a rough inclined plane of inclination 30°, then the minimum frictional force required for it to roll without slipping down the plane is?

A. 2N B. 5N C. 15N D. 18N

Solution: $\tau = fR = I\alpha = \frac{1}{2}mR^2\left(\frac{a}{R}\right) \Rightarrow f = \frac{1}{2}ma$

and, $mg\sin\theta - f = ma$, substituting, $f = \frac{mg\sin\theta}{3} = 5N$

Answer: B

161. A couple produces a

A. Pure linear motion

B. Pure rotational motion

C. Both linear and rotational motions

D. No motion **Answer: B**

162. A person sitting firmly on a rotating stool has his arms stretched. If he folds his arms, his angular momentum.

A. Increases C. Remains same

B. Decreases D. Doubles **Answer: C**

Centre of mass

163. A shell at rest explodes. The center of mass of the fragments

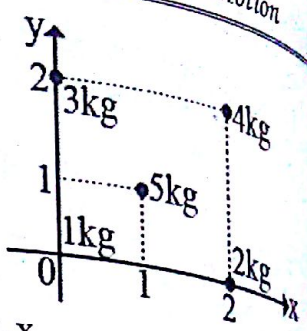
A. Moves along a parabolic path.

B. Moves along an elliptical path.

C. Moves along a straight line.

D. Remains at rest. **Answer: D**

164. Five masses are placed in a plane as shown. The co-ordinate of the center of mass are about



A. 1.2, 1.4 C. 1.1, 1.3
B. 1.3, 1.1 D. 1.0, 1.0

Solution: C.m = (x_{cm}, y_{cm})

$$X_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + m_5x_5}{m_1 + m_2 + m_3 + m_4 + m_5} = 1.1$$

$$Y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4 + m_5y_5}{m_1 + m_2 + m_3 + m_4 + m_5} = 1.3$$

Answer: C

165. If a gymnast, sitting on a rotating stool, with his arms outstretched, suddenly lowers his hands, then
A. The angular velocity decrease.
B. His moment of inertia decrease.
C. The angular velocity stays constant.
D. The angular momentum increases.

Answer: B

166. A solid sphere is rotating in free space. If the radius of the sphere is increased keeping mass same, which one of the following will not be affected?
A. Moment of inertia C. Angular velocity
B. Angular momentum D. Rotational kinetic energy

Answer: B

Unit 6: Conceptual Questions

1. If a car's wheels are replaced by wheels of a larger diameter, will their tangential velocity change?
2. When a wheel of radius R rotates about a fixed axis, do all points on the wheel have the same angular velocity?
3. What is the magnitude of the angular velocity of the second hand of a clock? What is the angular acceleration of the second hand?
4. If you see an object rotating, is there necessarily a net torque acting on it?
5. Explain why changing the axis of rotation of an object changes the object's moment of inertia.
6. A wheel rotates counter clockwise in the xy plane. What is the direction of the angular acceleration if the angular velocity is decreasing in time?
7. A turn table rotates at a constant rate of 45 rev/min. What is the magnitude of its angular acceleration?

8. If the torque acting on a particle about an arbitrary origin is zero, what can you say about its angular momentum about the origin?
9. What can you conclude about the direction of angular momentum of a body with respect to its direction of motion?
10. If a single force acts on a body and the torque due to the force is non-zero about some point, is there any point about which the torque is zero?
11. If a system of particles is in motion, is it possible for the total angular momentum to be zero about some origin? Explain
12. A ball is thrown in such a way that it does not spin about its own axis. Does this mean that the angular momentum is zero abut an arbitrary origin?
13. Why is it easier to keep your balance on a moving bicycle than on a bicycle that is at rest?
14. Three objects of uniform density: a solid sphere, a solid cylinder and a hollow cylinder are placed at the top of an incline. If they all are released from rest at the sometime and roll without slipping, which reaches the bottom first?
15. A mouse is initially at rest on a horizontal turn table mounted on a frictionless vertical axle. If the mouse begins to walk around the table, what happens to the table? Explain
16. Two balls have the same size and mass. One is hollow while the other is solid. How could you decide which is which without breaking them apart?
17. A pail of water can be whirled in a vertical circle such that no water is spilled, even when the pail is upside down. Why?
18. When an acrobat walks on a tight rope, he extends his arms out from his sides. How is stretching his arms become important while walking on the rope?

Unit 6: Problems

Rotation about a fixed axis

19. The linear velocity of a point on a CD is 1.2 m/s. At the start of the CD, this point is 23 mm from the axis of rotation. At the end it is 58 mm from the axis. What is the angular speed of the point at the start and at the end?
20. A turbine is rotating at 3000 rev/min.
    a) What is the angular velocity of the turbine in rad/s?

b) A turbine blade is 0.4m long. Calculate the linear velocity of a point on the end of a turbine blade and half ways down a blade.

21. A pulley of 6 cm radius is turning at 40rev/s and slows down uniformly to 25 rev/s in 3 sec. Calculate
   a) The angular acceleration
   b) the number of revolutions it makes
   c) the length of rope it winds

22. What are the angular velocities of the minute hand and the second hand of a clock?

23. The angular position of a body in rotational motion varies according to the equation $\theta(t) = 2t^2 + 4t + 6$. What is;
   a) its angular position at $t = 4$ sec
   b) its angular velocity at $t = 2$ sec

24. A wheel of 0.5m radius starts from rest and attains an angular velocity of 36 rad/s in 9 sec. Calculate
   a) the angular acceleration
   b) the angular displacement during this time

25. At what time will two discs have the same angular velocity, if one of them is initially at rest and accelerate at 10 rad/s^2, and the other is moving at 5 rad/s and accelerates to 15 rad/s^2. The second disc starts its motion 5 sec after the first.

26. A wheel revolving at 6 rev/s has an angular acceleration of 4 rad/s^2. Find the number of turns the wheel must make to acquire an angular speed of 26 rev/s?

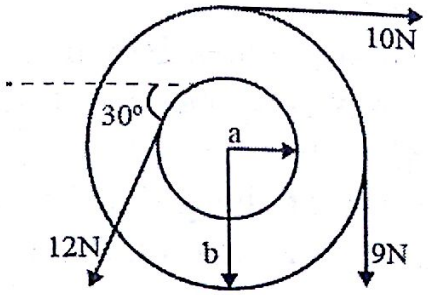
27. A diver makes 2.5 complete revolutions on the way from a 10m high platform to the water. Assuming zero initial velocity, find the diver's average velocity during a dive.

Torque and Angular Acceleration

28. A mechanic has two spanners, one 15 cm long and the other 20 cm long. He applies a force of 30N to the first and 20N to the second. If the forces are perpendicular to the spanner.
   a) With which spanner does he produce greater torque?
   b) How much work does he do with each spanner if he turns a nut one quarter of a turn?

29. A force $\begin{bmatrix} 2 \\ -5 \end{bmatrix}$ acts at a distance of $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ from a point. What is the torque exerted by the force?

30. A tap is a hospital has a 0.25m long rod attached to it. The rod is perpendicular to the axis of rotation of the tap. And it takes a force of 25N to turn the tap quarter turn.
    a) What is the torque exerted on the tap?
    b) How much work is done in turning on the tap?
31. Find the net torque on the wheel about the axle through 0 if a = 10 cm and b = 25 cm.



Rotational Kinetic Energy and Rational Inertia

32. The earth may be considered as a sphere of mass 6×10^{24} kg and radius 6.4×10^{6}m. Work out its kinetic energy if it spins on its axis once a day.
33. A starter cord for a generator is 1m long. It is wound onto a drum with diameter of 10 cm. If a person starts the generator by pulling the cord with 100N force.
    a) What torque does he apply to the engine?
    b) How much work done he do?
34. A vehicle is being designed, such that it has to run for at least 30 minutes and develop a steady power of 500W.
    a) How much energy will the flywheel need to supply?
    b) The largest flywheel that can be fitted has a moment of inertia of 20 kgm^2. Work out how fast it turn at the beginning.
35. A big wheel has a diameter of 5m and a mass of 1500 kg when fully laden with people.
    a) When the wheel rotates at full speed, a person has a linear velocity of 3 m/s. What is the rotational kinetic energy?
    b) If it takes 10 sec from rest to reach linear speed of 3 m/s. What is the power of the system?
36. A 7.27 kg ball with radius 9 cm rolls without slipping down a lane at 4.55 m/s. Calculate its kinetic energy.

4. No. A body can be in motion without external agents.
5. It is because, there is a direct relationship between them.
6. Using the right hand rule, it is in the negative z axis.
7. It is zero. 8. Constant. 9. Perpendicular
10. Yes. At the point of pivot.
11. Yes. It happens when the particles rotate in opposite sense.
12. No. The angular momentum requirement is, $\vec{r} \times \vec{v} \neq 0$.
13. In a moving bicycle, torque can easily be adjusted
14. From conservation of energy, $M_{Ei} = M_{Ef}$

$$P_E = K_{Et} + K_{Er} \Rightarrow mgh = \frac{1}{2}mv^2 + \frac{1}{2}CMR^2\left(\frac{V}{R}\right)^2$$

$\Rightarrow v = \sqrt{\frac{2gh}{1+C}}$, C is a number that depends on the shape of a body.

That is, for solid sphere, $C = \frac{2}{5}$, for hollow sphere, $C = \frac{2}{3}$, and for solid cylinder, $C = \frac{1}{2}$. And we can see that for small C, v is large. Therefore, of the three, the solid sphere reaches the bottom first.

15. Because of Newton's 3rd law, as the mouse walks clockwise, the turn table rotates counterclockwise.
16. When rotating with the same angular speed, the one with a higher kinetic energy is the hollow ball.
17. The normal force from the pail provides a radial force to keep the water moving in a circular path.
18. So as to balance the clockwise torque with the counterclockwise torque.

Unit 6: Solution to Problems

19. $\omega = v/r$. Hence, 52.2 rad/s and 20.7 rad/s respectively
20. a) 100π rad/s b) At end, $v = 40\pi$ m/s, at mid–way, $v = 20\pi$m/s
21. a) $\alpha = -5$ rev/s^2 b) $\theta = 97.5$ rev c) $s = 11.7\pi$m
22. $\omega_m = \frac{2\pi}{3600\,\text{sec}} = 0.002$ rad/s, $\omega_s = \frac{2\pi\text{rad}}{60\,\text{sec}} = 0.1$rad/s
23. a) $\theta = 54$ rad b) $\omega = 4t + 4$, at $t = 2$ sec, $\omega = 12$ rad/s
24. a) $\alpha = 4$ rad/s^2 b) $\theta = 36$ rad
25. $\omega_1 = 10t$ and $\omega_2 = 5 + 15(t-5)$, but $\omega_1 = \omega_2 \Rightarrow t = 14$ sec
26. 502 rev 27. 11 rad/s

28. a) $\tau = Fr$, for $r = 30$ cm, $\tau = 450$ N.m
b) $W = \tau\theta = 707$ J with the 30 cm and 628J with the 20cm

29. $\tau = r \times F = 17\text{N.m}\,\hat{k}$

30. a) $\tau = Fr = (25)(0.25) = 6.25\text{N.m}$
b) $W = \tau\theta = (6.25)(2\pi/4) = 3.125\pi\text{J}$

31. $\sum\tau = \tau_{CC} + \tau_C = (0.1)(6 + 10.4) - (0.25)(10+9) = 3.1\text{N.m}$ clockwise

32. $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{2\pi}{24\times 60\times 60}\right)^2 = 2.6\times 10^{29}\,\text{J}$

33. a) $\tau = Fr = 5\text{N.m}$ b) $\omega = \tau\theta = \tau\left(\frac{s}{r}\right) = 5\left(\frac{1}{0.05}\right) = 100\text{J}$

34. a) $W = pt = (500)(1800) = 9 \times 10^5\text{J}$
b) $W = \Delta kE_r \Rightarrow \omega = 2900$ rev/min

35. a) $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}(MR^2)\left(\frac{v}{R}\right)^2 = 6750\text{J}$

b) $p = \frac{w}{t} = \frac{k_E}{t} = 675\text{W}$

36. $kE = kE_t + kE_r = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = 105.5\text{J}$

37. a) $\vec{a} = 3.12\ \text{m/s}^2$ b) $T_1 = 26.7\text{N}, T_2 = 9.4\text{N}$

38. $I = \frac{1}{2}MR^2, \tau = Fr = I\alpha \Rightarrow \alpha = \frac{Fr}{I} = 50\ \text{rad/s}^2$

39. $20 - T_1 = \alpha$, $T_2 - 18 = 0.36\alpha$ and $0.5T_1 - 0.2T_2 = 1.7\alpha$
From these, $\alpha = 8\ \text{rad/s}^2$, $T_1 = 17.4\text{N}$ and $T_2 = 20.9\text{N}$

40. $L = I\omega = \tau\Delta t = (2)(10)(0.5)(2.5) = 25\ \text{kgm}^2/\text{s}$

41. $L_i = L_f = 10(I) = 2(I)(\omega) \Rightarrow \omega = 5\ \text{rad/s}$

42. $L = (\vec{r}\times\vec{v})\,m = 1.5\ \text{kgm}^2/\text{s}$

43. $I_i\omega_i = I_f\omega_f \Rightarrow I_f = 2\ \text{kgm}^2$

44. $L_i = L_f \Rightarrow mv_oR = m\sqrt{2ghR} = mvR + \frac{1}{2}MR^2\omega, v = R\omega$

$\Rightarrow \omega = \frac{\sqrt{2gh}}{R(1 + M/2m)}$

45. 8 rev/s 46. 0.712m 47. 10kg 48. (3, 2.5)

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Unit 7

Equilibrium

We use and feel forces all the time but we are not always consciously aware of them. Forces which push and pull and get things moving or slow things down are easy to spot, but forces which hold things still and keep things balanced are less obvious but very important in everyday life and in the design and analysis of structures in engineering.

Any building, from a multistory skyscraper to the humblest shed, must be designed so that it won't topple over. Similar concerns arise with a suspension bridge, ladder leaning against a wall, or a crane hoisting a bucket full of concrete.
Therefore, in many problems and real life situations, there are times when a body remains at rest or move with constant velocity.

Equilibrium: Is a condition in which the state of motion of a body doesn't change.

☛ **Important points to remember**

- A body is said to be in equilibrium when the net force and the net torque acting on its are zero.
- A body in equilibrium may be at rest or may move with constant velocity
- **Static equilibrium**: Is a type of equilibrium that occurs when a body is at rest and there is no net force or net torque acting on it.
- **Dynamic equilibrium**: Is a type of equilibrium that occurs when a body is moving at a steady velocity and there is no net force or net torque acting on it.

Conditions of Equilibrium

For a body to be in equilibrium, two conditions must be satisfied

1. **First condition or translational equilibrium** states that "The net force must be zero."

$$\sum \vec{F} = 0 \Rightarrow \sum F_x = 0, \sum F_y = 0, \sum F_z = 0$$

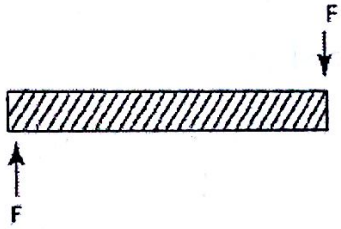
- The first condition of equilibrium is
  - A condition where linear momentum is constant
  - Another way of stating Newton's 1st law

2. **Second condition or rotational equilibrium** states that "The net torque must be zero." $\sum \tau = 0 \Rightarrow \sum \tau_c = \sum \tau_{cc}$

clockwise torque (τ_c) = counterclockwise torqe (τ_{cc})

- The second condition of equilibrium is
  - A condition where angular momentum is constant
  - A condition where a body does not rotate
- **Bear in mind that**
  - The two conditions are independent of one another, that is satisfying one does not guarantee that the other is satisfied.
  - In this book, we treat counter clockwise torque ( ) as positive and clockwise torqe ( ) as negative.

Couple: Is pair of forces with equal magnitude but opposite directions, which produce rotation, but not translational motion of a body



Example, a car steering wheel

Conceptual Examples

1. Can a body be in equilibrium if only one external force acts on it? Explain

   Explanation: No! For a body to be in equilibrium, the net external force acting on it must be zero. This is not possible when only one force acts on the body.

2. Can a body be in equilibrium if it is in motion? Explain

   Explanation: Yes! this is possible when the body is moving with constant velocity.

3. Give an example in which the net torque acting on an object is zero and yet the net force is non zero.

   Explanation: $\sum F = 2F \neq 0$

About pivot point, $\sum \tau = Fd - Fd = 0$

4. Give an example in which the net force acting on an object is zero and yet the net torque is non zero.

Explanation: $\sum F = F - F = 0$

About pivot point, $\sum \tau = -Fd - Fd = -2Fd = 2Fd$ clockwise

5. Can an object be in equilibrium if the only torque acting on it produces clockwise rotation?

Explanation: No! this is not possible.

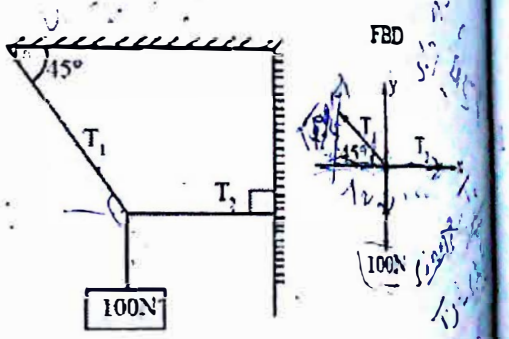
6. A ball thrown upward comes to rest when it reaches the highest point. Is the ball in equilibrium at this point?

Explanation: No! the ball's velocity at the lightest point is zero, however the ball is still accelerating downward at gravitational acceleration $g = 9.8\ m/s^2$. Hence, there is a net downward force acting on the ball.

Illustrative Examples

7. For the situation shown, find T_1, T_2 and T_3.

Solution:



$T_3 = W = 100N$

$\sum F_y = 0, T_1 \sin 45 = T_3$

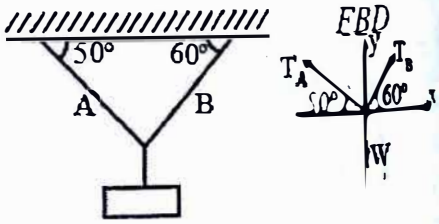
$\Rightarrow T_1 = 141.4N$

$\sum F_x = 0, T_1 \cos 45 = T_2$

$\Rightarrow T_2 = 100N$

8. The tension in cord A is 30N. Find the tension in cord B and the weight of the suspended block.

Solution:



$\sum F_x = 0, T_A \cos 50°$

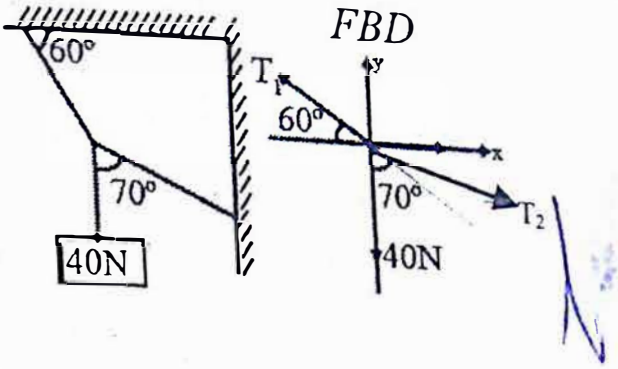
$= T_B \cos 60°$

$\Rightarrow T_B = 39N$

$\sum F_y = 0, T_A \sin 50 + T_B \sin 60 = W$

$\Rightarrow W = 56N$

9. If the weight of the block suspended is 40N. Find tensions T_1 and T_2.



Solution:

$\sum F_x = 0, T_1 \cos 60° = T_2 \sin 70°$

$\Rightarrow T_1 = 1.9T_2$

$\sum F_y = 0, T_1 \sin 60 + (-T_2 \cos 70)$

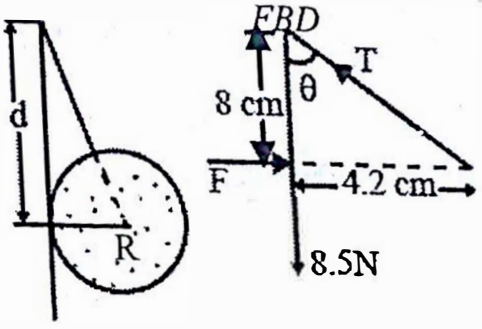
$= 40 \Rightarrow T_1 = 59\ N,\ T_2 = 31N$

10. A uniform sphere of mass 0.85 kg and radius 4.2cm is held in place by a mass less rope attached to a frictionless wall at d = 8 cm above the center of the sphere. Find

a) The tension of the rope and

b) The force of the wall on the sphere

Solution: $\tan\theta = \frac{4.2}{8}, \theta = 28°$



$\sum F_y = 0, T \cos 28 = 8.5 \Rightarrow T = 9.6N$

$\sum F_x = 0, F = T \sin 28 \Rightarrow F = 4.5N$

a) $T = 9.6N$ and

b) $F = 4.5N$

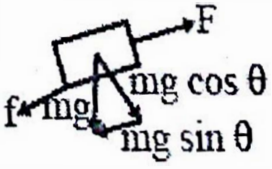
11. Assume that m = 6 kg, $\theta = 43°$ and $\mu_k = 0.3$. What force F is needed along the plane to move the block at constant speed?

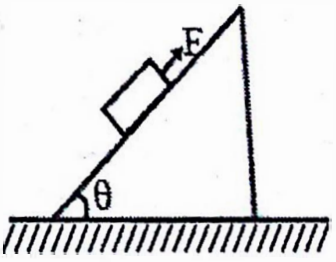
a) Up the plane

b) Down the plane

Solution:

FBD





a) $\sum F_y = 0$

$F_N = mg \cos\theta$

$\Rightarrow F_N = 43.88N$

$f_k = \mu_k F_N = (0.3)(43.88) = 13.16N$

$\sum F_x = 0$

$F - f_k - mg \sin\theta = 0$

$F = f_k + mg \sin\theta = 54.1N$

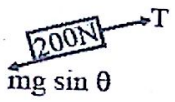
b. $\sum F_x = 0 \Rightarrow F + mg \sin\theta = F_k \Rightarrow F = 27.8N$

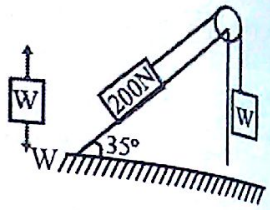
12. If the friction between the block and the incline is negligible, how much must the block hanging weigh if the 200N block is to remain at rest?

Solution: $\sum F_x = 0$ FB

$T = 200 \sin 35°$

$T = 114.7$ N

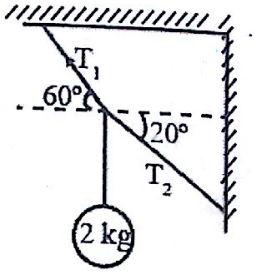


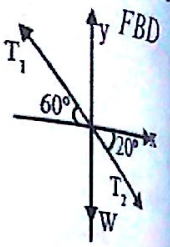


$\sum F_y = 0 \Rightarrow T = W \Rightarrow W = 14.7N$

13. The system shown is in equilibrium. What are the tensions T_1 and T_2?

Solution:





$\sum F_x = 0$

$T_1 \cos 60° = T_2 \cos 20° \ldots (*)$

$\sum F_y = 0, T_1 \sin 60°$

$= W + T_2 \sin 20° \ldots (**)$

From equation (*) and (**),

$0.5T_1 = 0.94\ T_2$ and $0.87T_1 = 20 + 0.3T_2 \Rightarrow T_1 = 28N$ and $T_2 = 15N$

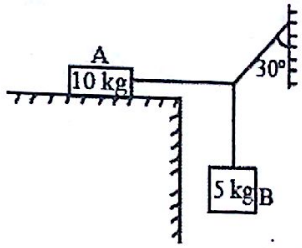
14. What should be the coefficient of static friction between block A and the surface so that the system remains at rest?

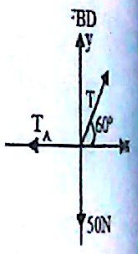
Solution: $\sum F_y = 0$

$T \sin 60° = 50$

$\Rightarrow T = 57.7N$

$\sum F_x = 0$





$T_A = T \cos 60 = (57.7)(0.5) = 28.9N$

Also, $T_A = \mu_s m_A g \Rightarrow \mu_s = \dfrac{T_A}{m_A g} = \dfrac{28.9}{(10)(10)} = 0.29$

15. If the system shown is in equilibrium. Find T_1, T_2, T_3 and θ.

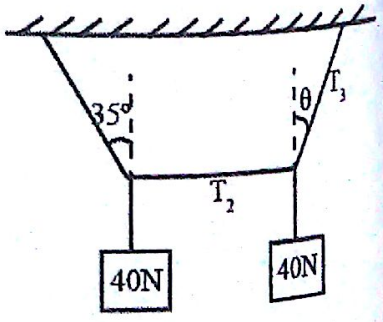
Solution: From FBD_1

$\sum F_y = 0$

$T_1 \sin 55° = 40$

$T_1 = 49N \sum F_x = 0$

$T_1 \cos 55° = T_2\ T_2 = 28N$

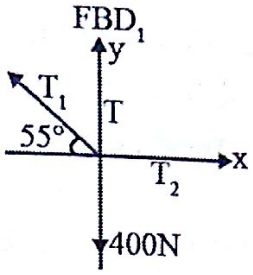


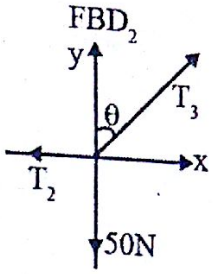
From FBD_2

$\sum F_y = 0 \Rightarrow T_3 \cos\theta = 50$

$\sum F_x = 0 \Rightarrow T_3 \sin\theta = T_2 = 28$

$\Rightarrow T_3 = \sqrt{(T_3\cos\theta) + (T_3\sin\theta)^2} = 57\,N$

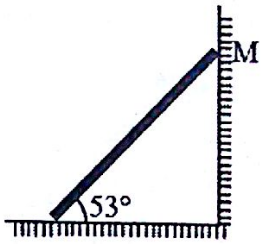




$$\tan\theta = \frac{T_3\sin\theta}{T_3\cos\theta} = \frac{28}{50}$$

$\Rightarrow \theta = 29°$

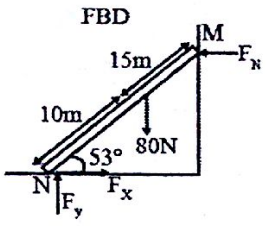
16. A 25m ladder rests against a frictionless wall at point M. The center of mass of the ladder is located at 10m from the bottom of the ladder. If the ladder weighs 80N, find the supporting force of the wall on the ladder.



Solution:

Taking moment at N, $\sum\tau = 0$, $(F_N)(25\sin 53) - (80)(10\cos 53°) = 0$

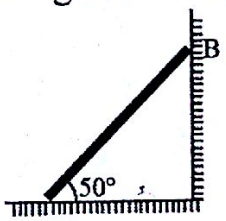
$\Rightarrow F_N = 24N$



17. A ladder of length L and weight 200N leans against a smooth wall. If its center of mass is located at 0.4L from the base,

a) How large a friction force must exist at the base if it is not to slip?

b) What is the necessary coefficient of static friction?



Solution:

Taking moment about point A, $\sum\tau = 0$

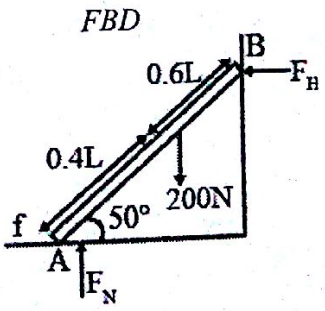
$(F_H)(L\sin 50°) - (200)(0.4L\cos 50°) = 0$

$\Rightarrow F_H = 67N$

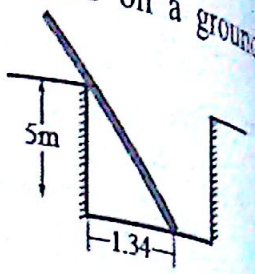
$\sum F_x = 0 \Rightarrow f = F_H = 67N$

a) $f = 67N$, and $\sum F_y = 0 \Rightarrow F_N = 200N$

b) $$\mu = \frac{f}{F_N} = \frac{67}{200} = 0.34$$



22. A uniform electric pole of 400N and 12m rests on a ground supported by a wall of 5m deep. The pole is in equilibrium as long as $x \leq 1.34$m but will slip if $x > 1.34$m. Find:



a) The supporting force of the wall.
b) The coefficient of friction between the pole and the ground.

Solution: $\sum F_x = 0 \Rightarrow F = f$

$\sum F_y = 0 \Rightarrow F_N = 400N$

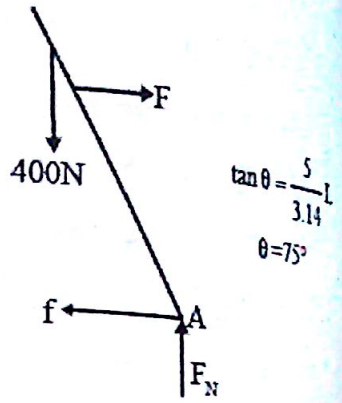
Taking moment about A

$\sum \tau = 0 \Rightarrow (400)(6 \cos 75°)$

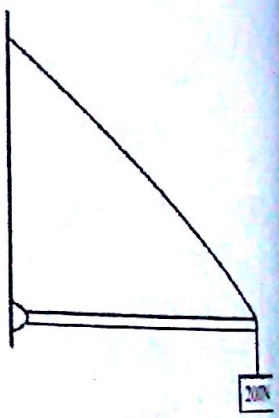
$= (5)(F) \Rightarrow F = 124.2N, = f$

a) $F = 124.2N$

b) $f = \mu F_N \Rightarrow \mu = \dfrac{f}{F_N} = \dfrac{124.2}{400} = 0.31$



23. A uniform pole of weight 400N at length 12m is attached with a rope and a 200N block is hung on it. Find



a) The tension of the rope.
b) The components of the force by the hing on the pole.

Solution: $\sum F_x = 0 \Rightarrow F_x = T \cos\theta$,

$\tan\theta = \dfrac{5}{12}$, $\theta = 23°$

$\sum F_y = 0$

$\Rightarrow F_y + T \sin\theta = 400 + 200 = 600N$

Taking moment about A,

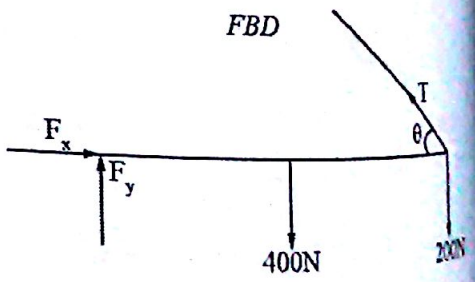
$\sum \tau = 0$

$(T \sin\theta)(12) = (400)(6) + (200)(12)$

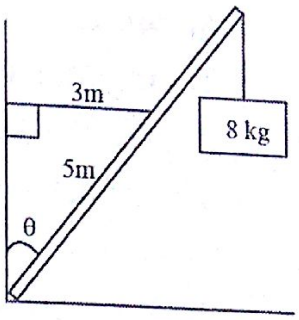
$\Rightarrow T = 1024N$, thus

a) $T = 1024N$

b) $F_x = T \cos\theta = 943N$, $F_y = 600 - T \sin\theta = 200N$



24. A uniform 5 kg rod has length of 12m. At one end an 8 kg block is suspended and the rod is supported by a horizontal rope. Find:
a) The tension in the rope.
b) The reaction of the wall corner on the rod.



Solution: $\sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$

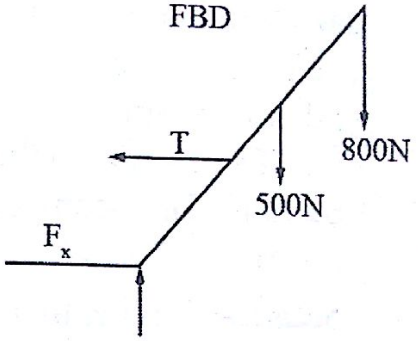
$\sum\tau = 0$, about A

$(T)(5\cos\theta) = (50)(6\cos\theta) + (80)(12\cos\theta) \Rightarrow T = 252N$ and,

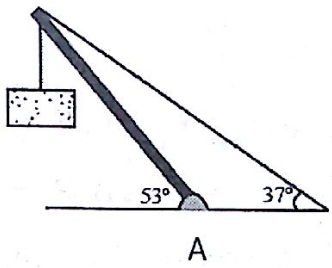
$\sum F_x = 0 \Rightarrow F_x = T = 252N$ &

$\sum F_y = 0 \Rightarrow F_y = 50 + 80 = 130N \Rightarrow$

$F = \sqrt{F_x^2 + F_y^2} = 284N$



25. A uniform rod of mass 15 kg and length 12m is tied with a rope. If a 20 kg mass is suspended on it. Find
a) The tension of the rope.
b) The force exerted by the hinge on the rod.



Solution: Taking moment about A

$\sum\tau = 0$, $(200)(12\cos 53°) + (150)(6\cos 53°) = (T\cos 37)(12\sin 53°) + (T\sin 37°)(12\cos 53°)$

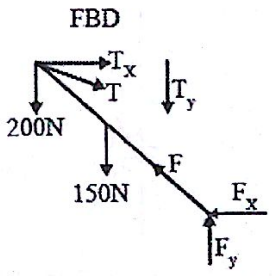
a) $T = 165.5N$

b) $\sum F_x = 0$, $T_x = F_x$

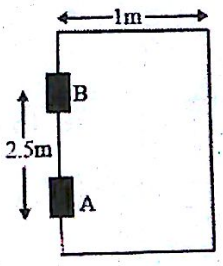
$\Rightarrow T\cos 37° = F_x = 132.2N$

$\sum F_y = 0$, $200 + 150 + T_y = F_y = 449.6N$

$F = \sqrt{F_x^2 + F_y^2} = 468.6N$



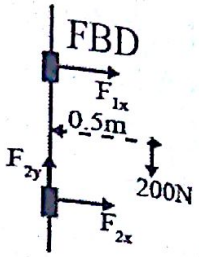
26. A 1m wide door with 200N weight has hinges 2.5m apart. If the weight of the door is supported by the lower hinge, find the forces exerted by the hinges on the door.



Solution:

Taking moment about A; $\sum\tau = 0$

$(F_{1x})(2.5) + (200)(0.5) = 0, F_{1x} = -40N$



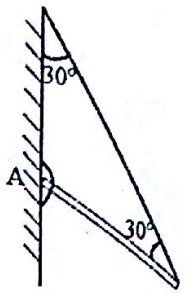
$\sum F_x = 0, F_{1x} + F_{2x} = 0,$

$F_{2x} = 40N$

$\sum F_y = 0, F_{1y} + F_{2y} = 200$, but $F_{1y} = 0$

$\Rightarrow F_{2y} = 200N$

27. A rod of 200N is hinged to a wall from one end and is supported by a string from the other end. Find

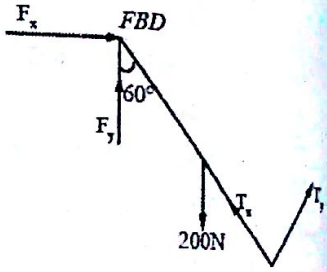


a) The tension in the string.

b) The force components by the wall on the rod.

Solution: Taking moment at point A $\sum\tau = 0$

a) $(200)\left(\frac{L}{2}\sin 60^0\right) = LT \sin 30° \Rightarrow T =$ 173N

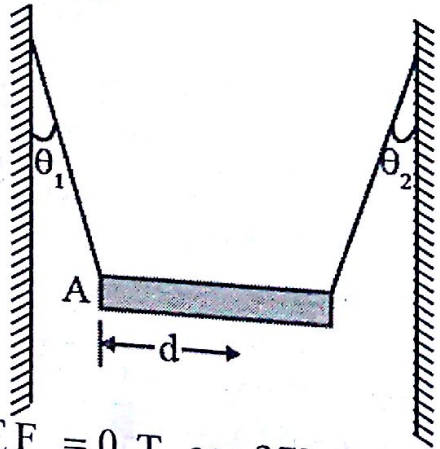


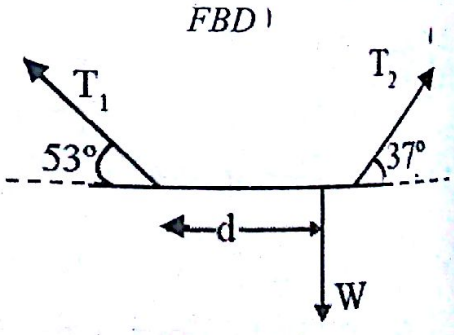
b) $\sum F_x = 0, F_x = T \cos 60° = 87N$

$\sum F_y = 0, F_y = 200 - T \sin 60 = 50N$

28. A non uniform 6m bar is suspended in a horizontal position by two cords. If one of the cords makes $\theta_1 = 37°$ and the other cord $\theta_2 = 53°$ from the vertical, find the distance d from the left end of the bar to its center of mass.

Solution:





$\sum F_x = 0, T_2 \cos 37° = T_1 \cos 53° \ldots (*)$

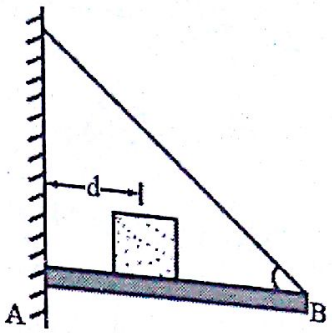
$\sum F_y = 0, T_1 \sin 53° + T_2 \sin 37° = W \ldots (**)$

Taking moment about point A

$\sum\tau = 0$, $(T_2 \sin 37)(6) = (W)(d)$ . . . (***)

Solving equations given, we get $d = 2.16m$

29. Suppose a uniform bar of 3m and 200N is supported by a wire at angle $\theta = 30°$. If a block of weight 300N is imposed on it and the tolerance of the wire is 500N, what is the maximum distance d the block can be positioned before the wire breaks?

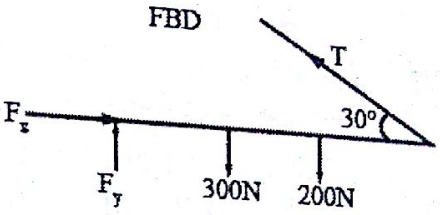


Solution:

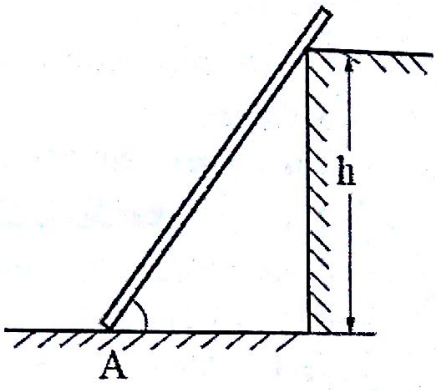
Taking moment about point A

$(200)(1.5)+(300)(d)=(500)(3)(\sin 30°)$

$d = 1.5m$



30. A uniform pole of weight 445N and length 6.1m rests on the ground supported by a wall edge at $h = 3.05m$. The pole remains in equilibrium if $\theta \geq 70°$ but slips if $\theta < 70°$. Find the coefficient of static friction between the pole and the ground.



Solution: Taking moment about A

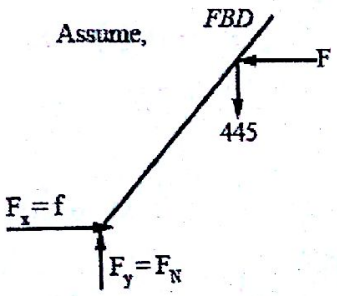
$\sum\tau = 0$, $(F)(3.05)$

$= (445)(3.05 \cos 70°)$

$\Rightarrow F = 152.2N = f$

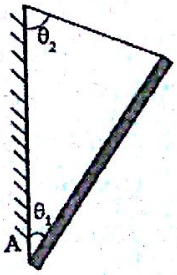
$\sum F_y = 0,$

$\Rightarrow F_N = 445$



$$f = \mu F_N,\ \mu = \frac{f}{F_N} = \frac{152.2}{445}\ \mu = 0.34$$

31. A uniform rod of mass m is hinged to a building with its upper end supported by a rope what is attached to the wall. If angle $\theta_1 = 60°$, what value must angle θ_2 have so that tension in the rope is equal to mg/2?



Solution:

Taking moment about point A, $\Sigma\tau = 0$

$$(mg)\left(\frac{L}{2}\sin 60\right)(T)(L)(\sin(120-\theta_2)$$

$$\frac{mg}{2}(\sin 60) = \frac{mg}{2}\sin(120-\theta_2)$$

$\sin 60 = \sin(120-\theta_2)$

$= \sin 120\cos\theta_2 - \cos 120\sin\theta_2$

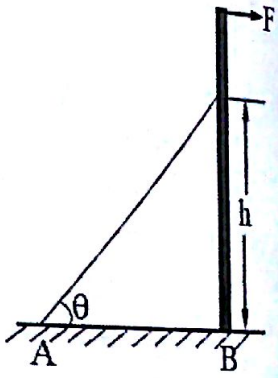
$0.87\cos\theta_2 + 0.5\sin\theta_2 = 0.87$

$\Rightarrow \theta_2 = 60°$

32. A uniform rod of 60N weight and 3.2m length is hinged at its lower end, and a horizontal force F = 50N acts at its upper end. The rod is held vertical by a cable that makes angle $\theta = 25°$ with the ground and is attached to the rod at height h = 2m. What is:

a) The tension in the cable.

b) The force on the rod from the hinge in unit vector notation.



Solution:

Taking moment about point B

$\Sigma\tau = 0,$

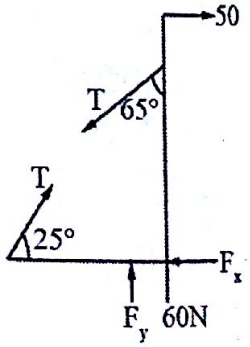
$(T\sin 65)(2)$

$= (50)(3.2)$

$\Rightarrow T = 88N$

$\Sigma F_x = 0, 50 - 88\sin 65 + F_x = 0$

$\Rightarrow F_x = 30N$

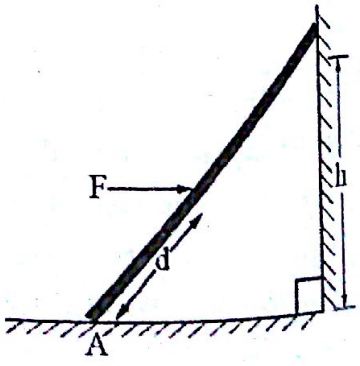


$\Sigma F_y = 0, F_y - 60 - 88\cos 65° = 0$

$\Rightarrow F_y = 97N$, thus

a) $T = 88N$ b) $(30i + 97j)N$

33. A uniform ladder is 10m long and 200N weight. The ladder leans against a wall at height h = 8m above the ground. A horizontal force Fis applied to the ladder at



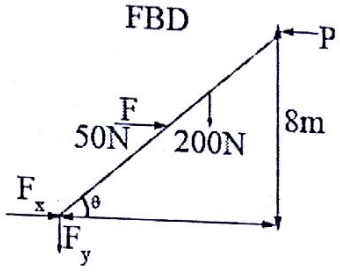
distance d = 2m from its base. If the force F = 50N, what is the force of the ground on the ladder in unit vector notation.

Solution: $\sin\theta = \frac{8}{10} \Rightarrow \theta = 53°$

taking moment about point A

$(50)(25 \sin 53°) + (200)(5 \cos 53°)$

$= (8)(P) \Rightarrow P = 85N$



$\sum F_x = 0 \Rightarrow F_x + F = P$

$F_x = 85 - 50 = 35N$

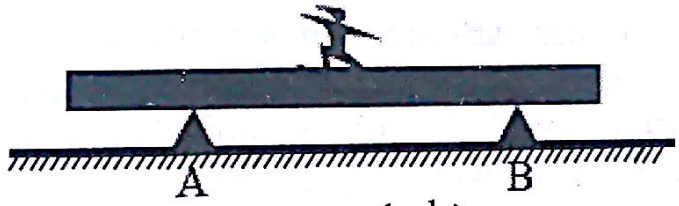
$\sum F_y = 0 \Rightarrow F_y = 200N$, thus $F' = (35i + 200j)N$

34. A uniform aluminum rod 9m long, weighing 300N rests symmetrically on two supports 5m apart. A boy weighing 600N starts at point A and walks toward the right.

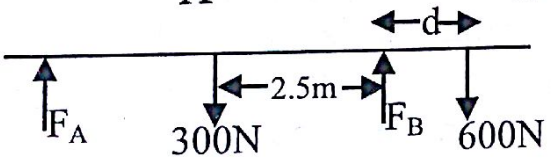
a) How far beyond point B can the boy walk before the rod tips?

b) How far from the right end of the rod should support B be placed so that the boy can walk just to the end of the rod without causing it to tip?

Solution:



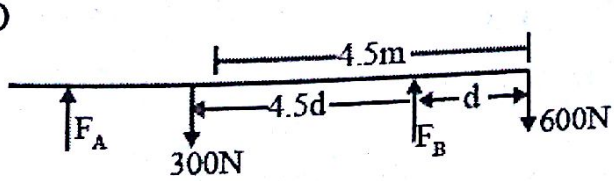
a) FBD



F_A & F_B are reactions of supports A & B.

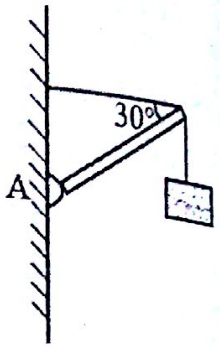
Taking moment about point B, and notice that as the boy moves from point A to B, F_A decreases, to zero and F_B increases. Thus, $\sum \tau = 0$, $(300)(2.5) = (600)(d) \Rightarrow d = 1.25m$ from B.

b) FBD



Taking moment at the new position of B, $\sum\tau = 0, (300)(4.5 - d) = (600)(d) \Rightarrow d = 1.5m$

35. Find the tension in the cable and the magnitude and direction of the force exerted on the rod by the pivot. Assume that the rod and the suspended block do have the same weight of value 400N.



Solution: Taking moment about point A

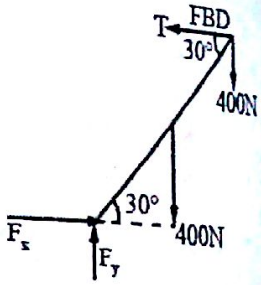
$$\sum\tau = 0, (400)\left(\frac{L}{2}\cos 30^0\right) + (400)(L \cos 30°) = (T)(L \sin 30°)$$

$\Rightarrow T = 1040N$ and $\sum F_x = 0, F_x = 1040,$

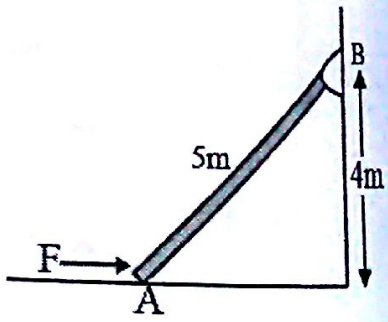
$\sum F_y = 0, F_y = 800N$

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(1040)^2 + (800)^2}$$

$= 1312N$ at $\theta = 37.6°$



36. End A of bar AB rests on smooth surface and end B is hinged. A horizontal force F = 120N is applied on end A. If the weight of the bar is ignored, what are the horizontal and vertical components of the force exerted by the bar on the hinge at B?



Solution: $\sin\theta = \frac{4}{5} \Rightarrow \theta = 53°$

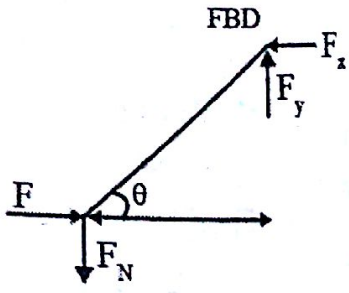
Taking moment about B,

$\sum\tau = 0, (120)(4) = (F_N)(5 \cos 53°)$

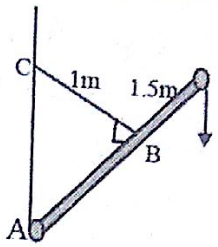
$F_N = 160N$

$\sum F_x = 0, F = F_x = 120N,$ and

$\sum F_y = 0, F_N = F_y = 160N$



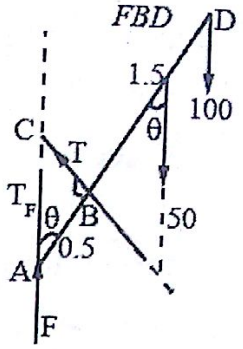
37. A uniform rod of length 2m and mass 5 kg is connected to a vertical wall by a smooth hinge at A and a wire CB. If a mass of 10 kg is attached to D. Find:



a) The tension in the wire.

b) The magnitude of the force at the hinge A.

Solution: $\tan\theta = \dfrac{1}{0.5} \Rightarrow \theta = 63.4°$



Taking moment about A

$0.5T = (50)(\sin 63.4) + (100)(2 \sin 63.4°)$

$\Rightarrow T = 447N$ and $\sum F_y = 0$,

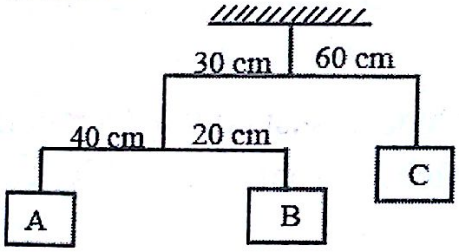
$\Rightarrow F = T_f + 50 + 100$

$= \dfrac{T}{\sin 63.4} + 50 + 100 = 650N$, thus

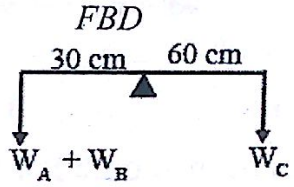
a) $T = 447N$ b) $F = 650N$

38. Three blocks A, B, C are connected by weightless rods. If block A is 10N and the system is in equilibrium, what are the weights of blocks B and C?

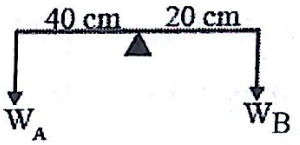
Solution:



ii)



i)



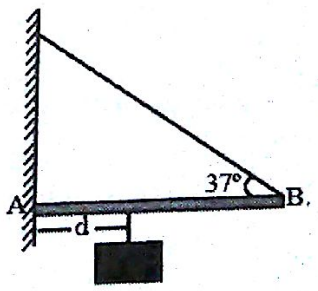
i. $\sum \tau = 0$, about pivot

$(40)(10) = (20)(W_B) \Rightarrow W_B = 20N$

ii. $\sum \tau = 0$, about pivot

$(30)(W_A + W_B) = (60)(W_C) \Rightarrow W_C = 15N$

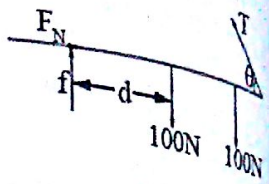
39. One end of a uniform 4m long rod of weight 100N is supported by a cable. The other end rests against a wall where it is held by friction. If the coefficient of static friction between the wall and the rod is 0.5, what is the minimum distance d from



point A at which a block of 100N weight can be hung without causing the rod to slip at point A.

Solution: $\sum F_x = 0, F_N = T\cos 37° = 0.8T$

$\sum F_y = 0, f + T\sin 37° = 100 + 100 = 200$



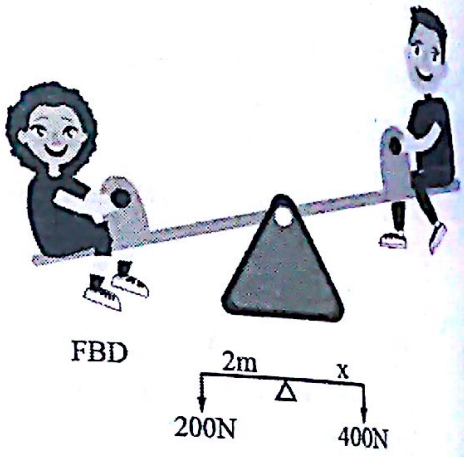
but, $f = \mu F_N = (0.5)(0.8)(T)$

$\Rightarrow (0.5)(0.8)(T) + 0.6T = 200, T = 200N$

And, taking moment about point A

$\sum \tau = 0, (d)(100) + (2)(100) = (T\sin 37°)(4) \Rightarrow d = 2.8m$

40. A see – saw of length 4m pivoted at its center rests in a horizontal position. A boy of mass 20kg sits on one end. How far FROM the center should his sister of mass 40 kg sit if the see – saw is to balance?



FBD

Solution: $\sum \tau_{pivot} = 0$

$-400(x) + 200(2) = 0$

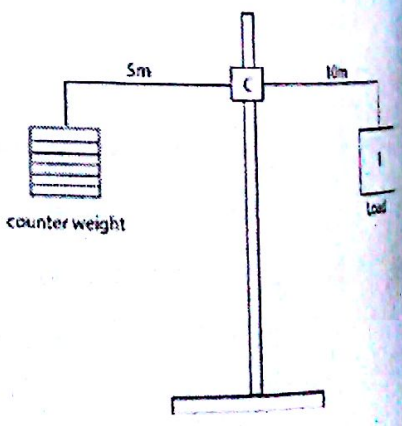
$\Rightarrow x = 1m$

41. In a tower crane, a 30 tonne of load is suspended determine the mass of the counter weight, which is usually layer of concrete slabs (1 tonne = 1000 kg)

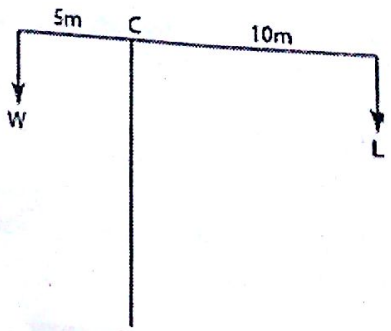
Solution: $\sum \tau_c = 0$

$5(W) - 10(3 \times 10^5) = 0$

$W = 60{,}000kg = 60$ tonne

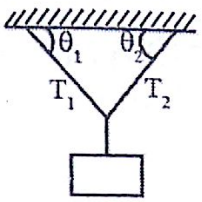


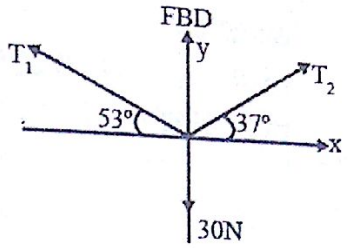
FBD



$\sum F_y = 0$, $T_1 \sin 53 + T_2 \sin 37 = 30 \ldots (*)$

$\sum F_x = 0$, $T_1 \cos 53 = T_2 \cos 37 \ldots (**)$





From these two, equations

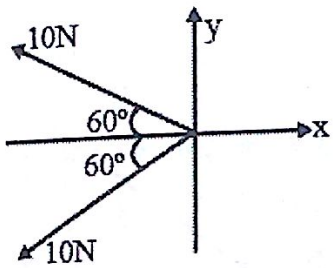
$0.8T_1 + 0.6T_2 = 30$, and $0.6T_1 = 0.8T_2$

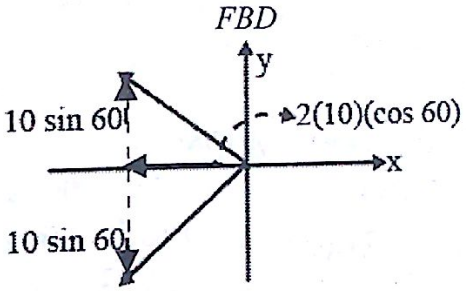
$\Rightarrow T_1 = 24N, T_2 = 18N$ **Answer: D**

49. A particle is set to move with constant velocity under the action of three forces. Two forces with magnitude 10N each are shown. What must be the third force magnitude and direction?

A. 10N to the right
B. 5N upward
C. 10N upward
D. 5N to the right

Solution:





$\sum F_x = 0, (2)(10)(\cos 60°) = F_x = 10N$

$\sum F_y = 0, (10)(\sin 60) - (10)(\sin 60°) = F_y = 0$ **Answer: A**

50. A body experiencing two forces of equal magnitude remains in translational equilibrium. Which of the following is the angle between the lines of actions of the forces?

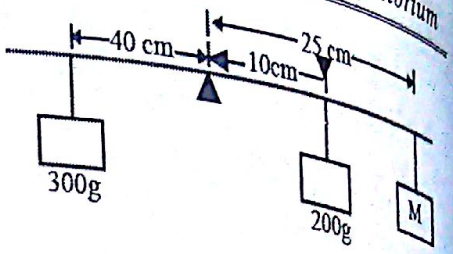
A. 0° B. 45° C. 90° D. 180°

Solution: $\sum F = 0 \Rightarrow F_1 + F_2 = 0, F_1 = -F_2 \Rightarrow \theta = 180°$ **Answer: D**

51. A uniform meter stick supported at 50 cm mark has 300g and a 200g masses hanging from it at the 10 cm and 60 cm marks respectively. What must be hung at the 75 cm mark to keep the meter stick balanced?

A. 200g B. 300g C. 400g D. 500g

Solution:
$(300)(40) = (10)(200) + (25)(m)$
$m = 400g$

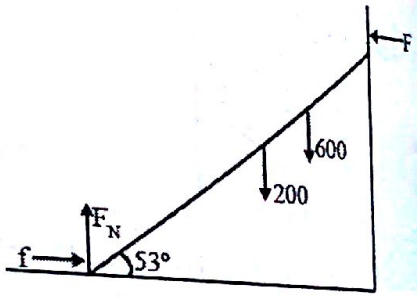


52. A 10 m uniform ladder weighing 200N rests against a frictionless wall. The ladder makes 53° with the horizontal. If the ladder is just on the verge of slipping when a 600N fire fighter is 6m along the ladder from the bottom, what is the coefficient of static friction between the ladder and the ground? (UEE 2006)
A. 0.3240 B. 0.4313 C. 0.5750 D. 0.7667 **Answer: C**

Solution: $\sum \tau = 0$, about point A
$(200)(5\cos 53°) + (600)(6\cos 53°) = (F)(10\sin 53°)$
$\Rightarrow F = 346.6N$

$F_N = 200 + 600 = 800N$
$f = F = 346.6$

$$\mu = \frac{f}{F_N} = \frac{346.6}{800} = 0.4313$$



A

53. Which one of the following statement is NOT correct about a particle that is in equilibrium? **Answer: B**
A. It can have a velocity different from zero.
B. Its velocity can be equal to zero.
C. It can be under the action of more than one force.
D. It can have an acceleration different form zero.
Explanation: A body is in equilibrium if the net forces and torques acting on it are both zero. Therefore, the body in equilibrium cannot have acceleration. **Answer: D**

Unit 7: Conceptual Questions

1. What should be the line of action of three forces of equal magnitude acting on a body, so that the body will be in a rotational equilibrium?

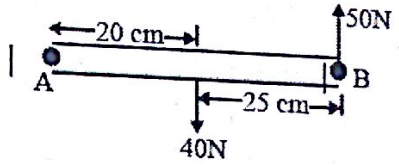
2. Which one is the safest way to climb up a ladder that stands on the ground leaning against to a wall when'
   a) The ground is frictionless and the wall is rough.
   b) The ground is rough and the wall is frictionless.
   c) The ground and the wall both are frictionless.

3. Are these concurrent forces in equilibrium?

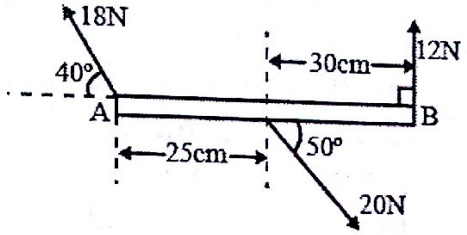
$$G\begin{bmatrix}-6\\2\end{bmatrix}, \quad H\begin{bmatrix}1\\1\end{bmatrix}, \quad I\begin{bmatrix}5\\-3\end{bmatrix}$$

Unit – 7: Problems

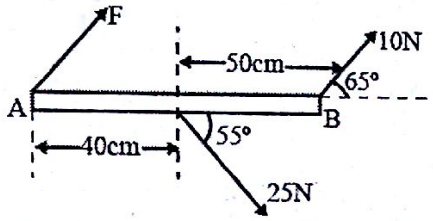
4. Two forces act on a bar as shown. What is the net torque about?
   a) Axis A
   b) Axis B



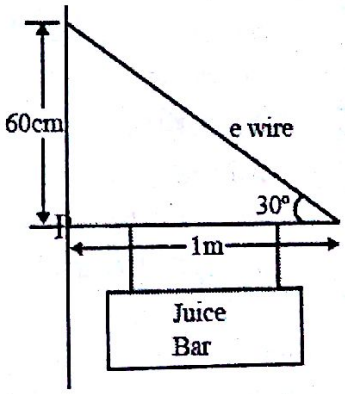
5. Three forces acts on a bar as shown. What is the net torque about?
   a) Axis A
   b) Axis B



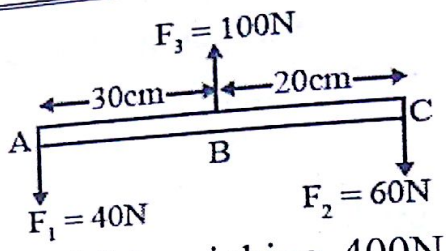
6. Three forces act on a bar as shown.
   a) Calculate the net torque about axis A.
   b) The net torque about axis B is zero. Calculate the vertical component of force F.



7. The sign shown is hanging from a horizontal bar. The bar is fixed to the wall at P and is free to rotate. There is a wire strung from the outer end of the bar to a point 60cm above the pivot. The sign and the bar have a mass 2 kg. If the wire is mass less and the system is in equilibrium, what is the size of the force in the wire?

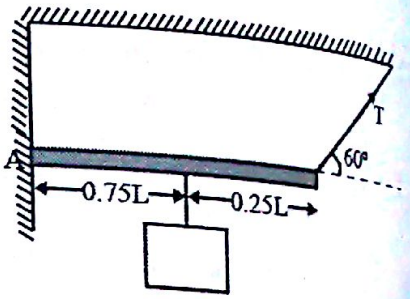


8. Three forces act on a bar as shown. Determine whether the bar is in equilibrium or not.



9. A child weighing 400N sits on one end of a see – saw that is pivoted 1.4m from the child. The see – saw is 3m long. If another child sit at the opposite end just balances, what is his weight? Neglect the weight of the see – saw.

10. A uniform beam of weight 60N is subjected to a block of 200N suspended on it. Find the tension in the supporting rope and the reaction force of the vertical wall.



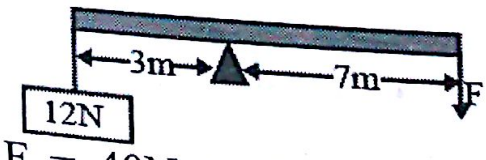
11. A ladder 4m long leans against a frictionless wall with its lower end 1m from the wall. The ladder weighs 250N and its to stay in place, what must be the minimum coefficient of friction between the bottom of the ladder and the ground?

12. Calculate the torque due to two forces 40N act as a couple. The forces act at a distance of 20cm from the point of rotation.

13. A block is pulled to the right at a constant velocity by a 10N force acting at 30° above the horizontal. If the coefficient of sliding friction is 0.5, what is the weight of the block?

14. If a block placed at an incline plane of angle 30° slides down at constant velocity, what is the coefficient of friction between the block and the incline?

15. A uniform bar 10m long and weighing 5N is supported by a fulcrum 3m from the left. If a 12N load is hung from the left end, what down ward pull at the right end is necessary to hold the bar in equilibrium?

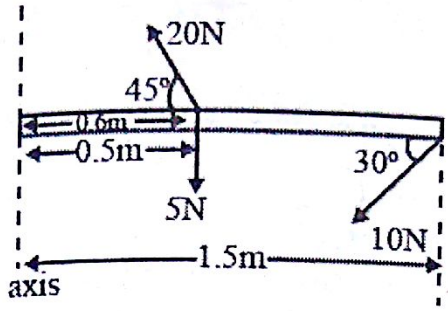


16. Three forces F = 40N to the east, F = 20N to the north and F = 40N to the West are applied on a body. What force should be applied on the body to keep it in equilibrium?

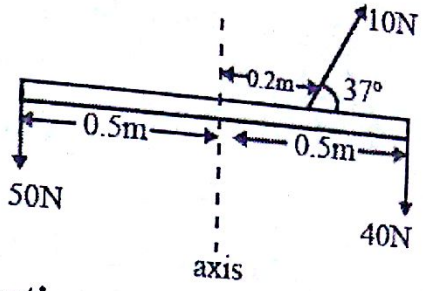
17. What force is needed to pull a 200 kg body up an incline of 37° at constant speed neglecting friction?

18. For the figures shown, find the torque about the given axis.

a)



b)

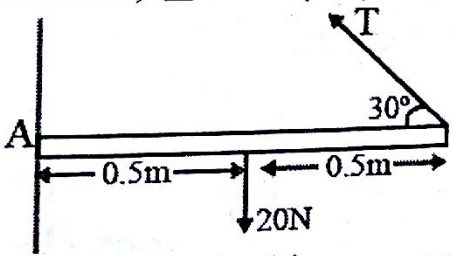


Unit 7: Explanation to Questions

1. The line of action of the forces should be parallel and two of the forces must act in the same direction at equal distances from the pivot point and the third may be placed at the pivot point or at any point perpendicular to the other two forces.
2. The safest way to climb up a ladder is that, there must be friction between the ground and the ladder so that it will not skid and fall.
   Answer: B
3. $\sum F_x = 0, -6 + 1 + 5 = 0$, and
   $\sum F_y = 0, 2 + 1 + (-3) = 0$. Thus, the first condition is satisfied and hence the concurrent forces are in equilibrium.

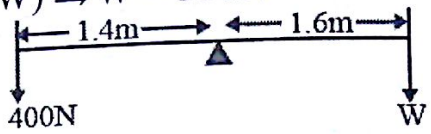
Unit 7: Solution to problems

4. a) $(0.45)(50) - (0.2)(40) = 14.5$N.m anti clock wise
   b) $(0.25)(40) = 10$ N.m anti clock wise
5. a) $(0.55)(12) - (0.25)(20 \sin 50°) = 2.77$N counter clock wise
   b) $(0.55)(18 \sin 40°) - (0.3)(20 \sin 50°) = 1.76$ N.m counter clock wise
6. a) $(0.9)(10 \sin 65°) - (0.4)(25 \sin 55°) = -0.035$N.m
   b) $(F_y)(0.9) + (0.5)(25 \sin 55°) = 0 \Rightarrow F_y = 11.4$N
7. Taking moment about A, $\sum \tau = 0\ (0.5)(20) = T \sin 30°$
   $\Rightarrow T = 20$N



8. $\sum F = -40 - 60 + 100 = 0$, and taking moment about any point
   $\sum \tau_A = (0.3)(100) - (0.5)(60) = 0$. So, the bar is in equilibrium.

9. Taking moment about the pivot point, $\sum\tau = 0 \Rightarrow (1.4)(400) = (1.6)(W) \Rightarrow W = 350N$

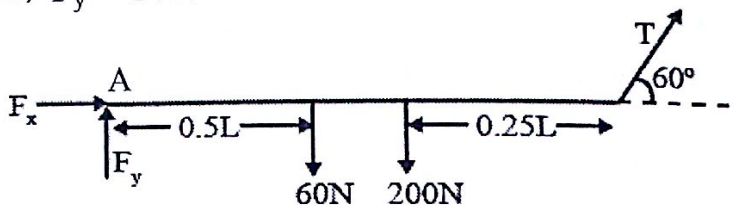


10. Taking moment about A. $(0.5L)(60) + (0.75L)(200) = (L)(T \sin 60°) \Rightarrow T = 207.85N$

$\sum F_x = 0 \Rightarrow F_x = T \cos 60 = 103.9N$

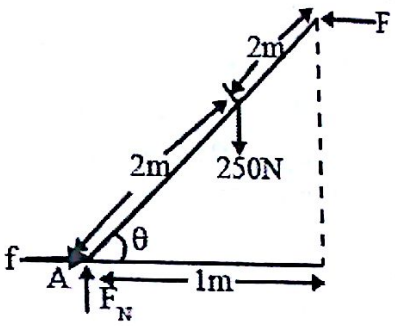
$\sum F_y = 0 \Rightarrow F_y + T \sin 60 - 60 - 200 = 0$

$\Rightarrow F_y = 80N$



11. Taking moment about A.
$(250)(2 \cos 76°) = (F)(4 \sin 76°)$,
$F = 31.2N$
$f = F = 31.2N$ and
$F_N = 250N$

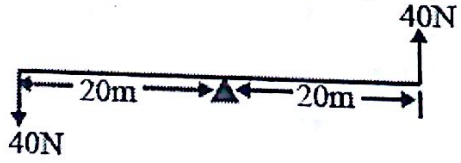
$$\Rightarrow \mu = \frac{f}{F_N} = 0.12$$



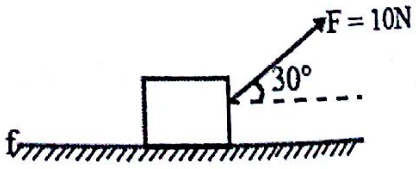
$\cos\theta = \frac{1}{4}$

$\Rightarrow \theta = 76°$

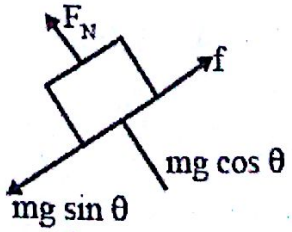
12. $\tau = \tau_1 + \tau_2 \Rightarrow (40)(0.2) + (40)(0.2) = 16$ N. m



13. $(F \cos\theta) - f = 0$
$(10)(\cos 30°) = \mu(mg - (10)(\sin 30°))$
$W = 22.3N$



14. $mg \sin\theta = \mu mg \cos\theta$
$\mu = \tan\theta = 0.57$



$f = \mu F = \mu mg \cos\theta$

15. Taking moment at the pivot point (12)(3) = (7)(F), F = 5.1N

16. $\sum F_x = 0 \Rightarrow 40 + F_x + (-40) + 0 = 0 \Rightarrow F_x = 0$

$\sum F_y = 0 \Rightarrow 0 + F_y + 20 + 0 = 0 \Rightarrow F_y = -20N$

Thus, a 20N force due south is needed.

17. $F = mg \sin\theta = (200)(10)(\sin 37°) = 1203.6N$

18. a) $\tau = (0.6)(20)(\sin 45) - (0\ .5)\ (5) - (1.5)(10)(\sin 30°)$

$\tau_{net} = -1.5$ N.m $\Rightarrow$ 1.5 N.m clock wise

b) $\tau = (0.5)(50) + (0.2)(10)(\sin 37) - (0.5)(40)$

$\tau_{net} = 6.2$N.m, counter clock wise

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Unit 8

Properties of Bulk Matter

Effective use of materials, first of all, requires knowledge of their properties. Any building, from a multistory skyscraper to the humblest shed, must be designed so that it won't topple over. Similar concerns arise with a suspension bridge, a ladder leaning against a wall or a crane hoisting a bucket full of concrete. What is more, it is hard to think of life without fluids. After all, we begin life as a fluid filled cell suspended in a fluid. In fact, fluids coursing through our circulatory system are literally the lifeblood of our existence. If it were not for the gases in our atmosphere and the liquid water on the Earth's surface, we could not exist. Moreover, heat being a form of energy, it is all around us. All living things need energy. Without energy, all living things would die.

8.1 Elastic Behavior

A wire or a rubber band that is stretched tends to come back to its original length when a force acting on it is removed. Such property of a body to regain its original condition, on the removal of the deforming force is called *elasticity*.

Elasticity: Is the property of a body to retain its original shape or size after the removal of a deforming force.

When a force is applied on a body, the body may be deformed, and the deformation is either *elastic* or *plastic* deformation.

Elastic deformation: Is a deformation where the body regains its original shape when the force that caused the deformation is removed

Plastic deformation: Is a deformation where the body do not regain its original shape after the force that caused the deformation is removed

Remember; materials are elastic until a certain limit and beyond that limit, the material will be permanently deformed.

Elastic limit: Is a point beyond which all materials are permanently deformed

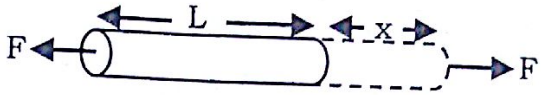
Note that; elastic limit depends on the internal structure of the material. Steel for example has a lower elastic limit than rubber band.

Stress – strain relation

Stress: Is force of resistance per unit area offered by a body against deformation, and

Strain: Is change in length per unit length

Tensile stress: Is tension force divided by cross sectional area of the body $\text{tensile stress} = \frac{F}{A}$



Its unit is N/m^2 (pascal)

Tensile strain: Is extension divided by the original length of the body.

$\text{tensile strain} = \frac{X}{L}$ Its unitless X – extension, L – original length

Important points to remember

- Tensile stress occurs when a body is subjected to two equal and opposite pulls as a result of which the body increases in length.
- For a tensile stress,, the force usually acts on the cross sectional area perpendicularly.

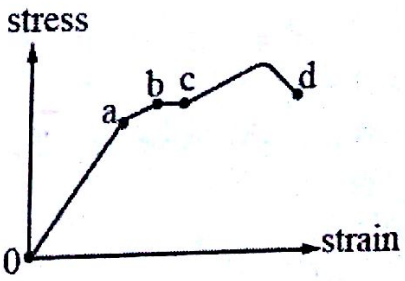
Hooke's Law

The relationship between stress and strain was first discovered by Robert Hooke.

Hooke's Law: states that; "The force applied to a material is directly proportional to its extension up to the elastic limit."

In other words, Hooke's law can be stated as, ""within its elastic limit, stress is directly proportional to the stain."

The behavior of materials under stress up to breaking point is visualized by the stress – strain curve as shown. In the first part of the curve, the stress and strain are proportional until the proportionality limit, point **a** is reached. From **a** to **b** stress and strain not proportional, nevertheless, if the load is removed at any point between **a** and **b** the material will return to its original shape. In the region **ob,** the material is said to be elastic and point **b** is called *elastic limit* or *yield point.*



If the material is stretched further, the strain increases rapidly, but when the load is removed at some point beyond b say c, the material do not come back to its original shape. Further increase of load beyond c

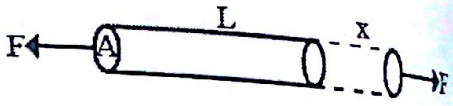
produces a large increase in strain even if the stress decreases until point d is reached at which *fracture* takes place. From b to d, the material is said to undergo *plastic deformation*.

- If a large plastic deformation takes place between the elastic limit and the fracture point, the material is said to be *ductile*. If, however, fracture occurs soon after the elastic limit is passed, the material is said to be *brittle*.

Young's Modulus: Is the ratio of longitudinal stress to the longitudinal strain up to the material's elastic limit.

Consider a wire of length L and cross – sectional area of A. If a force F is applied normal to the cross – section and produces a change in length of the wire by x, then

$$\text{Young's Modulus} = \frac{\text{longitudinal stress}}{\text{longitudinal strain}}$$



$$Y = \frac{F/A}{X/L} = \frac{FL}{Ax}$$

- **Important points to remember**
  - Since strain is unit less, the unit of young's modulus is the same as that of the stress, that is N/m^2 or Pascal.
  - For many materials, Young's modulus has the same value for both tensile and compressive stresses.
  - A material with a large value of young's modulus is relatively unstretchable.

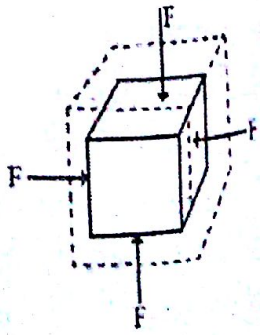
For example, the value of Y for steel is 2×10^{11} pa is much larger than that for rubber, 5×10^8 Pa.

Bulk Modulus: Is the ratio of normal stress to volume strain of a material.

If a body having volume v is subjected to a force F applied normal to the surface area A, leads the volume to change by, Δv then

$$\text{Bulk Modulus} = \frac{\text{normal stress}}{\text{volume strain}}$$

$$B = \frac{F/A}{\Delta V/V} = -\frac{\Delta P}{\Delta V/V}, \quad P = F/A = \text{Pressure}$$

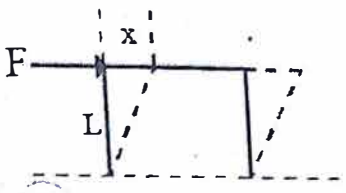


Important points to remember

- Bulk modulus of a body, in other terms is the measure of the ability of a body to resist changes in volume when exposed to increasing pressure from all sides.
- A minus sign is introduced in the equation, because an increase of pressure always causes a decrease in volume. After all, the bulk modulus is a positive quantity.

Shear modulus: Is the ratio of shear stress to shear strain of a body.

Consider a force F acting tangent to the surface of area A. If a body with original length of L on each side is displaced by distance x relative to its original position, then



$$\text{Shear Modulus} = \frac{\text{Shear stress}}{\text{Shear Strain}}$$

$$S = \frac{F/A}{X/L} = -\frac{FL}{AX}$$

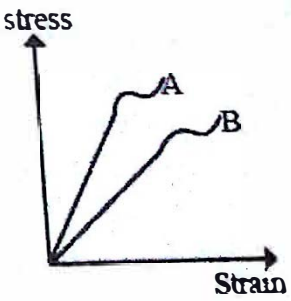
Strain Energy: Is the energy stored in a stretched or compressed body by a force F to an extension or compression x.

That is,

$$\text{Strain energy} = \frac{1}{2}Fx$$

Conceptual Examples

1. From the stress – strain graphs of two materials A and B, which one do have larger young's modulus? Explain



Explanation: $Y = \frac{\text{stress}}{\text{strain}}$ = slope of linear position of the graph

Therefore, Young's modulus of A is larger.

2. A material having greater young's modulus also possesses greater bulk modulus. True or False?

Explanation: True. Because, the elastic moduli are due to intermolecular forces existing in the material.

3. Rubber is more elastic than steel. True or False?
Explanation: False. Because, for the same stress, strain in rubber is more.

4. Water is more elastic than air. True or False?
Explanation: True. Because, for the same pressure applied, volume strain in water is less than air.

5. The breaking stress of a steel wire of radius r is s. The breaking stress of another steel wire of radius 2r is also s. True or False.
Explanation: True. Because, breaking stress is independent of length and radius. It depends only on material.

6. A wire can be broken by applying a load of 200N. The force required to break the wire of twice the diameter is 400N. True or False?
Explanation: False, Because, force = (stress)(Area)
Force = (stress)(πr^2) = 800N

Illustrative Examples

7. A metal wire 75cm long and 1.5mm in diameter stretches 0.4mm when a load of 10kg is hung on its end. Find
a) the stress b) the strain, and
c) the young's modulus of the wire

Solution: a) $\text{stress} = \frac{F}{A} = \frac{mg}{\pi\frac{d^2}{4}} = \frac{(10)(10)}{\pi\frac{(1.5\times10^{-3})^2}{4}} = 5.7\times10^7\,Pa$

b) $\text{strain} = \frac{X}{L} = \frac{0.4}{750} = 5.3\times10^{-4}$

c) $\text{Young's Modulus} = \frac{stress}{strain} = 106.96Pa$

8. The bulk modulus of water is 2.1G pa. Calculate the volume contraction of 100mL of water when subjected to a pressure of 1.5Mpa

Solution: $\text{Bulk modulus} = \frac{-\Delta P}{\Delta V/V} \Rightarrow \Delta V = -\frac{V\Delta P}{B}$

$\Rightarrow \Delta V = -100 \times \frac{1.5\times10^6}{2.1\times10^9} = -0.071\text{mL} \Rightarrow$ volume reduced

9. A chalk box has a top area of $15cm^2$ and a height of 3cm. when a shearing force of 0.5N is applied to the upper surface, the upper surface displaces 4mm relative to the bottom surface. Find
a) the shear stress
b) the shear strain
c) the shear modulus

Solution: a) Shear stress $= \frac{F}{A} = \frac{0.5}{15\times10^{-4}} = 333Pa$

b) Shear strain $= \frac{X}{L} = \frac{0.4}{3} = 0.133$

c) Shear Modulus $= \frac{\text{stress}}{\text{strain}} = \frac{333}{0.133} = 2.5kpa$

10. a) An iron rod 4m long and $0.5cm^2$ in cross section stretches 1mm when a mass of 225kg is hung from its lower end. Compute young's modulus for the iron.
b) If the rod is considered to be a spring, within its elastic limit, what is the force constant of the spring

Solution: a) $E = \frac{FL}{XA} = \frac{(225)(10)(4)}{(1\times10^{-4})(0.5\times10^{-4})} = 176000Mpa$

b) $F = \frac{EAX}{L} = KX \Rightarrow K = \frac{EA}{L} = 2.2\times106\ N/m$

11. A steel wire whose cross section is $4mm^2$ is stretched by 0.1mm by a certain weight. How long will the wire stretch if its cross sectional are is doubled keeping the other things constant?

Solution: $E = \frac{FL}{X_1A_1}$ ..... (*), also

$E = \frac{FL}{X_2A_2}$ .....(**),but $A_2 = 2A_1$

taking ratio of equation (**) to equation (*)

$\frac{1}{X_2A_2} = \frac{1}{X_1A_1} \Rightarrow X_2A_2 = X_1A_1 \Rightarrow X_2 = \frac{1}{2}X_1 = 0.05\ mm$

12. How large a force is required to stretch a 2cm diameter steel rod by 0.01 percent? E = 195000Mpa

Solution: $F = \frac{AEX}{L} = \pi(0.01)^2(195\times10^9)(10^{-4}) = 6100N$

13. A copper wire has a diameter of 1mm and is 3m long. Assume elastic limit stress = 1.6×10^8Pa, and ultimate stress = 3.4×10^8Pa. Find

a) the greatest load that can be supported by the wire without exceeding its elastic limit
b) the change in length of the wire under this load
c) the maximum load that can be supported without breaking the wire
d) the maximum elongation

Solution: a) $\frac{F}{A} = 1.6\times10^{8}\, N/m^{2}$

$$\Rightarrow F_{max} = \left(1.6\times10^{8}\right)\left(\pi\frac{(0.5\times10^{-3})^{2}}{4}\right) = 130.5\text{N}$$

b) $x = (F/A)(L/E) \approx 4\text{mm}.$ $\quad E = 0.1\times10^{11}\text{N/m}^2$

c) $\frac{F^{1}}{A} = 3.4\times10^{8}\,\text{N/m}^2$

$$\Rightarrow F' = \left(3.4\times10^{8}\right)\left(\pi\times\frac{(0.5\times10^{-3})^{2}}{4}\right) = 277\text{N}$$

d) $x' = (F'/A)(L/E) = 87.5\text{mm}$

14. Determine the fractional change in volume as the pressure of the atmosphere, 0.1Mpa around a metal block is reduced to zero by placing the block in vacuum. The Bulk modulus for the metal is 125000Mpa.

Solution: Bulk modulus $= \frac{-\Delta P}{\Delta V/V} = B$

$$\frac{\Delta V}{V} = \frac{-\Delta P}{B} = \frac{-(-0.1)}{125000} = 8\times10^{-7}$$

15. Calculate the volume change of a copper cube 40mm on each edge, when subjected to a pressure of 20Mpa. The bulk modulus for copper is 125000Mpa.

Solution: $\Delta V = \frac{-V\Delta P}{B} = \frac{-(40)^{3}(20)}{125000} = -10\text{mm}^3$

16. The pressure in an explosion chamber is 345Mpa. What would be the percent change in volume of a piece of copper subjected to this pressure. B = 138Gpa

Solution: $100\left|\frac{\Delta V}{V}\right| = 100\frac{\Delta P}{B} = 100\times\frac{345\times10^{6}}{138\times10^{9}} = 0.25\%$

17. a) How large a pressure must be applied to water if it is to be compressed by 0.1 percent.
b) what is the ratio of this pressure to atmospheric pressure, 101kpa? Bulk modulus of water is 2100Mpa

Solution: a) $B = \dfrac{\Delta P}{\Delta V/V} \Rightarrow \Delta P = B \times \dfrac{DV}{V} = (2.1\times10^{9})(10^{-3})$

$$\Delta P = 2100\text{kpa}$$

b) $\dfrac{\Delta P}{P} = \dfrac{2100}{101} = 21, \dfrac{\Delta V}{V} = 0.1\% = 10^{-3}$

18. What increase in pressure is required to decrease the volume of 200L of water by 0.004 percent? Take B = 2100Mpa

Solution: $\Delta P = B\left(\dfrac{-\Delta V}{V}\right), \Delta V = 0.00004\times200 = 0.008L$

$$\Delta P = 2100\times10^{6}\left(\frac{0.008}{200}\right) = 84\text{kpa}$$

19. Young's Modulus is the

A. ratio of stress to strain
B. force per unit area
C. ratio of strain to stress
D. force per unit elongation

Answer A

20. When stretched beyond its elastic limit, a metal rod such as steel

A. become elastic
B. has no energy
C. obeys Hook's law
D. becomes plastic

Answer D

21. When a load of 500kg is hanging from a steel wire of length 3m and cross sectional area 0.2cm^2, the wire stretches beyond its no-load length. If the young's modulus for the wire made of steel is 2×10^{11}Pa. what is the change in length of the wire beyond its no load length.

A. 3.75mm B. 2.75mm C. 1.75mm D. 0.75mm

Solution: $E = \dfrac{FL}{A\Delta L} \Rightarrow \Delta L = \dfrac{FL}{AE} = \dfrac{(500)(10)(3)}{(0.2\times10^{4})(2\times10^{11})}$

$\Delta L = 3.75\times10^{-3}\text{m}, \Delta L = 3.75\text{mm}$ **Answer: A**

8.2 Fluid Statics

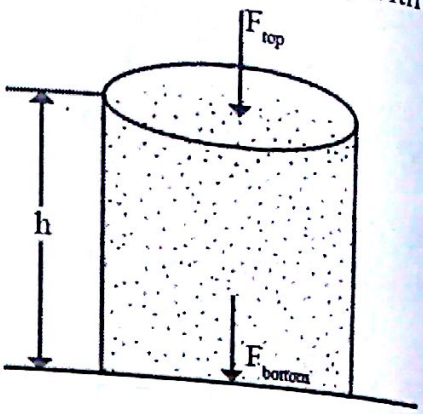
Fluids play a vital role in many aspects of everyday life. We drink them, breathe them, swim in them. They circulate through our bodies and control our weather. Airplanes fly through them, ships float in them.

- **Important points to remember**
  - **Fluid**: Is any substance that can flow. Both, liquids an gases are fluids.
  - **Fluid statics**: Is the study of fluids at rest.

Pressure due to a Fluid Column

Consider a cylindrical container filled to a height h with a fluid of density ρ. The top surface of the fluid is open to the atmosphere, with a pressure, P_{atm}. If the cross – sectional area of the container is A, the downward force exerted on the top surface by the atmosphere is

$$F_{top} = P_{atm}A$$



At the bottom of the container, the downward force is F_{top} plus the weight of the fluid. Recalling that, $M = \rho v$, $v = hA$, weight is,

$$W = Mg = \rho vg = \rho hAg$$

$$\Rightarrow F_{bottom} = F_{top} + W = P_{atm}A + \rho hAg$$

$$P_{bottom} = \frac{F_{bottom}}{A} = P_{atm} + \rho gh$$

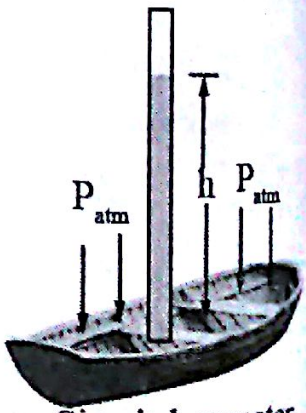
Therefore, the total pressure at a depth h below the fluid's surface is given by

$$\boxed{P = P_{atm} + \rho gh}$$

- **Important points of remember**
  - The pressure in a fluid acts equally in all direction, and acts at right angles to any surface.
  - Water seeks its own level. Hence, the pressure due to a fluid at the same depth is independent of the shape of the container.
  - When the fluid is in an open container, the total pressure at a depth h below the fluid's surface is given by;

    $\boxed{P = P_{atm} + \rho gh}$ ←Absolute pressure

  - **Absolute pressure**: Is the total pressure at a given point.
  - **Gauge pressure**: Is the amount by which the container pressure differs



Simple barometer

from atmospheric pressure.

$$P - P_{atm} = \rho gh \leftarrow \text{gauge pressure}$$

An interesting application of the variation of pressure with depth is the barometer, which was first proposed by Evangelista Torricelli in 1643. At the level of the fluid in the bowl, the pressure is atmospheric

$$P_{atm} = \rho gh$$

A fluid that is often used in a barometer is mercury (Hg), with a density of $\rho = 1.36 \times 10^4$ kg/m^3. The corresponding height for a column of mercury is $h = \frac{P_{atm}}{\rho g} = \frac{1.01 \times 10^5}{(1.36 \times 10^4)(9.81)} = 760\text{mm}$

In fact, atmospheric pressure is defined in terms of millimeters of mercury, mmHg.

1 atm = 760 mmHg

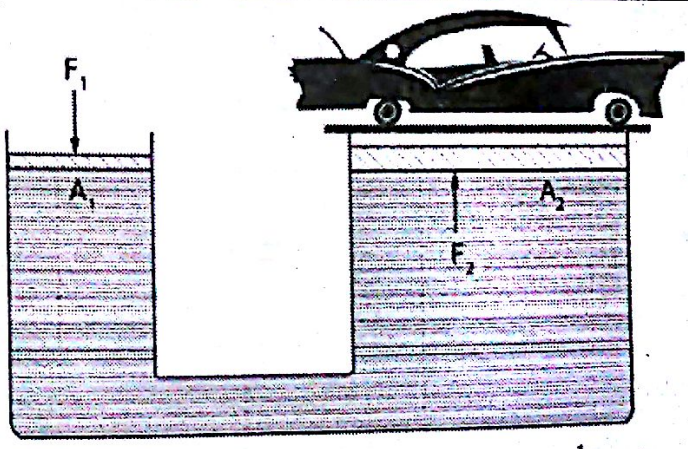
- The pressure at any point due to a fluid column depends on density and depth of the fluid.

Pascal's Principle

An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid. This fact was recognized in 1653 by the French scientist Blaise Pascal and is called *Pascal's principle.*

Pascal's principle states that, "pressure exerted anywhere in a confined fluid is transmitted equally and undiminished in all directions throughout the fluid."

That is, if a small force F_1 is applied to a piston with small cross – sectional area A_1 in a confined fluid, then the pressure is transmitted to the larger piston of cross – sectional area A_2, making it experience a larger force F_2. Since the pressure on the two pistons, according to Pascal's principle is the same, we have



$$P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \leftarrow \text{Pascal's principle}$$

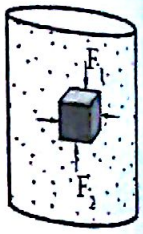
- **Important points to remember**
  - Dentist's chairs, car lifts, forklifts, hydraulics jacks, hydraulic press, elevators and many others use this principle.
  - Pascal's principle helps a larger load, such as a car to be lifted on the larger piston by applying a much smaller force on the smaller piston.

Archimedes' Principle

The fact that a body immersed in water seems to weigh less than when it is in air because of the upward force of the water called buoyant force was discovered by the Greek scientist Archimedes' so long ago.

Archimedes' principle states that; "when a body is completely or partially immersed in a fluid, the fluid exerts an upward force on the body equal to the weight of the fluid displaced by the body."

To see the origin of buoyancy, consider a cubical block immersed in a fluid of density ρ. Clearly, the horizontal forces on the block cancel out and has no effect.



Assume the cubical block is of length L and that the pressure on the top surface is P_1. The down ward force on the block, then, is

$$F_1 = P_1A = P_1L^2$$

The pressure at the bottom face is, $P_2 = P_1 + \rho gL$.

Therefore, the upward force exerted on the bottom surface is,

$$F_2 = P_2A = (P_1 + \rho gL)L^2 = P_1L^2 + \rho gL^3 = F_1 + \rho gL^3$$

The net vertical force exerted by the fluid on the block –that is, the buoyant force, F_b is,

$$F_b = F_2 - F_1 = \rho gL^3$$

- Therefore, the buoyant force is equal to the weight of the fluid that is displaced by the cube.

More generally, if a volume v of an object is immersed in a fluid of density ρ, the buoyant force can be given by:

$\boxed{F_b = \rho gv}$ ← buoyant force

- **Important points to remember**
  - If the weight of a body is less than the up thrust on the body, then the body floats with a part of it out-side the liquid.

- If the body floats completely immersed is a liquid then the weight of the body is equal to the up thrust on the body. In this case, the apparent weight of the body is zero.
- If the weight of a body is greater than the up thrust on the body, then the body sinks.
- Consider a piece of ice which floats in a liquid whose density is greater than that of water. When the ice melts completely, the fluid level rises.
- Consider a piece of ice which floats in a liquid whose density is less than that of water. When the ice melts completely, the level of fluid falls.

Conceptual Examples

22. A metal block sinks in water. But, when the metal is shaped like a bowl, it floats. Why? Explain

Explanation: In the second case, the volume is increased as a result, the buoyancy is larger.

23. A ship floats, but a coin dropped in water sink. How do you explain this?

Explanation: According to Archimedes' objects that are denser than the fluid they are immersed in will sink, those less dense will float. Steel is much denser that water, but ship is not made of solid steel all the way through. There are open spaces filled with air and other materials in the ship, hence the average density of the ship is less than that of water and thus it floats.

Illustrative Examples

24. When a submarine dives to a depth of 100m, to how large a total pressure is its exterior surface subjected? Take density of sea water, $1.05g/cm^3 = 1\times10^5 + (1000)(1.05)(10)(100)$

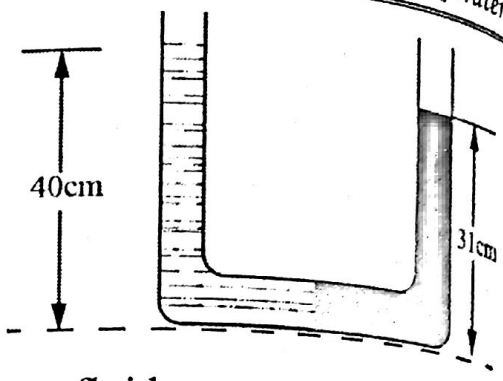
Solution: $P = P_{atm} + \rho gh, \quad P_{atm} = 1 \times 10^5 Pa$

$1g/cm^3 = 1000kg/m^3 = 1.15 \times 10^6 pa = 1.15Mpa$

25. In a hydraulic press, the larger piston has cross sectional area $A_2 = 200cm^2$ and the smaller piston has cross sectional area $A_1 = 5cm^2$. If a force of 250N is applied to the smaller piston, find the force on the larger piston.

Solution: $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = \frac{A_2}{A_1}F_1 = \frac{200}{5} \times 250 = 10,000N$

26. As shown below, a column of water, 40cm high supports a 31cm column of an unknown fluid. What is the density of the unknown fluid?

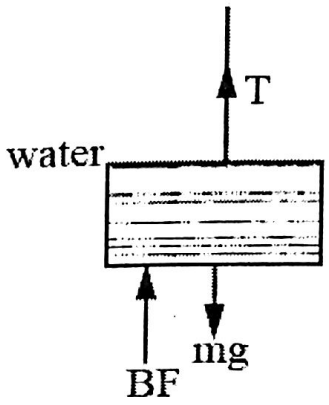


Solution: pressure due to water = pressure due to unknown fluid

$$\rho g h_w = \rho_u g h_u$$

$$\rho_u = \frac{h_w}{h_u}\rho_w = \frac{40}{31} \times 1000 = 1290 kg/m^3$$

27. What will be the tension in a string that suspends an aluminum block of mass 25g when the block is totally submerged in water. Take density of aluminum, $2700 kg/m^3$



T - tension of string
V - volume of block
ρ_w - density of water

Solution: $v = \frac{m}{\rho} = \frac{0.025}{2700} = 9.26 cm^3 = 9.26 \times 10^{-6} m^3$ Buoyant force = weight of displaced water and

$T = mg - BF = mg - v\rho_w g$

$= (0.025)(10) - (9.26 \times 10^{-6})(1000)(10) = 0.157N$

28. A piece of alloy weighs 0.86N in air and 0.73N when immersed in water find

a) the volume, and b) the density of the alloy

Solution: B_F = Weight in air – weight in water

$B_F \rightarrow$ Buoyant force, $B_F = 0.86 - 0.73 = 0.13N$, also

$V \rightarrow$ volume of alloy B_F = weight of displaced water

$0.13 = V(\rho_w)(g) = v(1000)(10)$

a) $V = 1.3 \times 10^{-5} m^3$

b) $\rho = \frac{m}{v} = \frac{0.086}{1.3 \times 10^{-5}} = 6600 kg/m^3$

29. A solid aluminum cylinder with density of $2700kg/m^3$ weighs 0.67N in air and 0.45N when immersed in a liquid. What is the density of the liquid?

Solution: $BF = 0.67 - 0.45 = 0.22N$

$$\text{Volume of liquid} = \text{volume of cylinder} = \frac{0.067}{2700} = 2.48\times10^{-5}\,m^3$$

$0.22 = BF = $ weight of displaced liquid

$\rho_\ell \rightarrow$ density of liquid, $0.22 = V\rho_\ell g = 2.48 \times 10^{-5} \times \rho_\ell \times 10$

$$\rho_\ell = 887kg/m^3$$

30. If a glass stopper weighs 0.025N in air, 0.015N in water and 0.007N in sulfuric acid, what is the density of the acid

Solution: B_F due to water $= 0.025 - 0.015 = 0.01N$

Volume of stopper = volume of displaced water.

$$V = \frac{B_F}{\rho g} = \frac{0.01}{(1000)(10)} = 1\times10^{-6}\,m^3$$

The buoyant force in acid is,

B_F due to acid $= 0.025 - 0.007 = 0.018N$

$$\text{Also, } V = \frac{B_F}{\rho_{acid} g} \Rightarrow \rho_{acid} = \frac{B_F}{V_g} = \frac{0.018}{(10^{-6})(10)} = 1800kg/m^3$$

31. The density of ice is $917kg/m^3$. What fraction of the volume of a piece of ice will be above water when floating in fresh water?

Solution: B_F = weight of displaced water = weight of ice

$$\rho_{ice} g V_{ice} = \rho_{water} g V_{\text{water displaced}}$$

$$V_{\text{water displaced}} = \frac{\rho_{ice}}{\rho_{water}} V_{ice} = \frac{917}{1000} V_{ice} = 0.917 V_{ice}$$

The fraction of the volume that is above water is then

$$\frac{V_{ice} - V_{\text{displace water}}}{V_{ice}} = \frac{V_{ice} - 0.917V_{ice}}{V_{ice}} = 0.083$$

32. A 60kg rectangular box, has base dimensions 1m by 0.8m and depth 0.5m. How deep will it sink in fresh water

Solution: B_F = weight of displaced water = weight of box

$$\rho vg = mg \Rightarrow (1000)(1\times0.8\times h)(10) = (60)(10)$$

$$\Rightarrow h = 0.075m$$

33. A foam plastic of density $580kg/m^3$ is to be used as a life preserver. What volume of plastic must be used if it is to keep 20

percent of an 80kg man above water in a lake? Take average density of the man to be $1040kg/m^3$.

Solution: At equilibrium we have

B_F on man + B_F on plastic = weight of man + weight of plastic

$(\rho_w)(0.8V_m)g + \rho_w V_p g = \rho_m V_m g + \rho_p v_p g$

$$(\rho_w - \rho_p)v_p = (\rho_m - 0.8\rho_v)\, v_m,\ V_m = \frac{m_m}{\rho_m} = \frac{80}{1040}$$

$$(1000 - 580)v_p = (1040 - 0.8 \times 1000)\frac{80}{1040} \Rightarrow v_p = 0.044m^3$$

34. A piece of pure gold of density $193000kg/m^3$ suspected to have a hole. If it weight 0.3825N in air and 0.3622N in water. What is the volume of the hole in the gold?

Solution: volume of pure gold $= \frac{m}{\rho} = \frac{0.03825}{19300} = 1.982 \times 10^{-6} m^3$

$\rho_w \times g \times$ volume of displaced water $= B_F = 0.3825 - 0.3622$

$V_{displaced\ water} = 2.03 \times 10^{-6} m^3$, and

Volume of hole $= (2.03 - 1.982)cm^3 = 0.048cm^3$

35. What must be the volume of a 5kg balloon filled with helium of density $0.178kg/m^3$, if it is to lift a 30kg load? Take $\rho_{air} = 1.29kg/m^3$

Solution: B_F = weight of balloon + weight of helium

$$v\rho_{air}g = m_b g + V\rho_{he}g$$

$$V = \frac{m_b}{\rho_{air} - \rho_{he}} = \frac{35}{1.29 - 0.178} = 31.5m^3$$

36. A solid wooden cube, 30cm on each edge, can be totally submerged in water if it is pushed downward with a force of 54N. What is the density of the wood?

Solution: $F = F_b - mg = \rho_w vg - \rho vg$

$$\rho vg = \rho_w vg - F \Rightarrow \rho = \frac{\rho_w\ vg - F}{vg}$$

$$\rho = \frac{(1000)(0.3)^3(10) - 54}{(0.3)^3(10)} = 800kg/m^3$$

37. Downward force of 45N and 15N respectively are required to keep a plastic block totally immersed in water and in oil. If the volume of the block is $8000cm^3$. Find the density of the oil

Solution: $F = F_b - mg \Rightarrow 45 = \rho_w gv - \rho_b gv$ .....(*)

$$15 = \rho_0 gv - \rho_b gv \quad \ldots\ldots\ldots\ldots\ldots\ldots (**)$$

Subtracting, eqn (**) from eqn (*) we have

$$30 = \rho_w v - \rho_0 v \Rightarrow \rho_0 = \rho_w - \frac{30}{v}$$

$$\rho_0 = 1000 - 30/8 \times 10^{-3} = 625 kg/m^3$$

38. a) Determine the unbalanced force acting on an iron ball (r = 1.5cm, $\rho = 7.8 g/cm^3$) when just released while totally immersed in water, in mercury ($\rho = 13.6 g/cm^3$)
b) what will be the initial acceleration of the ball in each case?

Solution: a) $F_1 = F_b - mg = \rho_w gv - \rho_b gv = (1000)(10)\frac{4}{3}\pi r^3 -$

$$(7800)(10)\frac{4}{3}\pi r^3 = 0.96N \text{ down}$$

$$F_2 = F_b - mg = \rho_w gv - \rho_b gv = (13600)(10)\frac{4}{3}\pi r^3 -$$

$$(7800)(10)\frac{4}{3}\pi r^3 = 0.82N \text{ up}$$

$$b)\ a_1 = \frac{F_1}{m_1} = \frac{0.96}{(7800)\left(\frac{4}{3}\pi r^3\right)} = 8.7 m/s^2 \text{ down}$$

$$a_2 = \frac{F_2}{m} = \frac{0.82}{(7800)\left(\frac{4}{3}\pi r^3\right)} = 7.4 m/s^2 \text{ up}$$

39. What is the minimum volume of a block of wood having density = $850 kg/m^3$ if it is to hold a 50kg woman entirely above the water when she stands on it?

Solution: Weight of woman + weight of wood = buoyant force

$$Mg + \rho_{wood} vg = \rho_{water} vg \Rightarrow 50g + 850vg = 1000vg$$

$$50 + 850v = 1000v,\ v = 0.33 m^3$$

40. A solid cube of 0.75cm on each edge floats in oil of density $800 kg/m^3$ with one-third of the block out of the oil.
a) what is the buoyant force on the cube?
b) what is the density of the cube?

Solution: a) $\rho_b v_b g = B_F = \rho_0 \left(\frac{2}{3} v_b\right) g = 2.21 \times 10^{-3} N$

b) $B_F = \rho_b v_b g \Rightarrow \rho_b \frac{2}{3} \rho_{oil} = 533 kg/m^3$

41. A cubical copper block is 1.5cm on each edge
a) What is the buoyant force on it when it is submerged in oil of $\rho = 820 kg/m^3$
b) What is the tension in the string that supported the block when submerged? $\rho_{cu} = 8920 kg/m^3$

Solution: a) $B_F = \rho_0 V_{cu} g = (820)(1.5\times10^{-2})^3(10) = 0.027N$
b) $T = mg - B_F = \rho_{cu} V_{cu} g - 0.027$
$T = (8920)(1.5 \times 10^{-2})^3(10) - 0.027 = 0.268N$

42. The weight of a galloon and the gas it contains is 11.12kN. if the balloon displaces $1132m^3$ of air and the weight of $1m^3$ of air is 12.3N, what is the acceleration with which the balloon begins to rise?

Solution: $B_F - w = ma \Rightarrow a = \frac{B_F - w}{m}$

$$a = \frac{(1132)(12.3) - (1.112\times10^4)}{1.112\times10^4/10} = 2.5 m/s^2$$

43. A small block of wood of density $0.4 \times 10^3 kg/m^3$, is submerged in water at a depth of 2.9m.
Find a) the acceleration of the block toward the surface when the block is released, and
b) the time for the block to reach the surface

Solution: a) $\rho_w g v = \rho_{wo} g v = ma$, $m = \rho_{wood} V$

$$a = \frac{\rho_w g v - \rho_{wo} g v}{\rho_{wo} V} = \left(\frac{1}{0.4} - 1\right)(10) = 15\, m/s^2$$

b) $S = \frac{1}{2} at^2 \Rightarrow t = \sqrt{\frac{25}{a}} = \sqrt{\frac{2(2.9)}{15}} = 0.63 sec$

44. The pressure at the surface of an ocean is $10^5 N/m^2$. At what depth in the ocean is the absolute pressure three times the atmospheric pressure?

Solution: $P = P_{atm} + \rho gh \Rightarrow 3P_{atm} = P_{am} + \rho gh$

$$h = \frac{2P_{atm}}{\rho g} = \frac{2(10^5)}{1000(10)} = 20m$$

45. How far would a stone of density $2.5g/cm^3$ starting from rest, sink in water in two seconds?

Solution: $-B_F + mg = ma \Rightarrow a = g\frac{1-\rho_w)}{\rho_s}$

$$\Rightarrow a = 10\left(1-\frac{1}{2.5}\right) = 6m/s^2$$

and, $S = \frac{1}{2}at^2 = \frac{1}{2}(6)(2)^2 = 12m$

46. A table – tennis ball has a diameter of 3.8cm and average density of $0.084g/cm^3$. What force is required to hold it completely submerged under water?

A. 0.521N B. 0.258N C. 0.3N D. 0.510N

Solution: $F = F_b - mg = \rho_w gv - \rho_b gv$, $v = \frac{4}{3}\pi r^3$

$$F = (1000-84)(10)\left(\frac{4}{3}\pi(1.9\times10^{-2})^3\right) = 0.258N$$

Answer B

47. The pressure at the bottom of a cylindrical tube filled with water was measured to be 5000pa. If the water in the tube were replaced with ethyl alcohol, what would be the new pressure at the bottom of the tube? Density of ethyl alcohol is $0.8g/cm^3$

A. 4000pa B. 4800pa C. 5000pa D. 6250pa

Solution: $P_1 = \rho gh \Rightarrow h = \frac{P}{\rho g} = \frac{5000}{(1000)(10)} = 0.5m$

$P_2 = \rho gh = (800)(10)(0.5) = 4000Pa$

Note: $1g/cm^3 = 1000kg/m^3$

Answer: A

48. A very large open tank is full of oil whose density is $800kg/m^3$. If the absolute pressure at the bottom of the tank is 1.81×10^5Pa, then what is the depth of the oil? $P_{atm} = 1.01\times10^5$pa

A. 100m B. 75m C. 10m D. 9.5m

Solution: $P = P_{atm} + \rho gh \Rightarrow \rho gh = P - P_{atm}$, $h = \frac{P-P_{atm}}{\rho g}$

$$h = \frac{1.8\times10^5 - 1.01\times10^5}{(1000)(10)} = 10m$$

Answer: C

49. A container resting on the floor of an elevator contains an incompressible fluid of density (ρ), when the elevator has an upward acceleration of (a) then what will be the pressure difference between two points in the fluid sparated by a vertical distance Δh?

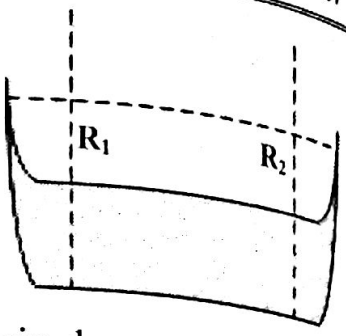
A. $\rho a\Delta h$ B. $\rho(g+a)\Delta h$ C. $\rho g\Delta h$ D. $\rho ga\Delta h$

Solution: $F - mg = ma \Rightarrow F = ma + mg$, $m = \rho v = \rho(A\Delta h)$

$$P = \frac{F}{A} = \frac{\rho A\Delta h(a+a)}{A} = \rho(g+a)\Delta h$$

Answer: B

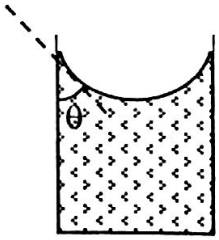
In symbols; $\boxed{\Delta P = f\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$



This equation can help us to find the shape of rain drops, soap bubbles, menisci .....etc

Angle of contact and capillary action

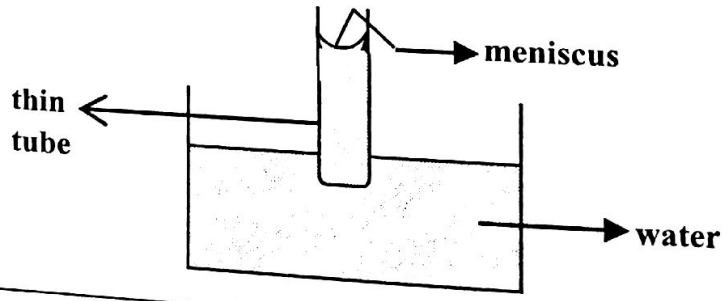
Contact angle (θ): Is the angle at which a liquid surface meets a solid surface.



Remember, for different liquid-solid interaction, the contact angle is different. For instant, the angle between water and silver container is about 90^0 and the contact angle between water and lead glass is 0^0 see? Good!

Capillary Action: Is the movement of a liquid along the surface of a solid caused by the attraction of molecules of the liquid to molecules of the solid.

Meniscus: Is a curve in the surface of a liquid caused by the relative attraction of the liquid molecules to the surface of the container.



Keep in mind that for water, liquid; the thiner the tube, the more the capillary rise

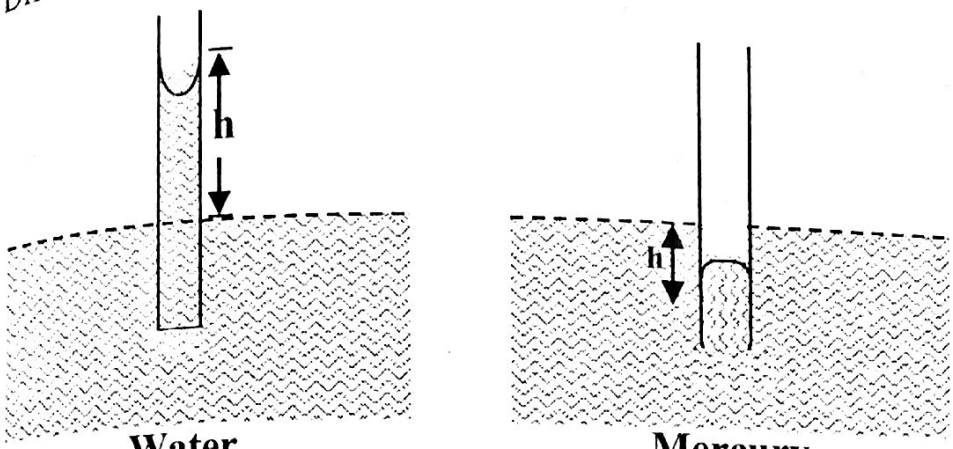
In a more general sense, capillary action explains two phenomena

1. The movement of liquids in thin tubes
2. The flow of liquids through porous materials

Such as the follow of water through soil and rock

Explaining capillary action in thin tubes

Different liquids do have different capillary action



Water **Mercury**

Note that, the height h, the liquid rise or drop through the tube from the surface of the liquid is given by;

$h = \frac{2f\cos\theta}{\rho g r}$ where; θ = contact angle

f = surface tension

ρ = density of liquid

g = gravitational acceleration

r = radius of tube

Conceptual Examples

52. The spherical shape of a rain drop is due to

A. Density of liquid C. Atmospheric pressure

B. Gravity D. Surface tension

Answer: D

53. A needle floats on the surface of water because of

A. Lighter weight C. Viscosity

B. Adhesive force D. Surface tension

Answer: D

Illustrative Examples

54. 8.40. A vertical force of 1.6×10^{-2}N is required to lift a wire ring of radius 1.5cm from the surface of milk. What is the surface tension of the milk?

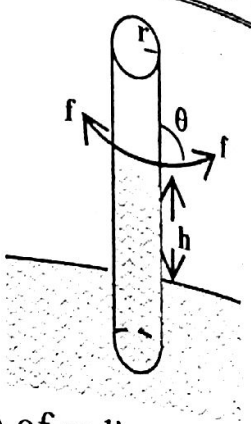
Solution: $= \frac{F}{\ell} = \frac{F}{2\pi r} = \frac{1.6\times10^{-2}}{(2)(3.14)(1.5\times10^{-2})}$, ℓ - contact length

$$f = 0.17N/m$$

55. Show that the height a liquid can rise through a thin tube is given by

$h = \frac{2f\cos\theta}{\rho g r}$

Solution: let the liquid rise to a height h is a tube of radius r.



$$\sum F_Y = 0 \Rightarrow mg = F_y$$

but $m = \rho v = \rho\pi r^2 h$, $v = \pi r^2 h$

and $F_y = F\ell\cos\theta = f2\pi r\cos\theta$, $\ell = 2\pi r$

$$\Rightarrow \rho\pi r^2 hg = 2\pi fr\cos\theta$$

$$\Rightarrow h = \frac{2f\cos\theta}{\rho gr}$$

56. Find the height of methyl iodide in glass tube of radius 20mm, if the surface tension is 0.26, the contact angle is 29^0 and the density is 2.28

Solution: $h = \frac{2f\cos\theta}{\rho gr} = \frac{2\times0.26\times\cos(29^0)}{(2.28)(10)(0.02)} \approx 1m$

57. A certain fluid has density of $1080kg/m^3$ and is observed to rise to a height of 2cm in a 2mm diameter tube. The contact angle between the wall and the fluid is zero. Calculate the surface tension of the fluid.

Solution: $h = \frac{2f\cos\theta}{\rho gr} \Rightarrow f = \frac{\rho grh}{2\cos\theta}$

$$f = \frac{(1080)(10)(0.001)(0.02)}{(2)(\cos 0^0)}$$

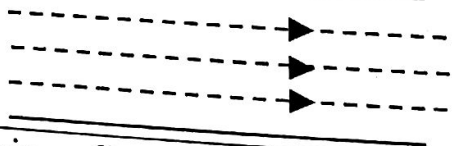
$$f \approx 0.11N/m$$

8.4 Fluid Dynamics

Fluids can move or flow in many ways. Water may flow smoothly and slowly in a quiet stream or violently over a waterfall. The air may form a gentle breeze or a raging tornado.

Streamline and Turbulent Flow

Streamline (laminar flow): Is a fluid flow where the fluid particles move smoothly in regular layers.

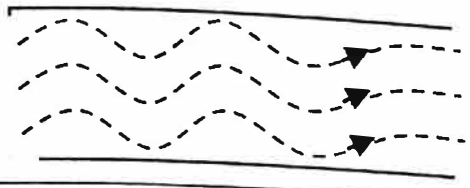


In simple terms; laminar flow;

- Is one in which the particles of the fluid *do not cross* one another but move along well defined paths.
- do have constant velocity and pressure at every point in the fluid
- can be observed in;
  - flow through capillary tube
  - flow of blood in veins and arteries

- ground water flow

Turbulent flow: Is a fluid flow where there is a disruption to the layers of the fluid.



In simple terms; turbulent flow;

- Is that flow in which fluid particles move in *a zig zag* way.
- do have a continuously changing velocity both in magnitude and direction
- can be observed in
  - high speed flow in a conduit of large size
  - nearly all fluid flow problems in Engineering

In fluid mechanics, the basic equations are;

- *continuity equation*
- *energy equation, and*
- *impulse – momentum equation*

- The basic fluid equations are derived by considering the fluid as "ideal fluid". That is, the fluid is:

→ **non – viscous:** there is *no internal friction* between fluid layers

→ **incompressible:** the fluid *density* is constant

→ **steady:** the *velocity*, *density* and *pressure* at each point in the fluid do not change with time

→ **laminar:** each element of the fluid has *zero angular velocity*.

Rate of flow (Discharge): Is the quantity of liquid flowing per second through a section of pipe or a channel.

$$\text{discharge} = \frac{volume}{time} = Area \times velocity$$

$\boxed{Q = A\vec{V}}$, Its unit is m^3/s

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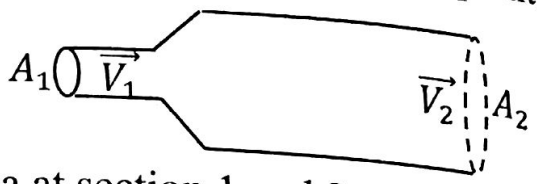
Equation of continuity

Equation of continuity states that, "the volume flow rate of an ideal fluid, flowing through a closed pipe is the same at every point."

$$\frac{V_1}{t_1} = \frac{V_2}{t_2}$$

$$\Rightarrow \boxed{A_1\vec{V_1} = A_2\vec{V_2}}$$



Where: A = cross sectional area at section 1 and 2

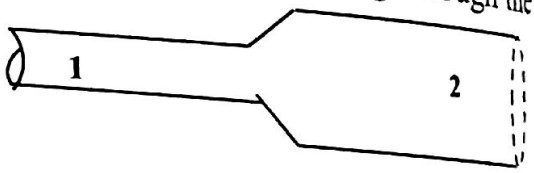
$\vec{V}$ = velocity of flow at section 1 and 2

$\frac{v}{t}$ = volume flow rate at section 1 and 2

Remember; continuity equation is based on the principle of *conservation of mass*. That is, if no fluid is added or removed from the pipe, then the mass passing across different section is the same.

Illustrative Examples

58. The diameters of a pipe at section 1 and 2 are 200mm and 300mm respectively. If the velocity of water flowing through the pipe at section 1 is 4m/s. Find



a) Discharge through the pipe, and

b) Velocity of water at section 2

Solution: a) $Q = AV = A_1V_1 = A_2V_2$

$$Q = A_1\vec{V_1} = \pi\frac{d_1^2}{4}\vec{V_1}$$

$$Q = \frac{\pi}{4}(0.2)^2(4) = 0.1256m^3/s$$

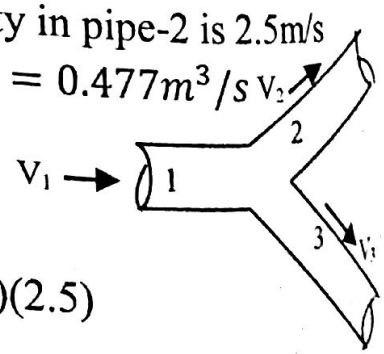
b) $A_1V_1 = A_2V_2 \Rightarrow V_2 = \frac{A_1V_1}{A_2} = \frac{0.0314\times4}{0.0707} = 1.77m/s$

59. 8.45. Pipe -1, 450mm in diameter branches into two pipes, pipe-2 of diameter 300mm and pipe -3 of diameter 200mm. If the average velocity in the 450mm diameter pipe is 3m/s. find;

a) Discharge through pipe -1

b) Velocity in pipe -3 if the velocity in pipe-2 is 2.5m/s

Solution: a) $Q_1 = A_1V_1 = \frac{\pi}{4}(0.45^2)(3) = 0.477m^3/s$



b) $Q_1 = Q_2 + Q_3$

$Q_3 = Q_1 - Q_2$

$A_3V_3 = A_1V_1 - A_2V_2$

$= (0.159)(3) - (0.0707)(2.5)$

$0.0314V_3 = 0.3 \Rightarrow V_3 = 9.55m/s$

60. If a 9000kg water flows in a minute through a pipe of cross sectional area $0.3m^2$, what is the speed of water in the pipe?

Solution: $V = \frac{m}{\rho} = \frac{9000}{1000} = 9m^3$

but $V = A\ell \Rightarrow \ell = \frac{V}{A} = \frac{9}{0.3} = 30m$

and $\vec{V} = \frac{\ell}{t} = \frac{30}{60} = 0.5\ m/s$

61. For an ideal fluid, show that the relation, $A_1\vec{V_1} = A_2\vec{V_2}$ holds

Solution: From conservation of mass:

$\frac{m_1}{t_1} = \frac{m_2}{t_2}$, but $m = \rho v = \rho A\ell$

$\frac{\rho A_1 \ell_1}{t_1} = \frac{\rho A_2 \ell_2}{t_2}$, ρ constant and $\frac{\ell}{t} = \vec{V}$

$$\boxed{A_1V_1 = A_2V_2}$$

Bernoull's Equation

For steady flow, the speed, pressure and elevation of an incompressible and non-viscous fluid are related by the equation discovered by the Danish physicist Daniel Bernoulli in 1738.

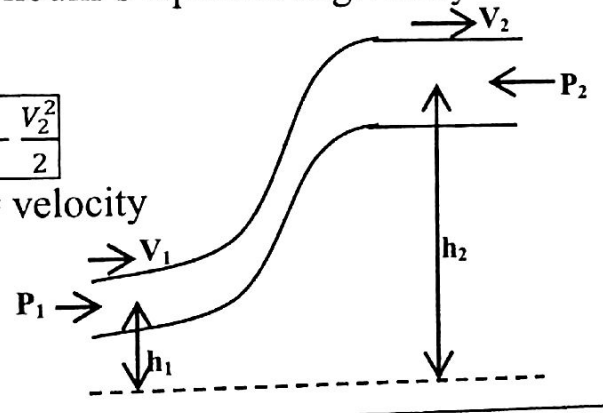
Bernoulli's equation states that: "In an ideal, incompressible fluid when the flow is steady and continuous, the sum of pressure energy, kinetic energy and potential energy is constant along a stream line."

In its common form, Bernoulli's equation is given by

$\frac{P}{\rho} + gh + \frac{v^2}{2} = \text{constant}$

$$\Rightarrow \boxed{\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}}$$

P = pressure, h = elevation, v = velocity



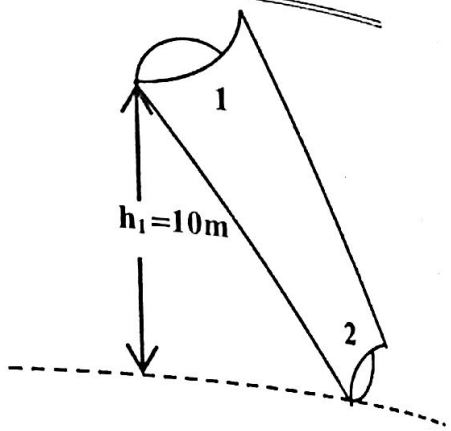
Note that; Bernoulli's equation is

- Based on the principle of *conservation of energy*. That is, total pressure is constant along a streamline.
- Written in simplified form as: $P_0 = p + q$, that is;

Total pressure (P_0) = static pressure (P) + dynamic pressure (q)

Illustrative Examples

62. ater flows in a pipe. At one section, the diameter is 0.3m, the static pressure is 260kpa, the velocity is 3m/s and the elevation is 10m above ground level. The elevation at a section downstream is 0m and the pipe diameter is 0.15m. Find the static pressure at downstream section



Solution: From Bernoulli's;

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$$

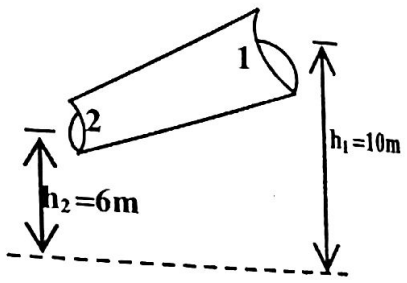
$$\frac{260\times1000}{1000} + (10)(10) + \frac{3^2}{2} = \frac{P_2}{1000} + 0 + \frac{v_2^2}{2} \quad \ldots\ldots(*)$$

$$A_1V_1 = A_2V_2 \Rightarrow V_2 = \frac{A_1}{A_2}V_1 = \left(\frac{0.3}{0.15}\right)^2 \times 3 = 12m/s$$

Substituting for V_2 in to eqn $(*)$

$$260 + 100 + \frac{9}{2} = \frac{P_2}{1000} + \frac{12^2}{2} \Rightarrow P_2 = 292.5kpa$$

63. 8.49. Water is flowing through a pipe having diameters 300mm and 150mm at sections 1 and 2 respectively. The discharge through the pipe is 40liters/sec. section 1 is 10m above datum and section 2 is 6m above datum. Find the intensity of pressure at section 2 if that at section 1 is 400kN/m^2



Solution:

$$Q = \frac{40\ell}{s} = \frac{40}{1000} = 0.04m^3/s$$

$$A_1 = \frac{\pi}{4}0.3^2 = 0.0707$$

$$Q = A_1V_1 = A_2V_2 \rightarrow \text{continuity}$$

$$V_1 = \frac{Q}{A_1} = \frac{0.04}{0.0707} = 0.566m/s$$

$$V_2 = \frac{Q}{A_2} = \frac{0.04}{0.01767} = 2.264m/s$$

$$A_2 = \frac{\pi}{4}0.15^2 = 0.01767$$

From Bernoulli's

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$$

$$\Rightarrow \frac{400000}{1000} + (10)(10) + \frac{0.566^2}{2}$$

$$= \frac{P_2}{1000} + (10)(6) + \frac{2.264^2}{2}$$

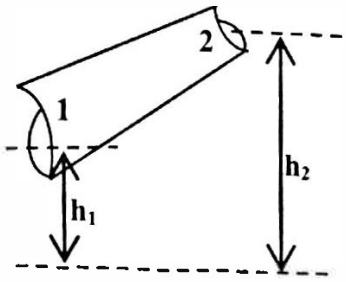
$$\Rightarrow P_2 = 4316 kN/m^2$$

64. Water is flowing through a pipe having diameters 600mm and 400mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is $350kN/m^2$ and the pressure at the upper end is $100kN/m^2$. Determine the difference in elevation if the rate of flow is 60 liters/sec

Solution: $A_1 = \frac{\pi}{4}0.6^2 = 0.283m^2$

$A_2 = \frac{\pi}{4}0.4^2 = 0.1257m^2$

$Q = 60\ell/s = \frac{60}{1000} = 0.06m^3/s$



From continuity:

$$Q = A_1V_1 = A_2V_2,\ V_1 = \frac{Q}{A_1} = \frac{0.06}{0.283} = 0.212m/s$$

$$V_2 = \frac{Q}{A_2} = \frac{0.06}{0.1257} = 0.477m/s$$

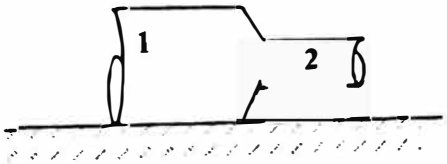
Applying Bernoulli's eqn

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$$

$$g(h_2 - h_1) = \frac{P_1}{\rho} - \frac{P_2}{\rho} + \frac{V_1^2}{2} - \frac{V_2^2}{2} = 350 - 100 + (-0.031)$$

$h_2 - h_1 = 25m$

65. Water flows steadily through a horizontal pipe of varying cross section. At section 1 the pressure is 130kpa and the speed is 0.6m/s. determine the pressure at section 2 where the water speed is 9m/s.



Solution: $\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$

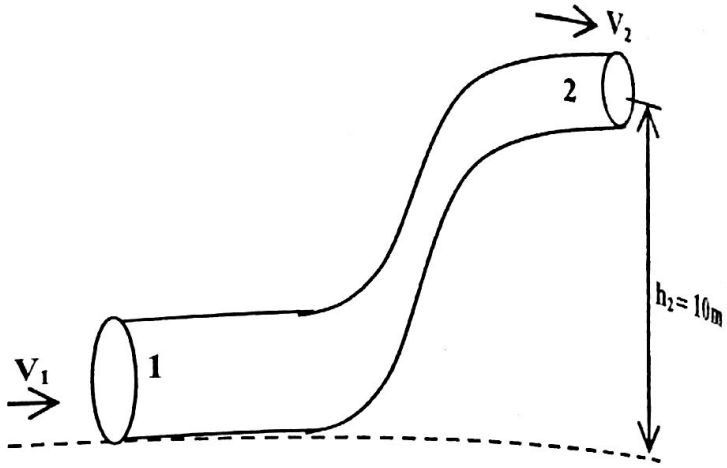
but, $h_1 = h_2 = 0$

$$\frac{P_1}{\rho} + \frac{V_1^2}{2} = \frac{P_2}{\rho} + \frac{V_2^2}{2}$$

$$\Rightarrow P_2 = \rho\left(\frac{P_1}{\rho} + \frac{V_1^2}{2} - \frac{V_2^2}{2}\right) = 1000\left(\frac{130000}{1000} + \frac{0.6^2}{2} + \frac{9^2}{2}\right)$$

$P_2 = 89.68kpa$

66. The cross sectional area of a pipe at point 1 is twice that at point 2 which is 10m above point 1. The speed of water in the pipe at point 1 is 5m/s and the pressure at this point is 150kpa. What are the speed and pressure at point 2?



Solution: $A_1 = 2A_2$, $h_1 = 0$, $h_2 = 10m$

$P_1 = 150kpa$, $V_2 = ?$, $P_2 = ?$

Using equation of continuity

$$A_1V_1 = A_2V_2 \Rightarrow V_2 = \frac{A_1v_1}{A_2} = \frac{2A_2(5)}{A_2} = 10m/s$$

Using Bernoulli's equation

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$$

$$\frac{150{,}000}{1000} + 0 + \frac{5^2}{2} = \frac{P_2}{1000} + (10)(10) + \frac{(10)^2}{2}$$

$P_2 = 12.5kpa$

67. Wind stream horizontally past a small air plane's wing such that the speed is 100m/s over the top surface and 50m/s past the bottom surface. If the plane has a mass of 800kg and the wing has area of $10m^2$, what is the net vertical force on the plane. Take density of air $1.2kg/m^3$

Solution: From Bernoulli's equation;

$$\frac{P_b}{\rho} + gh_b + \frac{V_b^2}{2} = \frac{P_T}{\rho} + gh_T + \frac{V_T^2}{2}$$

$$\Rightarrow \frac{P_T}{\rho} - \frac{P_b}{\rho} = \frac{V_b^2}{2} - \frac{V_T^2}{2} \Rightarrow P_T - P_b = \Delta P = \frac{\rho}{2}(v_b^2 - v_T^2)$$

$$\Rightarrow \Delta P = \frac{1.2}{2}(50^2 - 100^2) = \frac{4500N}{m^2}$$ and the pressure force is

$$\Delta P = \frac{F}{A} \Rightarrow F = \Delta P \times A = (4500)(10) = 45000N$$

The net vertical force is;

$F_{net} = F - mg = 45000 - 8000 = 37000N = 37kN$

68. A pipe of cross sectional area $4 \times 10^{-4}m^2$ carries water in to the ground floor at a velocity of 6m/s and pressure of 300kpa. What are the velocity and pressure when it reaches the 2nd floor which is 8m above the ground floor. Assume the cross section area of the pipe at the 2nd floor is half that of the ground floor.

Solution: using equation of continuity.

$A_1V_1 = A_2V_2 \Rightarrow V_2 = \frac{A_1}{A_2}V_1 = \frac{(4\times10^{-4})(6)}{2\times10^{-4}} = \frac{12m}{s}$

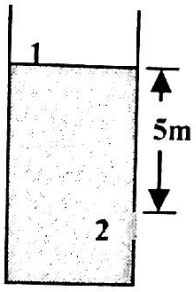
Using Bernoulli's equation

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$$

$$\frac{300{,}000}{1000} + 0 + \frac{(6)^2}{2} = \frac{P_2}{1000} + (10)(8) + \frac{(12)^2}{2}$$

$P_2 = 166kpa$

69. 8.55. What volume of water will escape per minute from a hole of 3cm in diameter that is 5m below the water level in the tank?



Solution: from Bernoulli's equation;

$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$, $v_1 = 0$

$\Rightarrow V_2 = \sqrt{2gh_2}$ $\quad h_1 = 0, h_2 = 5m$

$V_2 = \sqrt{2(10)(5)} = 10m/s$ $\quad P_1 = P_2 = P_{atm}$

and $Q = V_2A_2 = (10)(\pi\times1.5\times10^{-2})^2 = 7 \times 10^{-3}m^3/s$

70. A tank is filled with water to a depth H. If a hole is made at a depth h from the water surface, at what horizontal distance R from the foot of the wall does the stream strike the floor?

Solution: From conservation of energy between point 1 and 2

$\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh}$

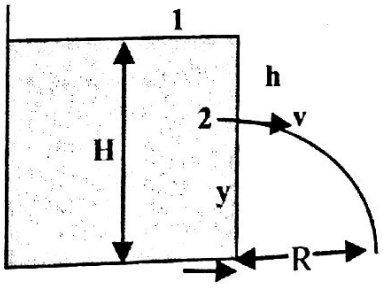
and $y = u_yt + \frac{1}{2}gt^2$, but $u_y = 0$,

$y = H - h$

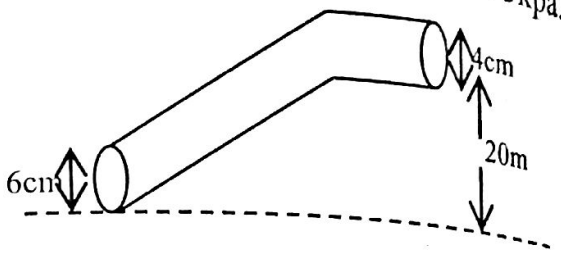
$t = \sqrt{2(H-h)/g}$ and

$R = vt = \left(\sqrt{2gh}\right)\left(\sqrt{2(H-h)/g}\right)$

$\Rightarrow R = 2\sqrt{h(H-h)}$



71. Water flows through a pipe as shown below. The water speed through the lower pipe is 5m/s and a pressure guage reads 75kpa. What is the reading of the guage on the upper pipe?



A. 75kpa
B. 55kpa
C. 24.6kpa
D. 4.6kpa

Solution: from continuity equation

$$A_1V_1 = A_2V_2 \Rightarrow V_2\frac{A_1}{A_2}V_1 = \frac{\pi(3)^2\times(5)}{\pi(2)^2} = 11.25m/s$$

From Bernoulli's equation

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}, h_1 = 0$$

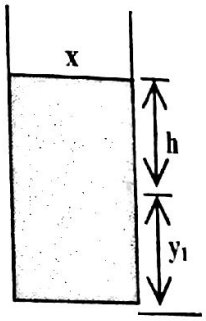
$$\frac{75000}{1000} + 0 + \frac{(5)^2}{2} = \frac{P_2}{1000} + (10)(20) + \frac{(11.25)^2}{2}$$

$\Rightarrow P_2 = 175.78$kpa

$P_{gange} = P - P_{atm} = 175.78 - 101 \approx 75$kpa.

Answer: A

72. An open tank containing a liquid of density ρ has a small hole in its side at a distance y_1 from the bottom. What is the speed at which the fluid leaves the whole. When the liquid level is a distance h above the hole?



A. $\sqrt{2gh}$ B. $\sqrt{2gy_1}$
C. $\sqrt{gh}$ D. $\sqrt{gy_1}$

Solution: $mgh = \frac{1}{2}mv^2$

$$V = \sqrt{2gh}$$

Answer: A

73. Water flows through a pipe whose cross-sectional are is 50cm2 into another pipe whose cross-sectional area is 10cm2. Both pipes are horizontal. If the velocity of water in the larger pipe is 12m/s, the velocity in the smaller pipe is

A. 0.24m/s B. 1.2m/s C. 60m/s D. 30m/s

Solution: From continuity eqn

$$A_1V_1 = A_2V_2 \Rightarrow V_2 = \frac{A_1}{A_2}V_1 = \frac{50}{10}\times 12 = 60m/s$$

Answer: C

74. A table – tennis ball has a diameter of 3.8cm and average density of $0.084g/cm^3$. What force is required to hold it completely submerged under water?

A. 0.521N B. 0.258N C. 0.3N D. 0.510N

Solution: $F = F_b - mg = \rho_w gv - \rho_b gv$, $v = \frac{4}{3}\pi r^3$

$$F = (1000 - 84)(10)\left(\frac{4}{3}\pi(1.9 \times 10^{-2})^3\right) = 0.258N$$

75. A container resting on the floor of an elevator contains an incompressible fluid of density (ρ), when the elevator has an upward acceleration of (a) then what will be the pressure difference between two points in the fluid separated by a vertical distance Δh? **Answer: B**

A. ρaΔh B. ρ(g+a)Δh C. ρgΔh D. ρgaΔh

Solution: $F - mg = ma \Rightarrow F = ma + mg$, $m = \rho v = \rho(A\Delta h)$

$$P = \frac{F}{A} = \frac{\rho A\Delta h(a+a)}{A} = \rho(g+a)\Delta h$$

Answer: B

76. A horizontal pipe of 10cm in diameter has a smooth reduction to a pipe of 5cm is diameter. If the pressure of water in the larger pipe in 80kpa and the pressure in the smaller pipe is 60kpa, at what rate does the water flow through the smaller pipe?

A. 1.63m/s B. 6.53m/s C. 13.6m/s D. 42.67m/s

Solution: from continuity. Eqn

$$A_1V_1 = A_2V_2 = \pi\frac{d_1^2}{4}V_1 = \pi\frac{d_2^2}{4}V_2 \Rightarrow (0.1)^2V_2 = (0.05)^2V_2$$

$V_2 = 4V_1$

From Bernoulli's eqn

$$\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}, \quad h_1 = h,\ V_1 = \frac{V_2}{4}$$

$$\frac{80000}{1000} + \frac{V_2^2}{32} = \frac{60000}{1000} + \frac{V_2^2}{2}$$

$V_2 = 42.67$m/s **Answer: D**

Viscosity

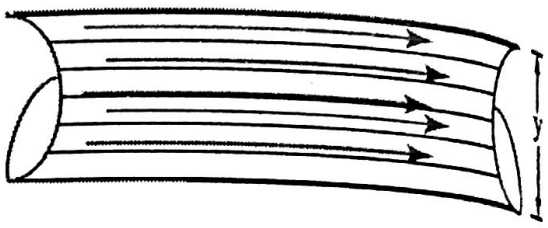
When a block slides across a rough floor, it experiences a frictional force opposing the motion. Similarly, a fluid flowing past a stationary surface experiences a force opposing the flow. This tendency to resist flow is referred to as the *viscosity* of a fluid. Fluids like air have low viscosities, thicker fluids like water are more viscous, and fluids like honey and molasses are characterized by their high viscosity.

Viscosity: Is the internal friction in a fluid and it is a measure of fluids resistance to flow.

In a fluid flow, its different parts slide past each other as if in layers, Each layer exerts a tangential forces on the other layer

which is adjacent to it. These forces between layers, which are resistive to the sliding of one layer over the other are called *viscous forces*.

Now, consider the flow of viscous fluid through a pipe. Let A be the area of the layer considered and $\frac{\Delta v}{\Delta y}$, the velocity



per depth called the *velocity gradient*. Then, the force required to drive the fluid through the pipe is given by the formula,

$$F = \eta A \frac{\Delta v}{\Delta y}$$

Where, η (eta) is called the *coefficient of viscosity*.

☛ **Important points to remember**

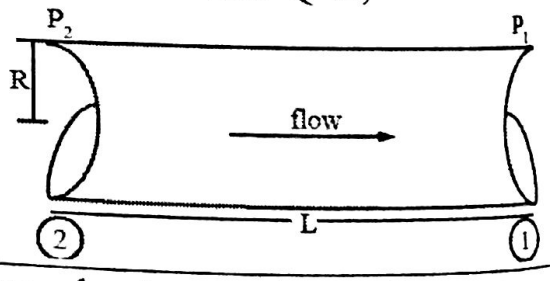
- The larger the viscosity, the slower the fluid flow.
- Usually, the viscosity of liquids decreases as the temperature is increased. In contrast, the viscosities of gases increase as the temperature is raised.

Poiseuille's Law

Experimentally, poiseuille found that, for viscous flow, the difference in pressures, the radius and length of the pipe and the viscosity of the flow influence the volume flow rate.

Poiseuille's law states that; "A fluid whose viscosity is η; flowing through a pipe of radius R and length L, with pressures of P_1 and P_2 at the ends of the pipe has a volume flow rate Q of;"

$$Q = \frac{\pi R^4 (P_2 - P_1)}{8 \eta L}$$



☛ A greater pressure difference leads to a larger flow rate, a long pipe offers greater resistance to the flow, and high viscous fluids have low flow rate, and the flow rate decreases as the radius of the pipe is decreased.

Stoke's Law and Terminal Velocity

An object moving through a viscous fluid is acted on by a resistive force that arises due to the fluid viscosity. This force is called *drag*

force. This force arises due to the velocity gradient between the points at the boundary layer of the moving object and distant points in the viscous fluid. The force due to the fluid drag increases with the speed of the moving object.

Stokes's law states that; "The drag force F on a sphere of radius r moving with a speed V relative to the fluid is given by;

$$F = 6\pi\eta rv$$

If a ball is dropped through a fluid, the force of gravity causes it to go faster and faster. As its velocity increases, the drag force also increases. Eventually, the drag force becomes as large as the weight of the ball and therefore cancels it out. The ball no longer accelerates, instead, it goes at a constant velocity.

Terminal velocity: Is the maximum constant velocity reached by a falling body when the drag force acting on it is equal to the weight of the body.

Mathematically: $V_T = \frac{2}{9}\frac{r^2 g}{\eta}(\rho - \rho_f)$

Where: V_T = terminal velocity
r = radius of the body
g = gravitational acceleration
η = viscosity
ρ = density of the body
ρ_f = density of the fluid

Reynolds Numbers

Reynolds number: Is number that helps us to determine whether a fluid flow is laminar or turbulent. And is given by: $N_R = \frac{\rho VD}{\eta}$

Where: N_R = Reyndolds number
ρ = density of fluid
D = diameter of pipe
V = speed of flow
η = viscosity

Reynolds numbers of less than 2300 are generally considered to be of a laminar type. However, the Reynolds number at which laminar flows become turbulent is dependent upon the flow geometry.

Illustrative Examples

77. What is the horizontal force required to move a metal plate of area $10^{-2}m^2$ with a velocity of 4m/s when it rests on a layer of oil 1.2×10^{-2}m thick. If the viscosity of oil is 0.2pa.s?

Solution: $F = \eta A\frac{\Delta v}{\Delta y} = (0.2)(10^{-2})\frac{(4)}{1.2\times10^{-2}} = 0.67N$

78. Calculate the pressure difference between two ends of 1km section of a pipe 50cm in diameter if it is to transport oil at a rate of $0.004m^3/s$. It is given that, the viscosity of oil is 0.2pa.s

Solution: using Poiseville's law

$$\Delta P = \frac{Q(8\eta L)}{\pi r^4} = \frac{(0.004)(8)(0.2)(1000)}{(3.14)(0.25)^4} = 521.8Pa$$

79. According to poiseville's law, by what factor does the blood flowing in the artery reduced if the radius of the artery is reduced to half of its original value?

Solution: From poiseville's law

$Q \sim r^4$

$\Rightarrow \frac{Q_1}{r_1^4} = \frac{Q_2}{r_2^4}$, but $r_2 = \frac{1}{2}r_1$

$$Q_2 = \frac{r_2^4}{r_1^4}Q_1 = \frac{r_2^4}{(2r_2)^4}Q_1 = \frac{1}{16}Q_1$$

80. Find the viscous force on a spherical oil droplet that is 2×10^{-6}m in radius and falling with a speed of 0.5mm/s in air. The coefficient of viscosity of air is 1.8×10^{-5}Pa.s

Solution: From stokes's law:

$F = 6\pi\eta rv = (6)(3.14)(1.8\times10^{-5})(2\times10^{-6})(5\times10^{-4})$

$F = 3.4\times10^{-13}N$

8.5 Concepts of Heat and Temperature

The terms heat and temperature are often used interchangeably is everyday language. In physics, however, these two terms have very different meanings.

Heat: Is the energy transferred between objects because of a temperature difference.

The SI unit of heat is the Joule (J)

Temperature: Is the measure of average kinetic energy of particles of a body.

Important points to remember

- The SI unit of temperature is the kelvin (k).

- Temperature is not a measure of heat, for a body possessing a large quantity of heat may be cold while a body possessing small quantity of heat may be quite hot.
- It is NOT correct to say that a body contains heat. The body has internal energy not heat. The word heat is used only when referring to the energy actually in transit from hot to cold.

Specific heat capacity (c): Is the quantity of heat required to raise the temperature of 1 kg of a body by 1k or 1°C.

- **Note that**: The SI unit of specific heat capacity is J/kgk or J/kg°C and different substances do have different value of specific heat capacities.

For example, the heat required to raise the temperatures of 1kg of water by 1°C is about 4200J, whereas, the heat required to raises the temperature of 1 kg of Gold by 1°C is about 130J and so on.

Heat capacity, C: Is the amount of heat energy required to raise the temperature of the whole body by 1k or 1°C.

- **Note that**, the SI unit of heat capacity is J/k or J/°C and the heat capacity equals the product of the specific heat capacity and the given mass of a body. $C = cm$

Heat gained or Heat lost by a body

The amount of heat gained or lost by a body depends upon the mass of the body, temperature rise of the body and specific heat capacity of the substance the body is made from.

Therefore, the heat Q that must be supplied or removed to change the temperature of a body of mass m with specific heat capacity c, by an amount ΔT is

$$Q = mc\Delta T, \quad \Delta T = T_2 - T_1$$

- ΔT is positive when heat flows into the system and it is negative when heat flows out of the system.

Latent or Hidden Heat: Is the amount of heat supplied or removed to cause change of state without a change in temperature.

- Latent heat causes a phase change at constant temperature.

Specific Latent Heat of Fusion, L_f: Is the amount of heat required to change 1 kg of a substance from solid to liquid at its melting point without any rise in temperature.

- The heat energy required to melt a body of mass m from solid to liquid at its melting point is given by;
  $\boxed{Q = mL_f}$, the unit of L_f is J/kg

Specific Latent Heat of Vaporization, L_v: Is the amount of heat required to change 1 kg of a substance from liquid to gas at its boiling point.

- The heat energy required to evaporate a body of mass m from liquid to gas at its boiling point is given by,
  $\boxed{Q = mL_v}$, the unit of L_v is J/kg

Illustrative Examples

81. Calculate the amount of heat energy needed to rise the temperature of 2kg water from 40^0C to 70^0C, take C = 4200J/kgk
Solution: $Q = mC\Delta T = (2)(4200)(70 - 40) = 252kJ$

82. The heat energy required to raise the temperature of 2kg of water from 20^0C to 50^0C is equal to the heat required to raise the temperature of 10kg of unknown liquid from 20^0C to 80^0C. What is:
    a) The specific heat capacity of the unknown liquid
    b) How much heat is required to raise the temperature of the unknown liquid by 40^0C? (C_{water} = 4200J/kgk)

Solution: a) $Q_w = Q_u \Rightarrow m_wC_w\Delta T_w = m_uC_u\Delta T_u$
U = unknown $\Rightarrow (2)(4200)(50 - 20) = (10)(C_u)(80 - 20)$
W = water $\Rightarrow C_u = 420J/kgk$
b) $Q = m_uC_u\Delta T_u = (10)(420)(40) = 168kJ$

83. The following data was recorded in an experiment to determine the specific heat capacity of water
    - mass of water = 100g
    - initial temperature = 22^0C
    - steady current through heater = 1.5A
    - steady potential difference across heater = 12v
    - total time of heating = 600sec
    - final temperature of water = 47^0C

From the experiment, what is the specific heat capacity of water?
Solution: $P = IV = (1.5)(12) = 18W$
$W = pt = (18)(600) = 10800J$
$W = Q = mC\Delta T \Rightarrow (0.1)(C)(47 - 22) = 10800$
$C = 4320J/kg.k \neq 4200J/kgk$

Note: The variation is due to error in the experiment due to different factors.

84. A 500g lead bullet traveling with a speed of 300m/s is stopped by a large tree. If half of the kinetic energy of the bullet is transformed in to heat and remains with the bullet while the other half is transmitted to the tree. What is the increase in temperature of the bullet? Take $C_{lead} = 128J/kgk$

Solution: $k_E = \frac{1}{2}mv^2 = \frac{1}{2}(0.5)(300)^2 = 22500J$

Energy changed to heat the bullet

$Q = \frac{1}{2}K_E = 11250J$

But $Q = mC\Delta T \Rightarrow \Delta T = \frac{11250}{(0.5)(128)} = 176k$

85. Calculate the total heat energy required to change the temperature of 5kg of water from 50^0C to 150^0C. Given that,

$C_w = 4.2kJ/kg^0C$, $C_{steam} = 2.08kJ/kg^0C$, $L_v = 2250kJ/kg$

Solution: $Q = Q_1 + Q_2 + Q_3$

$Q = \text{heat} = mC_w\Delta T_1 + mL_v + mC_s\Delta T_2$

$= 5\times4200\times(10-50)+5\times2250000+5\times2080\times(150-100)$

$= 12.82MJ$

86. An Aluminum can of mass 500g contains 120g of water at a temperature of 20^0C. If a 200g block of iron at 75^0C is dropped in to the can, find the final temperature of the system

$C_{iron} = 470$, $C_{Almunium} = 910$, $C_{water} = 4200\ J/kgC^0$

Solution: $Q_{\text{lost by iron}} = Q_{\text{gained by can}} + Q_{\text{gained by water}}$

$mC\Delta T_1 = mC\Delta T_2 + mC\Delta T_3$, $\Delta T_2 = \Delta T_3 = (T_f - 20)$

$\Delta T_1 = (75 - T_f)$

$(0.2)(470)(75-T_f) = (0.5)(910)(T_f-20)+0.12(4200)(T_f-20)$

$94(75 - T_f) = 959(T_f - 20) \Rightarrow T_f = 25^0C$

87. The heat required to raise the temperature of an object from 0°C to 25°C is 125kCal. How much heat is needed to raise the temperature of the same object from 40^0C to 60^0C?

Solution: $Q = mC\Delta T \Rightarrow mC = \frac{Q}{\Delta T} = \frac{125KCAL}{25^0C}$, and the heat needed to raise the temperature from 40^0C to 60^0C is

$Q = mC\Delta T = \frac{125}{25}(20) = 100kcal$

88. How much steam, with heat of vaporization 540cal/g at 100^0C must be added to 150g of ice of heat of fusion 80cal/g at 0^0C to produce water at 40^0C? $C_w = 1cal/g^0C$

Solution:

heat lost to change steam to water + heat lost to lower temperature from 100 to 40

= heat gained by ice + heat gained to increse temperature

$Q_1 + Q_2 = Q_1' + Q_2'$

$mL_v + mC\Delta T_1 = mL_f + mC\Delta T_2$, $\Delta T_1 = 100^0 - 40^0$

$\Delta T_2 = 40 - 0^0$

$(m)(540)+(m)(1)(100-40) = (150)(80) + (150)(1)(40-0^0)$

$\Rightarrow m = 30g$

89. A 0.7kg metal dish contains 1kg of water at 20^0C. A 0.5kg iron bar at 120^0C is dropped into the water, and the final temperature become 24.9^0C. What is the material of the dish?

$C_{iron} = 0.11$ cal/g^{0}C, $C_w = 1$cal/g^{0}C

Solution: $Q_{lost\ by\ iron} = Q_{aganed\ by\ dish} + Q_{ganed\ by\ water}$

$m_i C_i \Delta T_1 = m_d C_d \Delta T_2 + m_w C_w \Delta T_2$

$(0.5)(0.11)(120\text{-}24.9) = 0.7\times C_d\times(24.9\text{-}20)+(1)(1)(24.9\text{-}20)$

$C_d = 0.096$kcal/kgC^0 → the dish is probably made of copper

90. A student heated a 10g iron nail for sometime in a Bunsen - burner flame and then plunged the nail in to 100g of water at 10^0C. The water temperature rose to 20^0C. What was the temperature of the flame? $C_{iron} = 0.11$cal/gC^0, $C_{water} = 1$cal/gC^0

Solution: $Q_{lost\ by\ nail} = Q_{gained\ by\ water}$

$m_n C_n \Delta T_n = m_w C_w \Delta T_w \Rightarrow (10)(0.11)(T\ -20) = (100)(1)(20-10)$

$\Rightarrow T = 929°C$

91. How many kilograms of ice at 0^0C must be added to 0.6kg of water at 100^0C in an insulated 0.1kg copper container in order to cool the container and its contents to 30^0C?

$C_w = 4.2$kJ/kgk $C_{cupper} = 0.39$kJ/kgk, $L_f = 335$kJ/kg

Solution:

$Q_{lost\ by\ water} + Q_{lost\ by\ copper} = Q_{gained\ by\ ice\ to\ melt} + Q_{gained\ to\ increase\ temp}$

$m_w C_w(100\text{-}30) + m_c C_c(100-30) = m_i l_i + m_i C_w(30-0$

$(0.6)(4.2)(70)+(0.1)(0.39)(70) = m_i(335) + m_i(4.2)(30)$

$\Rightarrow m_{ice} = 0.39$kg

92. A 0.02kg of ice and a 0.1kg of water at 0^0C are in a container. If steam at 100^0C is passed in until all the ice is just melted. What is the mass of total water

$L_v = 2.3\times10^{-6}$J/kg,, $L_f = 3.4\times10^5$J/kg, $C_w = 4.2\times10^4$J/kgk

Solution: $m_s L_v + m_s C_w \Delta T = m_i L_f$

$m_s(2.3\times10^6+4.2\times10^4\times100)=0.02\times3.4\times10^5 \Rightarrow m_s = 0.00105$

$m = m_i + m_w + m_s = 0.02 + 0.1 + 0.00105 = 0.12105kg$

93. Temperature of a body is a measure of the
A. heat content of a body
B. Average molecular kinetic energy
C. Average pressure in a gas
D. heat of vaporization of a liquid

Answer: B

94. Which of the following combinations of properties would be most desirable for a thermal insulator?
A. high specific heat and low conductivity
B. low specific heat and high conductivity
C. high specific heat and high conductivity
D. low specific heat and low conductivity

Answer: A

95. A cup of coffee at a temperature of 80^0C is placed on a table in a 20^0C room. What happens to the thermal energy and molecular motion of the coffee when it is left on the table?
A. The thermal energy decreases and the molecules move more slowly
B. The thermal energy increases and the molecules move more slowly
C. The thermal energy decreases and the molecules move faster
D. The thermal energy increases and the molecules move faster

Answer: A

96. The amount of heat required to raise the temperature of an object of mass m = 1kg by 1 kelvin is known as? (UEE: 2007)
A. Specific heat capacity
B. Heat capacity
C. Expansion coefficient
D. Internal energy

Answer: A

Heat Transfer

Heat can be exchanged in a variety of ways. The sun, for example, warms the Earth from across 93 million miles of empty space by a process known as *radiation*. As the sunlight strikes the ground and raises its temperature, the ground level air gets warmer and begins to rise, producing a further exchange of heat by means of *convection*. Finally, if you walk across the ground in bare feet, you will feel the warming effect of heat entering your body by *conduction*.

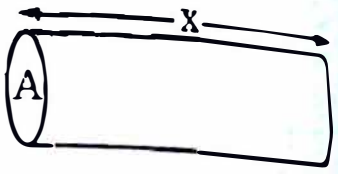
Now, let's see these three mechanisms of heat exchange in detail.

Conduction: Is the transfer of heat through a material due to vibration and collisions of molecules of the body without actual motion or flow of molecules.

For example, if you hold one end of a metal rod and put the other end in a fire, it doesn't take long before you begin to feel warmth on your end. The heat you feel is transported along the rod by *conduction.*

☛ The rate at which heat will be conducted through a bar of length x and cross – sectional area A, is given by

$$\frac{Q}{t} = kA\frac{\Delta\theta}{x}$$



Where, $\frac{Q}{t}$ = rate of heat flow

k = constant, which depends of the material the bar made from

$\frac{\Delta\theta}{x}$ = temperate gradient

$\Delta\theta$ = change in temperature between ends of the bar, $\Delta\theta = \theta_2 - \theta_1$

Convection: Is the transfer of heat that takes place due to the actual movement of molecules of the system.

For example, to heat a small room, you turn on an electric heater. As the heater coils get red hot, the air in the vicinity warms, it expands, becoming less dense. Because of its low density, the warm air rises, replaced by cold dense air from overhead. This sets up a circulating flow of air that transport heat from the coil to the air throughout the room. Heat exchange of this type is referred to as *convection.*

☛ The rate of heat flow by convection is given by

$$\frac{Q}{t} = hA\Delta T$$

Where: A = contact area

ΔT = change in temperature

h = convection constant

In forced convection, a system is heated or cooled by means of pumping a hot or cold fluid from place to place. For instance, in countries where the weather is very cold, hot water is pumped from hot source and passes through the pipe lines that are fixed for this purpose in the building.

Radiation: Is the way of heat transfer that takes place with the help of electromagnetic waves.

Remember: when charged particles vibrate, they emit electromagnetic wave with frequencies equal to their own frequency of vibration, and material bodies emit or absorb electromagnetic waves in the infra red region at normal temperature.

- It is a common experience that, when you stand near a source of heat, you feel warm due to the thermal energy reached to your body by means of radiation. Also,
- The heat energy from the sun reaches the earth through radiation.

The rate of radiation is explained by a law called *Stefan – Boltzman law*

Stefan – Boltzmann law states that; "The power radiated over all wavelengths from a surface is proportional to the forth power of the absolute temperature."

$P = \sigma e A T^4$ $\sigma = 5.67\times 10^{-8} \frac{N}{m^2k^4}$ constant.

- **Black body**: Is a theoretical object that absorbs all the electromagnetic radiation that hits it.

Newton's law of cooling: states that, "the rate of loss of heat from a body is proportional to its excess temperature above its surrounding

$\frac{\Delta Q}{\Delta t} = kA(\theta_{sur} - \theta_{obj})$

θ_{sur} – surrounding temperature

θ_{obj} – object's temperature

Conceptual Example

97. At night, is it "see breeze" or "land breeze". Explain?

Explanation: During the day, the sun warms the land more rapidly than the water. This is because, the land, which is mostly rocks, has a lower specific heat than the water. The warm land heats the air above it, which becomes less dense and rises. Cooler air from over the water flows into take its place, producing a "sea breeze". At night, the land cools off more rapidly than the water – again because of its lower specific heat. Now it is the air above the relatively warm water that rises and is replaced by cooler air from over the land, producing a "land breeze."

Illustrative Examples

98. What is the rate of heat loss through a glass window of cross sectional area $1m^2$ where its thickness in 4mm and the outside temperature is -20^0C and the inside temperature is 30^0C. Take thermal conductivity of glass, $k = 0.84W/mk$.

Solution: $\frac{Q}{t} = kA\frac{\Delta\theta}{\Delta x} = \frac{(0.84)(1)(30-(-20))}{0.004} = 10.5kw$

99. In a domestic heating system, a room is warmed by a radiator through which water passes at a rate of 0.12kg. The difference between the inlet and the outlet temperature of the water is 6k, and the radiator is made of iron of thermal conductivity 80W/mk and has an effective surface area of $1.5m^2$ with walls 2mm thick
    a) At what rate is heat supplied to the room?
    b) What is the mean temperature difference between the inner and outer surface of the radiator?

Solution: a) $\frac{Q}{t} = \frac{m_w C_w \Delta T_w}{t} = \frac{(0.12)(4200)(6)}{1} = 3024W$

b) $\frac{Q}{t} = \frac{kA\Delta\theta}{\Delta L} \Rightarrow \Delta\theta = \frac{\left(\frac{Q}{t}\right)(\Delta L)}{(k)(A)}$

$\Delta\theta = \frac{(3024)(0.002)}{(80)(1.5)} = 0.05k$

100. An immersion heater whose temperature is 180^0C is wholly immersed into cold water of temperature 20^0C. If heat flows at a rate of 42W from the heater, and the surface area of the heater is $25cm^2$, what is the convective surface coefficient of the heater?

Solution: $\frac{Q}{t} = hA\Delta T \Rightarrow h = \frac{\frac{Q}{t}}{A\Delta T} = \frac{42}{(2.5\times10^{-3})(160)} = 105\frac{w}{m^2k}$

101. By what factor does the rate of radiation changed when the absolute temperature of a source is doubled

Solution: $P_1 = \sigma AT_1^4$ .... (*), $P_2 = \sigma AT_1^4$ .... (**)

Taking ratio, $\frac{P_2}{P_1} = \frac{T_2^4}{T_1^4}, T_2 = 2T_1 \Rightarrow P_2 = 16P_1$

102. A 60w light bulb is painted black so that it emit no light. Its total power is released as heat. How long does the bulb take to heat 2000g of water by 30^0C, if it is completely submerged

Solution: $\frac{Q}{t} = \frac{mC\Delta T}{t} = 60 \Rightarrow t = \frac{mC\Delta T}{60} = \frac{(2)(4200)(30)}{60} = 4200sec$

103. A sphere made of aluminum has a mass of 12g and surface area of $12.5cm^2$. Its temperature is 427^0C. How much will it cool in one second, if it loses heat only through radiation? Take e = 0.6 for sphere.

Solution: $P = Ae\sigma T^4 = (1.25\times10^{-3})(0.6)(5.67\times10^{-8})(700)^4$

$P = 1.46W$, $T = 425^0C = 700k$

But $Q = pt = (1.46)(1) = 1.46J$

104. A certain body has a heat transfer coefficient of 24s/mk. If the temperature of the object is 40^0C and the temperature of the

surrounding is 35^0C, and surface area of the object is $1.2m^2$, what is the rate of heat loses from the body?

Solution: From Newton's law of cooling

$$\frac{dQ}{dt} = \frac{\Delta Q}{\Delta t} = kA(\theta_{sur} - \theta_{obj})$$

$$\frac{\Delta Q}{\Delta t} = (24)1.2)(35 - 40) = -144w$$

The negative sign indicates heat lose

105. Given the following information, the body surface area is $12.8m^2$, the clothing is 1cm thick, skin surface temperature is 33^0C, whereas the outer surface of the clothing is 1^0C, the thermal conductivity of the clothing is 0.04w/mk. What is the rate at which the body heat flows out through the clothing of man?

A. 22.04W B. 23.04W C. 2.3W D. 230.4W

Solution: $\frac{Q}{t} = kA\frac{\Delta\theta}{\Delta L} = (0.04)(1.8)\frac{(33-1)}{0.01} = 230.4W$ **Answer: D**

Additional Examples

Elasticity

106. A wire fixed at the upper end stretched by length ℓ by applying a force F. The work done in stretching is:

A. $2F\ell$ B. $F\ell$ C. $F/2\ell$ D. $F\ell/2$

Solution: $W = \frac{1}{2}$ (stress)(strain)(volume) $= \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{\ell}{\ell}\right)A\ell$

$$W = \frac{1}{2}F\ell$$

Answer: D

107. Which of the following is correct?

A. Hooke's law is applicable only within elastic limit.
B. Hooke's law is applicable only within elastic limit.
C. Young's modulus is dimensionless.
D. Stress multiplied by strain gives stored energy.

Answer: A

108. The elasticity limit for a gas

A. Exists
B. Exists at absolute zero
C. Exists for ideal gas
D. Does not exist

Answer: D

109. A rope, made of steel, has a diameter of 5 cm. The breaking strength of the rope is 2×10^8N. The breaking strength of a similar rope of double diameter is?

A. 2×10^8N B. 4×10^8N C. 1×10^8N D. 8×10^8N

Explanation: Diameter doubled cross – sectional area quadrupled breaking strength ~ cross sectional area. **Answer: D**

110. The Young's modulus for a liquid is?
A. 0 B. 1 C. ∞ D. non – zero constant **Answer: A**

111. The change in the shape of a regular body is due to
A. Bulk strain C. Shearing strain
B. Longitudinal strain D. Metallic strain **Answer: C**

112. A wire of area of cross – section $10^{-6}m^2$ is increased in length by 0.1%. The tension produced is 1000N. The Young's modulus of wire is
A. $10^{12}N/m^2$ B. $10^{11}N/m^2$ C. $10^{10}N/m^2$ D. $10^{9}N/m^2$

Solution: $Y = \frac{F\ell}{a\Delta\ell} = \frac{(1000)(100)}{(10^{-6})(0.1)} = 10^{12}N/m^2$ **Answer: A**

113. A and B are two wires. The radius of A is twice that of B. They are stretched by the same load. Then, the stress on B is
A. Equal to that on A C. Two times on A
B. Four times that on B D. Half that on A

Solution: $\frac{\text{stress on B}}{\text{stress on A}} = \frac{r_A^2}{r_B^2} = \frac{(2r_B)^2}{r_B^2} = 4$ **Answer: B**

114. The breaking stress of a wire depends upon
A. Length of wire C. Shape of cross – section
B. Radius of wire D. Material of wire **Answer: D**

115. Bulk modulus of water is 2×10^9 N/m^2. The change in pressure required to increase the density of water by 0.1% is
A. $2 \times 10^9 N/m^2$ C. $2 \times 10^6 N/m^2$
B. $2 \times 10^8 N/m^2$ D. 2×10^4 N/m^2

Solution: The density would increase by 0.1% if the volume decreases by 0.1%

$$k = \frac{\Delta P}{\Delta v/v} \Rightarrow \Delta P = \frac{k\Delta v}{v} = 2\times10^9\left(\frac{0.1}{100}\right) = 2\times10^6 N/m^2$$

Answer: C

116. The extension of a wire by the application of a load is 3mm. The extension in a wire of the same material and length but half the radius by the same load is?
A. 12mm B. 0.75mm C. 6mm D. 15mm

Solution: $\frac{\Delta\ell_2}{\Delta\ell_1} = \frac{a_1}{a_2} = \frac{a_1}{a_1/4} = 4 \Rightarrow \Delta\ell_2 = 4\Delta\ell_1$ **Answer: A**

117. For steel, $E = 2 \times 10^{11}$ N/m^2. The force required to double the length of steel wire of area 1 cm^2 is

A. 2×10^{11}N C. 2×10^{7}N

B. 2×10^{9}N D. 2×10^{5}N

Solution:

Strain = 1, $Y = F/A \Rightarrow F = YA = (2 \times 10^{11})(10^{-4}) = 2 \times 10^{7}$N

Answer: C

118. In an experiment for the determination of Young's modulus of the material of a wire, the length of the wire and the suspended mass are both doubled. Then, the Young's modulus

A. Remains unchanged C. Becomes four times

B. Becomes double D. Becomes sixteen times

Hint: Young's modulus does not depend on geometery.

Answer: A

119. The work done per unit volume in stretching a metallic wire is

A. $\frac{1}{2}$(force)(extension) C. (force)(extension)

B. $\frac{1}{2}$(stress)(strain) D. (stress)(strain)

Solution: Work done $= \frac{1}{2}F\Delta\ell$

$\frac{\text{work done}}{\text{volume}} = \frac{F\Delta\ell}{2(a\ell)} = \frac{1}{2}$(stress)(strain) **Answer: B**

120. A wire of length L and cross – sectional area A is made of a material of Young's modulus Y. If the wire is stretched by an amount x, what is the work done?

A. $\frac{YA}{2L}$ B. $\frac{YAX}{2L}$ C. $\frac{YAX^2}{L}$ D. $\frac{YAX^2}{2L}$

Solution: Work done $= \frac{1}{2}Fx$, but $Y = \frac{F/A}{X/L} = \frac{FL}{AX} \Rightarrow F = \frac{YAX}{L}$

Work done $= \frac{1}{2}\left(\frac{YAX}{L}\right)(X) = \frac{YAX^2}{2L}$ **Answer: D**

121. When a body undergoes a linear tensile strain, it experiences a lateral contraction also. The ratio of lateral contraction to longitudinal strain is known as

A. Young's modulus
B. Bulk modulus
C. Poisson's ratio
D. Hooke's law

Answer: C

122. A spring is made of steel and not a copper because

A. Elasticity of steel is greater than that of copper.
B. Elasticity of steel is less than that of copper.
C. Plasticity of copper is greater than that of steel.
D. Plasticity of steel is greater than that of copper.

Answer: A

123. If the density of the material increases, the value of Young's modulus

A. Increases
B. Decrease
C. First increases, then decreases
D. First decreases, then increases

Answer: A

124. A copper wire and steel wire of the same diameter and length are connected end to end and a force is applied which stretched their combined length by 1 cm. Then the two wires will have

A. The same stress and strain.
B. The same stress but different strain.
C. The same strain but different stresses.
D. Different stresses and strains.

Hint: same force, same area, same stress

Answer; B

125. Young's modulus for a perfectly plastic body is

A. Zero B. 1 C. infinite D. finite

Answer: A

126. What is the Yong's modulus of elasticity for a perfectly rigid body?

A. Infinity B. Zero C. 1 D. −1

Hint: Young's modulus= $\frac{\text{stress}}{\text{strain}}$, strain = 0 for rigid body

Answer: A

127. A force F is needed to break a copper wire having radius R. the force needed to break a copper wire of same length and radius 2R will be

A. $F/2$ B. 2F C. 4F D. $F/4$

Hint: breaking force ~ cross – sectional area **Answer: C**

128. A wire can sustain a weight of 10 kg before breaking. If the wire is cut into two equal parts, then each part can sustain a weight of
A. 2.5 kg B. 5 kg C. 10kg D. 15kg

Hint: maximum load is not related to length. **Answer: C**

129. On stretching a wire, the elastic energy stored per unit volume is. (letters have usual meaning)

A. $\frac{F\Delta\ell}{2A\ell}$ B. $\frac{FA}{2\ell}$ C. $\frac{F\ell}{2A}$ D. $\frac{F\ell}{2A}$

Solution: Energy $= \frac{1}{2}$ (stress)(strain) $= \frac{1}{2}\frac{F}{A}\left(\frac{\Delta\ell}{\ell}\right) = \frac{F\Delta\ell}{2A\ell}$

Answer: A

130. The young's modulus of a wire of length L and radius r is Y. If the length is reduced to $\frac{L}{2}$ and radius is reduced to $\frac{r}{2}$, then the young's modulus will be

A. $\frac{Y}{2}$ B. Y C. 2Y D. 4Y

Explanation: Y depends up on nature of material and not on geometric dimensions.

Answer: B

131. If the stress applied is equal to the Young's modulus of the material of the wire, the final length of the wire is equal to:
A. Original length of wire.
B. Half the original length of wire.
C. Twice the original length of the wire.
D. Four times the original length of the wire.

Hint: $Y = \frac{\text{stress}}{\frac{\Delta\ell}{\ell}}$, when Y = stress, $\Delta\ell = \ell$ **Answer: C**

132. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200N to the lower elastic energy stored in the wire is?
A. 0.1J B. 0.2J C. 10J D. 20J

Solution: $u = \frac{1}{2}(F)(\Delta\ell) = \frac{1}{2}(200)(10^{-3}) = 0.1J$ **Answer: A**

133. The dimensions of four wires of the same material are given below. In which wire the increase in length will be maximum?
A. Length 100cm, Diameter 1mm
B. Length 200cm, Diameter 2mm
C. Length 300cm, Diameter 3mm
D. Length 50 cm, Diameter 0.5mm

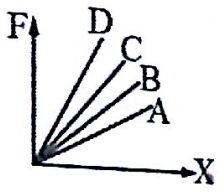
Solution: $Y = \frac{F/a}{\Delta\ell/\ell} = \frac{F\ell}{a\Delta\ell} \Rightarrow \Delta\ell \sim \frac{1}{D^2}$

$\frac{100}{1^2} = 100, \frac{200}{4} = 50, \frac{300}{9} = 33.33, \frac{50}{(0.5)^2} = 200$ **Answer: D**

134. On increasing the length by 0.5mm in a steel wire of length 2m and area $2mm^2$, the force required is? ($Y_{steel} = 2.2 \times 10^{11} N/m^2$)
A. $1.1 \times 10^5 N$ C. $1.1 \times 10^3 N$
B. $1.1 \times 10^4 N$ D. $1.1 \times 10^2 N$

Solution: $Y = \frac{F\ell}{A\Delta\ell} \Rightarrow F = 1.1 \times 10^2 N$ **Answer: D**

135. The load extension curves for four wires A, B, C and D are given. The dimensions of all the four wires are identical but materials of wires are different. Which wire has highest value of Young's modulus of elasticity?



A. A B. B C. C D. D
Hint: For the same load, extension is least in wire D. So, Y is largest. **Answer: D**

136. A cable is replaced by another one of the same length and material but of twice the diameter. The elongation would now become
A. Double C. One – fourth
B. One–half D. One – sixth

Solution: $Y = \frac{F\ell}{a\Delta\ell} \Rightarrow \Delta\ell \sim \frac{1}{a}$ or $\Delta\ell \sim \frac{1}{D^2}$

∴ If D is doubled, $\Delta\ell$ becomes one – fourth **Answer: C**

137. Two wires of the same material and length are stretched by the same force. Their masses are in the ratio 3: 2. Their elongations are in the ratio.
A. 3: 2 B. 9: 4 C. 2: 3 D. 4: 9

Solution: $Y = \frac{F\ell}{a\Delta\ell} \Rightarrow \Delta\ell \sim \frac{1}{a}$, again $m = a\ell\rho$ or $m \sim a$

$$\therefore \Delta\ell \sim \frac{1}{m} \Rightarrow \frac{\Delta\ell}{\Delta\ell} = \frac{m_2}{m_1} = \frac{2}{3}$$

Answer: C

138. A wire extends by 1mm when a force is applied. Double the force is applied to another wire of same material and length but half the radius of cross – section. The elongation of the wire in mm will be

A. 8 B. 4 C. 2 D. 1

Solution: $Y = \frac{F\ell}{a\Delta\ell} \Rightarrow \Delta\ell \sim \frac{F}{r^2}$

$$\frac{\Delta\ell_2}{\Delta\ell_1} = \left(\frac{F_2}{F_1}\right)\left(\frac{r_1^2}{r_2^2}\right) = (2)(2)(2) = 8 \Rightarrow \Delta\ell_2 = 8\Delta\ell_1 = 8$$ **Answer: A**

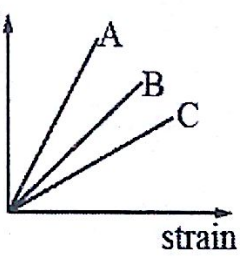
139. The stress – strain lines for brass, steel and rubber are given. The lines A, B and C are for

A. Steel, brass and rubber respectively.

B. Brass, steel and rubber respectively.

C. Steel, rubber and brass respectively.

D. Rubber, steel and brass respectively.



Hint: In the case of wire A, strain is least, so, Y is maximum.

Answer: A

140. Two wires of the same radius and material have their lengths in the ratio 1: 2. If these are stretched by the same force, then the strains produced in the two wires will be in the ratio

A. 1:1 B. 1: 2 C. 2: 1 D. 1: 4

Explanation: $Y = \frac{\text{stress}}{\text{strain}} \Rightarrow \text{strain} = \frac{\text{stress}}{Y}$. Both stress and Y remain unchanged in this case. **Answer: A**

141. A metal rod of Young's modulus $2\times10^{10}N/m^2$ under goes an elastic strain of 0.06%. The energy per unit volume stored in J/m^3 is?

A. 3600 B. 7200 C. 10800 D. 14400

Solution: $\frac{\text{Energy}}{\text{volume}} = \frac{1}{2}(\text{stress})(\text{strain}) = \frac{1}{2}Y(\text{strain})(\text{strain})$

$= \frac{1}{2}(2 \times 10^{10})(0.06 \times 10^{-2})^2 = 3600\ J/m^3$ **Answer: A**

142. A copper bar of length L and area A is placed in a chamber at atmospheric pressure. If the chamber is evacuated the percentage change in volume will be?

A. 8×10^{-7} C. 1.25×10^{-4}

B. 8×10^{-5} D. 1.25×10^{-5}

Solution: $\frac{1}{k} = \frac{\Delta v/v}{\Delta P} \Rightarrow \frac{\Delta v}{v} = \frac{\Delta P}{k} = (10^5)\ (8\times10^{-12})100 = 8\times10^{-5}$

Answer: B

143. What stress would cause a wire to increase in length by one – tenth of one percent if the young's modulus for the wire is $12 \times 10^{10}\ N/m^2$?

A. $12 \times 10^3 N/m^2$ C. $12 \times 10^7\ N/m^2$

B. $12 \times 10^5\ N/m^2$ D. $12 \times 10^9\ N/m^2$

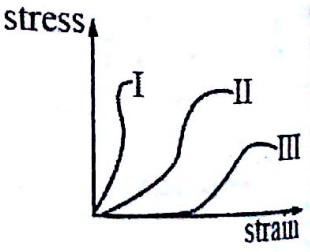
Solution: $12\times10^{10} = \frac{\text{stress}}{0.1\%} \Rightarrow \text{stress} = 12\times10^7 N/m^2$ **Answer: C**

144. A cube of aluminum of side 0.1m is subjected to a shearing force of 100N. The top face of the cube is displaced through 0.02cm with respect to the bottom face. The shearing strain would be

A. 0.02 B. 0.1 C. 0.005 D. 0.002

Solution: shearing strain $= \frac{0.02\times10^{-2}}{0.1} = 0.002$ **Answer: D**

145. From the stress – strain curves, which curve best shows the behavior of copper wire, and which one best shows the behavior of glass fiber respectively.



A. I, II B. II, I C. I, III D. II, III

Hint: Copper is ductile and glass is brittle. **Answer: B**

Fluids at rest

146. A ball floats on the surface of water in a container exposed to the atmosphere. When the container is covered and the air is partially removed, then the ball
A. Rises
B. Gets immersed more in water
C. Remains immersed at its former depth
D. May rise or sink
Explanation: The loss in up thrust due to air needs to be compensated.
Answer: B

147. A ship in passing from a river to a sea
A. Rises higher
B. Sinks lower
C. Neither rises nor sinks
D. Rises or sinks depending on the ship material
Hint: Sea water is denser.
Answer: A

148. If the weight of a body is more than the upward thrust, then the body
A. Sinks
B. Just floats with no portion outside it.
C. Is completely outside the liquid.
D. Is partly inside and parity outside.
Hint: There is a net downward force.
Answer: A

149. A block of wood weights 4N in air and 3N when immersed in a liquid. The buoyant force is
A. 1N B. 0N C. 3N D. 4N
Solution: Buoyant force = Loss in weight
Answer: A

150. We have two different liquids A and B whose relative densities are 0.75 and 1 respectively. If we dip solid objects P and Q having relative densities 0.6 and 0.9 in these liquids then
A. P floats in A and Q sinks in B.
B. P sinks in A and Q floats in B.
C. P floats in B and Q sinks in A.
D. P sinks in B and Q floats in A.
Answer: C

151. Which of the following would a hydrogen balloon find easiest to lift?
A. 1 kg oil C. 1 kg iron
B. 1 kg water D. 1 kg feather
Hint: volume of feather is maximum
Answer: D

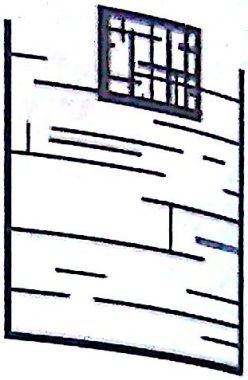
152. A wooden cylinder floats in water with half of its volume inside water. The density of wood in kg/m^3 is

A. 125 B. 0.8 C. 500 D. 800

Solution: $v\rho g = \frac{v}{2}(1000)g \Rightarrow \rho = 500\ kg/m^3$

Answer: C

153. A cubical block of wood of edge 3 cm floats in water. If density of wood is 800 kg/m=3, the height of cube outside water is?



A. 3cm B. 2.4cm

C. 1.2cm D. 0.6cm

Solution: If h is height of block inside

$h(9)(1)g = (27)(0.8g) \Rightarrow h = 2.4cm$

$\therefore$Height outside water = 0.6cm

Answer: D

154. A heavy stone is tied to a balloon which is partially inflated with air. This system is released into water. As the system sinks towards the bottom, the buoyant force on the balloon?

A. Decreases C. Remains constant

B. Increases D. Nothing can be said

Explanation: As the balloon sinks, the pressure of water on the balloon increases. The volume of balloon decreases. The volume of water displaced also decrease. So, the buoyant force decreases.

Answer: A

155. A sample of metal weighs 210g in air, 180g in water and 120g in an unknown liquid. Then, the density of the metal in g/cm^3 is

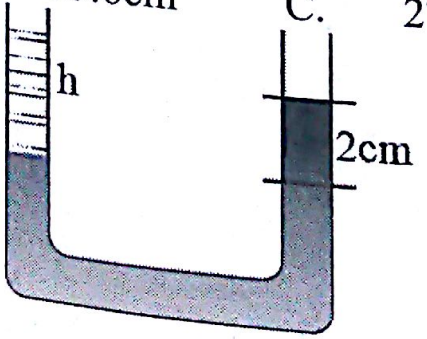
A. 7 B. 17 C. 2 D. 1.8

Solution: Specific gravity $= \frac{210}{210-180} = 7$ in cgs system density = specific gravity

Answer: A

156. Water and mercury are in a U – tube. If the mercury is 2 cm, then the water level, h is

A. 6.8cm B. 13.6cm C. 27.2cm D. 54cm



Solution: h (1000) g = 0.02 (13600) g ⇒ h = 27.2cm

Answer: C

156. An object is just floating on the surface of a liquid. The density of the object is the same as that of the liquid. If the object is slightly pushed down, it will

A. sink
B. rise
C. come back to its original position
D. begin to oscillate

Explanation: The weight of liquid above the object would move the object downwards.

Answer: A

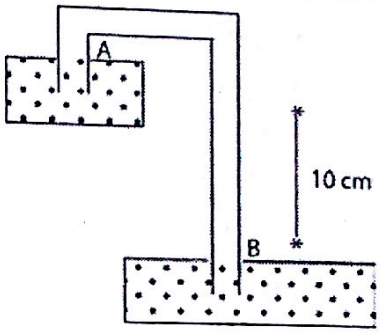
157. "A lump of ice floats in a vessel of water because the density of ice is less than that of water" This statement and reasoning is

A. True
B. False

Answer: A

158. A siphon is given as shown, the pressure difference $P_B - P_A$ is ?

A. 400 N/m^2
B. 1000 N/m^2
C. 3000 N/m^2
D. zero



Explanation: At A and B, the pressure is atmospheric.

Answer: D

159. An inflated balloon rises to a certain height and then cannot rise any further. This is because at that height

A. the atmospheric pressure reduces to zero
B. the gravity disappears
C. the apparent weight of balloon is zero
D. density of balloon is zero

Answer: C

160. The system shown is allowed to fall freely under gravity. The up thrust on the immersed body is

A. zero
B. infinity
C. equal to weight of liquid displaced
D. greater than the weight of liquid displaced

Explanation: In a state of free fall, the weight of liquid displaced is zero.

Answer: A

161. A piece of solid weighs 120g in air, 80g in water and 60g in a liquid, then the relative density of the solid and that of liquid are
A. 3, 2 B. 2, 4/3 C. 3, 3/2 D. 4, 3

Solution: Relative density of solid $= \frac{120}{120-80} = 3$

Relative density of liquid $= \frac{120-60}{120-80} = \frac{3}{2}$

Answer: C

162. A sphere of wax with relative density = 0.9 and volume = 18 cm^3 has some iron nails pierced into it. It floats on water. The mass of iron nails is
A. 0.9 g B. 1.8 g C. 2.7 g D. 3.6 g

Solution: Up thrust = 18g and weight of wax = 18 (0.9) g = 16.2g
weight of iron = 18g – 16.2 g = 1.8g

Answer: B

163. A 0.5 kg block of brass with density 8×10^3 kg/m^3 is suspended from a string. What is the tension in the string if the block is completely immersed in water:
A. 5N B. $\frac{0.5}{8\times10^3}$N
C. $\frac{5}{8}$N D. $\left(5-\frac{5}{8}\right)$N

Solution: volume of block $= \frac{0.5}{8\times10^3}$ m^3

Volume of displaced water $= \frac{0.5}{8\times10^3} m^3$

Weight of displaced water $= \frac{0.5}{8\times10^3}(10^3)(10) = \frac{5}{8}$N = buoyant force Tension = w- buoyant force $= \left(5-\frac{5}{8}\right)$N

Answer: D

164. A cubical block of wood of specific gravity 0.5 and chunk of concrete of specific gravity 2.5 are fastened together. The ratio of the mass of wood to the mass of concrete which makes the combination to float with its entire volume submerged under water is

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A. $\frac{3}{5}$ B. $\frac{4}{5}$ C. $\frac{3}{7}$ D. $\frac{2}{7}$

Solution: Let m_w and m_c be the masses of wood and concrete respectively. Then

$$(m_w + m_c)g = \left(\frac{m_w}{0.5} + \frac{m_c}{2.5}\right)(1)(9) \Rightarrow \frac{m_w}{m_c} = \frac{3}{5}$$ **Answer: A**

165. A steel block having an internal cavity weight 234g in air and 197g in water. If the density of steel is 7.8 g/cm^3, then the volume of the cavity is

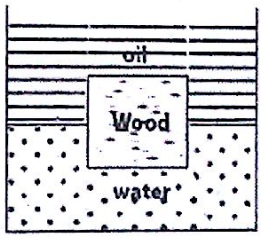
A. 5 cm^3 B. 7 cm^3 C. 9 cm^3 D. 11 cm^3

Solution: weight of water displaced by steel block = loss of weight of block in water = (234 – 197)g = 37g
Volume of water displaced = 37 cm^3
This is the volume of the block with cavity

Volume of block without cavity $= \frac{234}{7.8} = 30\text{cm}^3$

Volume of cavity = 37 – 30 = 7cm^3 **Answer: B**

166. A cubical block of wood 10 cm on a side floats at the interface between oil and water, with its lower face 2cm below the interface. The density of the oil is 0.6 g/cm^3. The mass of the block is



A. 340 g C. 80 g
B. 680 g D. 10 g

Solution: mg = (100× 2×1+ 100 × 8× 0.6) g = 680g **Answer: B**

167. A 700g solid cube having an edge of length 10 cm floats in water. The volume of cube outside water is

A. 50 cm^3 B. 100 cm^3
C. 300 cm^3 D. 500 cm^3

Solution: If v is the volume of cube inside water, then v (1000) (10) = 0.7 (10) $\Rightarrow$ v = 700 cm^3
Total volume = (10 cm)^3 = 1000 cm^3
Volume outside water = 1000 – 700 = 300 cm^3
Answer: C

168. One gram of iron and 1cm^3 of aluminum are dropped into water. Which of the following statements is correct?

A. Aluminum will experience larger buoyant force than iron

B. Iron will experience larger buoyant
C. Both iron and aluminum will experience the same buoyant force.
D. Aluminum shall experience zero buoyant force

Explanation: Volume of 1g of iron is less than $1cm^3$. So, the volume of water displaced by iron will be less than the volume of water displaced by aluminum. So, iron shall experience less buoyant force than aluminum

Answer: A

169. Ice of density $0.9\ g/cm^3$ floats in a glass of water. When the ice melts, the water level in the glass
A. goes down B. goes up
C. stays the same D. unknown

Explanation: The water level remain unchanged because the volume of water displaced by the immersed part of ice will be equal to the volume of water when the ice melts. **Answer: C**

170. Two solids A and B float in water. It is observed that A floats with half of its volume immersed and B floats with $\frac{2}{3}$ of its volume immersed. The ratio of density of A and B is
A. 2: 3 B. 3: 2 C. 3: 4 D. 4: 3

Solution: $V_A\rho_A = \frac{1}{2}V_A(1) \Rightarrow \rho_A = \frac{1}{2}$

Again, $V_B\ \rho_B = \frac{2}{3}V_B(1) \Rightarrow \rho_B = \frac{2}{3} \Rightarrow \frac{\rho_A}{\rho_B} = \frac{3}{4}$

Answer: C

171. The weight of a body is 120 N in air. When immersed completely in water, the weight becomes 80N. The relative density of the material of the block is?
A. 1 B. 2 C. 3 D. 4

Solution: Relative density $= \frac{\text{Weight in air}}{\text{loss of weight}} = \frac{120}{120-80} = 3$

Answer: C

172. A block floats in water with 0.1 its volume above water. The density of the material of the block is
A. $0.6\ g/cm^3$ B. $0.7\ g/cm^3$ C. $0.8\ g/cm^3$ D. $0.9\ g/cm^3$

Solution: $0.9v\ (1)\ (g) = v\rho g \Rightarrow \rho = 0.9\ g/cm^3$ **Answer: D**

173. A piece of ice having a stone in it floats in a vessel filled with water. When the ice melts, the level of water in the vessel
A. falls B. rises C. remain same D. unknown
Explanation: When stone in ice, weight of displaced water = weight of ice + weight of stone.
When the ice melts, the stone sinks. Now, the volume of water displaced by the stone is less than before. Water level falls
Answer: A

174. A piece of gold and a piece of aluminum, both having the same weight in air, are immersed in water. Then
A. both suffer the same loss of weight
B. aluminum suffers more loss of weight
C. gold suffers more loss of weight
D. both do not suffer any loss of weight
Explanation: Aluminum would have larger size a compared to gold. So, larger volume of water will be displaced. Thus, up thrust will be more on aluminum
Answer: B

175. A cube of wood supporting a 200 g mass just floats in water. The length of each edge of the cube is 10 cm. when the mass is removed, the cube rises by
A. 1 cm B. 2 cm C. 3 cm D. 4 cm
Solution: $200 = v\,(1) \Rightarrow v = 200\ cm^3$
Area of base $= 100\ cm^2$
Required height $= \frac{200}{100} = 2cm$ **Answer: B**

176. A body floats with $\frac{1}{3}$ of its volume outside water and $\frac{3}{4}$ of its volume outside another liquid. The density of other liquid in g/cm^3 is
A. $\frac{9}{4}$ B. $\frac{4}{9}$ C. $\frac{8}{3}$ D. $\frac{3}{9}$
Solution: $\frac{2v}{3}(1)(g) = mg,\ \frac{v}{4}\rho g = mg$
Comparing, $\frac{v}{4}\rho g = \frac{2v}{3}(1)(g) \Rightarrow \rho = \frac{8}{3}g/cm^3$ **Answer: C**

177. A boat 3m long 2m wide is floating in a lake. When a man climbs over it, sinks 1cm into the lake. The mass of the man is

A. 60 kg B. 64 kg C. 70 kg D. 72 kg

Solution: volume displaced $= (3)(2)\left(\frac{1}{100}\right) = \frac{3}{50}\text{m}^3$

Weight displaced $= \frac{3}{50}(1000) = 60\text{kg}$

Answer: A

178. A boat with scrap iron is floating in a lake. If the scrap iron is thrown in the lake, the water level will

A. go up B. go down
C. remain unchanged D. un known

Hint: $\rho_{iron} > \rho_{water}$

Answer: B

179. A piece of ice is floating in a glass of water and another identical piece in a glass of milk. When ice melts, the level

A. in two glasses will rise
B. of milk will rise and that of water will fall
C. of milk will rise and that of water remain same
D. in two glasses will fall

Explanation: volume of milk displaced $= \frac{m}{\rho_m}$

Volume of water formed $= \frac{m}{\rho_w}$

But, $\frac{m}{\rho_w} > \frac{m}{\rho_m}$

Answer: C

180. A piece of ice is dropped in a vessel containing kerosene. When ice melts, the level of kerosene will

A. rise B. fall
C. remain same D. unknown

Hint: $\frac{m}{\rho_w} < \frac{m}{\rho_k}$, so there will be a fall in level

Answer: B

181. A man is sitting in a boat which is floating in a pond. If the man drinks some water from the pond, the level of water in the pond will?

A. rise a little B. fall a little
C. remain same D. unknown

Hint: The water removed from the pond and the water displaced are equal.

Answer: C

Surface Tension

182. A soap bubble assumes a spherical surface. Which of the following statement is wrong?
   A. The soap film consists of two surface layers of molecules back to back
   B. The bubble encloses air inside it
   C. The pressure of air inside the bubble is less than the atmospheric pressure and atmospheric pressure compressed it equally from all sides to give it a spherical shape.
   D. Because of the elastic property of the film. **Answer: C**

183. The hight up to which water will rise in a capillary tube will be
   A. maximum when water temperature is $4^{o}c$
   B. maximum when water temperature is $0^{o}c$
   C. minimum when water temperature is $4^{o}c$
   D. same at all temperatures

   Explanation: Water has maximum density at $4^{o}c$. Since $h = \frac{2\sigma\cos\theta}{r\rho g} \Rightarrow h \sim \frac{1}{\rho}$. Therefore, h will be least. **Answer: C**

184. In the state of weight lessness, a capillary tube is dipped in water. The water
   A. will not rise at all
   B. will rise to same height as at atmospheric pressure
   C. will rise to less height than at atmospheric pressure
   D. will rise to the full length of the given capillary tube

   Explanation: There will be no force to balance the upward force due to surface tension **Answer: D**

185. The meniscus of mercury in the glass capillary tube is
   A. convex B. concave C. plane D. uncertain

   Answer: A

186. Water rises against gravity in a capillary tube when its one end is dipped into water because
   A. pressure below the meniscus is less than atmospheric pressure
   B. pressure below the meniscus is more than atmospheric pressure
   C. capillary attracts water
   D. of viscosity **Answer: A**

187. Soap helps in better cleaning of clothes because
A. it reduces the surface tension of Solution
B. it gives strength to Solution
C. it absorbs the dirt
D. of some chemical action of soap
Answer: A

188. Water from the soil reaches the plants by
A. atmosphere B. evaporation
C. capillary action D. gravity
Answer: C

189. Two glass plates, one upon the other with a little water in between them cannot be separated easily because of
A. surface tension B. viscosity
C. pressure D. inertia
Answer: A

190. At the boiling point of a liquid, surface tension
A. is zero C. is same as that at any temperature
B. is infinite D. unknown
Answer: A

191. Small liquid drops assume spherical shape because
A. gravity acts on the drop
B. atmospheric pressure acts on the drop
C. liquid tends to have minimum surface area due to surface tension
D. unknown
Answer: C

192. The role of a water – proofing agent is to change the angle of contact
A. from acute to obtuse C. to zero
B. from obtuse to acute D. to infinity
Answer: A

193. An oil drop placed on the surface of water spreads as a thin layer. This is because
A. surface tension of water is greater than that of oil
B. surface tension of water is less than that of oil
C. viscosity of oil is greater than viscosity of water
D. of large difference in the densities of oil and water
Answer: A

194. Soap and detergents make water easier to wash clothes with because they
A. make it lather C. destroy its hardness
B. increase its density D. decrease its surface tension
Answer: D

195. A liquid wets the containing vessel when
A. Cohesive force is more than adhesive force

B. Cohesive force is less than the adhesive force
C. Cohesion is absent
D. Adhesion is absent

Answer: B

196. When two capillary tubes of different diameters are dipped vertically, the rise of the liquid is
A. same in both the tubes
B. more in the tube of larger diameter
C. less in the tube of smaller diameter
D. more in the tube of smaller diameter

Hint: $H \sim \frac{1}{D}$

Answer: D

197. Water rise in a capillary tube when its one end is dipped vertically in it is 3 cm. If the surface tension of water is 75×10^{-3} N/m, then the diameter of capillary will be
A. 0.1mm B. 0.5 mm
C. 1.0 mm D. 2.0 mm

Solution:

$$h = \frac{2\sigma\cos\theta}{r\rho g} \Rightarrow r = \frac{2\sigma\cos\theta}{h\rho g} = \frac{2\left(75\times10^{-3}\right)\cos0°}{\left(3\times10^{-2}\right)\left(10^{3}\right)(10)} = 0.5 \text{ mm}$$

Answer: B

Fluids in Motion

198. Bernoulli's equation is a consequence of conservation of
A. Energy B. Linear Momentum
C. Angular Momentum D. Mass **Answer: A**

199. In old age, arteries carrying blood in the human body become narrow resulting in an increase in the blood pressure. This follows from
A. Pascal's principle B. Stoke's law
C. Bernoulli'e principle D. Archimedes principle

Answer: C

200. Viscosity is the property by virtue of which a liquid
A. becomes spherical in shape
B. occupies minimum surface area
C. oppose relative motion between its layers
D. tends to regain its deformed position **Answer: C**

201. Water is flowing through a pipe under constant pressure. At some place, the pipe becomes narrow. The pressure of water at this place

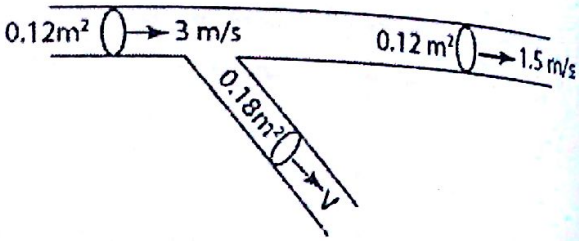
A. increases
B. decreases
C. remains same
D. unknown

Answer: B

202. The lift of an airplane is based on
A. Torricelli's theorem
B. Bernovlli's theorem
C. law of gravitation
D. conservation of momentum

Answer: B

203. What is the speed of water in the lower branch?
A. 1 m/s
B. 2. 25m/s
C. 1.5 m/s
D. 3 m/s

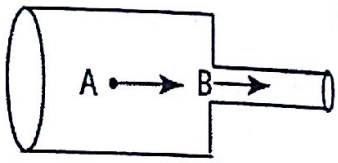


Solution: $0.18\ v + (0.12)(1.5) = (0.12)(3)$

$\Rightarrow v = 1$ m/s

Answer: A

204. Water is flowing in a pipe as shown, pressure is?
A. more at A than that at B
B. equal at A and at B
C. less at A than that at B
D. unknown



Explanation: At A, area is large, speed is less and pressure is more

Answer: A

205. Blood is flowing at the rate of 100 cm^3/s in a capillary of cross-sectional area 0.25 m^2. The velocity of flow is?
A. 0.1 mm/s
B. 0.2 mm/s
C. 0.3 mm/s
D. 0.4 mm/s

Solution: $V = \dfrac{100 \times 10^{-6}}{0.25} = 0.4$ mm/s

Answer: D

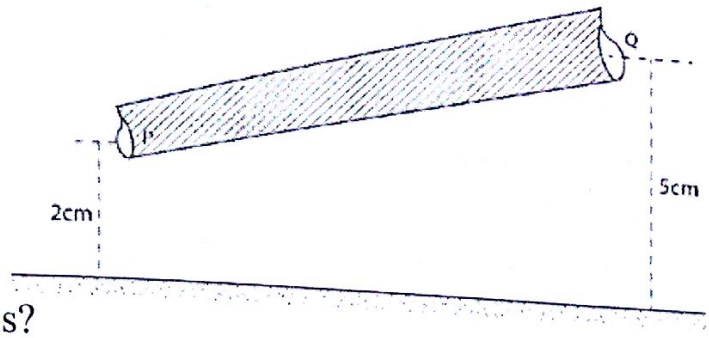
206. The reading of a manometer fitted to a closed tap is 2.5×10^5pa. If the valve is opened, the reading of the manometer falls to 2×10^5 pa. The velocity of water is
A. 0.5 m/s
B. 1 m/s
C. 10 m/s
D. 10 m/s

Solution: $P_1 = P_2 + \dfrac{1}{2}\rho v^2 \Rightarrow 2 \times 10^5 + \dfrac{1}{2}(1000)(V^2) = 2.5 \times 10^5$

$V = 10$ m/s

Answer: C

207. Water is flowing through a tube shown. If the area of cross – section of the tube at P and Q is 4×10^{-3} m^2 and 8×10^{-3} m^2 respectively and the velocity of the water at P is 1 m/s , then its velocity at Q is?



A. 0.5 m/s B. 1 m/s C. 1.5 m/s D. 2 m/s

Solution: $A_PV_P = A_QV_Q$

$$\Rightarrow V_Q = \frac{A_PV_P}{A_Q} = \frac{(4\times10^{-3})(1)}{8\times10^{-3}} = 0.5\text{m/s}$$

Answer: A

The Heat Concept

208. Melting point of ice
A. increases with increasing pressure
B. decrease with increasing pressure
C. is independent of pressure
D. is proportional to pressure

Answer: B

209. Latent heat of vaporization is the energy required to
A. Separate the molecules of the liquid
B. increase the average molecular speed in the liquid phase to that in the gas phase
C. separate the molecules and to force back the atmosphere
D. Separate the molecules and to increase their average molecular speed to that in the gas phase.

Explanation: The specific latent heat of vaporization of a liquid is the heat required to convert unit mass of it at its boiling point into vapor at the same temperature.

Answer: C

210. During melting process, the heat given to a body is utilized in
A. increasing the temperature
B. increasing the density
C. increasing the molecular distance
D. decreasing the mass

Answer: C

211. According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$. Where $\Delta\theta$ is the difference of the temperature of the body and the surroundings, and n is equal to
A. 1 B. 2 C. 3 D. 4

Hint: $\frac{d\theta}{dt} \sim \Delta\theta$

Answer: A

212. A bucket full of hot water is kept in a room. It cools from 75°c to 70°c in t_1 minute, from 70°c to 65°c in t_2 minute and from 65°c to 60°c in t_3 minute. Then
A. $t_1 < t_2 < t_3$ B. $t_1 = t_2 = t_3$
C. $t_1 < t_2 > t_3$ D. $t_1 > t_2 > t_3$

Hint: Use law of cooling

Answer: A

213. Metals are good conductors of heat because, they
A. contain free electrons
B. are relatively far apart
C. have atoms colliding frequently
D. have a reflecting surface

Answer: A

214. Which of the following statements regarding thermal radiation is not correct?
A. A good absorber of thermal radiation is also a good emitter
B. As the temperature of black body rise, the frequency, at maximum energy decreases.
C. The energy emitted by an opening in a wall of enclosure is same with that emitted by a black body at same temperature
D. When a body is in equilibrium with its surrounding, the rate of thermal radiation is equal to the rate of absorption

Answer: A

215. The rate of radiation of a black body at 0° c is E. The rate of radiation of this black body at 273°c will be?
A. 16 E B. 8E C. 4 E D. E

Solution: $T_1 = 273$ k, $T_2 = 546$ k

$$\frac{E^2}{E^1} = \frac{(2T)^4}{T^4} \Rightarrow E_2 = 16\,E_1$$

Answer: B

216. On heating one end of a rod, the temperature of the whole rod will be uniform when
A. $k = 1$ B. $k = 0$ C. $k = 100$ D. $k = \infty$

Answer: A

Hint: $\frac{d\theta}{dx} \sim \frac{1}{k}$, when $k = \infty$, $\frac{d\theta}{dx} = 0$

Answer: D

217. Under steady state, the temperature of a body
A. increases with time
B. decreases with time
C. does not change with time and is same at all points
D. does not change with time but is different at different points

Explanation: In steady state, temperature is time- independent but position – dependent.

Answer: D

218. For a perfectly black body, its absorptive power is?
A. 1 B. 0.5 C. 0 D. infinity

Explanation: A perfect black body absorbs all radiations incident on it.

Answer: A

219. The most suitable material for cooking is the one having
A. high specific heat and low conductivity
B. low specific heat and low conductivity
C. low specific heat and high conductivity
D. high specific heat and low conductivity

Answer: C

220. A container contains hot water at 100°c. If in time t_1 temperature falls to 80°c and in time t_2 temperature falls to 60°c from 80°c, then
A. $t_1 = t_2$ B. $t_1 > t_2$ C. $t_1 < t_2$ D. unknown

Explanation: In the first case, the temperature difference is greater as compared to the second case. So, the rate of loss of heat is quicker.

Answer: C

221. If the end of a metal rod is heated. Then the rate of flow of heat does not depend upon
A. area of rod
B. mass of rod
C. time
D. temperature gradient

Answer: B

222. The coefficient of thermal conductivity depends upon
A. temperature difference
B. area of plate
C. thickness of plate
D. material of plate

Answer: D

223. Heat travels through vacuum by
A. conduction
B. convection
C. radiation
D. unknown

Hint: Conduction and convection are not possible in vacuum

Answer: C

224. Woollen clothes are used in winter season because woolen cloths
A. are good sources for producing heat
B. absorb heat from surroundings
C. are bad conductors of heat
D. provide heat to body continuously

Answer: C

225. Black body is defined as the body
A. Which absorbs all the radiations of any wavelength.
B. Which does not absorb any radiation.
C. Which absorbs radiations in ultraviolet region.
D. Which absorbs radiations in infrared region.

Answer: A

226. The amount of radiation emitted by a perfectly black body is proportional to
A. Temperature on ideal gas scale.
B. Fourth root of temperature on ideal gas scale.
C. Fourth power of temperature on ideal gas scale.
D. Square of temperature on ideal gas scales.

Hint: Think of Stefan's law.

Answer: C

227. A hot body is placed in a room. After a time, its temperature becomes constant. At constants temperature,
A. The body does not radiate heat.
B. The body does not receive heat by radiation from surrounding.
C. Body is radiating heat and also takes heat from the surrounding.
D. Nothing is possible.

Answer: C

228. Green houses are provided with glass roofing's and walls, because glass
A. Transmits the radiation from a source of high temperature and absorbs the radiation from a source of low temperature.
B. Transmits the radiation from a source of low temperature and absorbs the radiation from a source of high temperature.
C. Absorbs the radiation from both types of sources at low and high temperature.
D. Transmits the radiation from both types of sources.

Answer: A

229. As the temperature of a black body is increased, the wavelength corresponding to maximum energy.
A. Increases C. Does not change
B. Decreases D. Increase then decrease **Answer: B**

230. Two thin blankets keep us more warm than one blanket of thickness equal to these two. The reason is
A. Their surface area increases.
B. A layer of air is formed between two blankets, which is bad conductor.
C. These have more wool.
D. They absorb more heat from outside. **Answer: B**

231. Which statement about convection is correct?
A. Brownian motion is a form of convection.
B. Convection occurs only in gases.
C. Convection results from a density change.
D. Evaporation is a form of convection. **Answer: C**

232. Thermal conduction in metals differs from thermal conduction in insulators. The reason for this is that, in metals, heat can be transported by
A. Electrons C. Photons
B. Lattice vibrations D. Positive ions
Explanation: Thermal conduction in insulators is by lattice vibration. Thermal conduction in metals is by free electrons. **Answer: A**

233. Which of the following process does not occur through convection
A. Boiling of water.
B. Land breeze and sea breeze.
C. Circulation of air around furnace.
D. Heating of glass bulb through filament.
Hint: bulb heats up by radiation process **Answer: D**

234. At what temperature is the Fahrenheit scale reading equal to twice of Celsius?
A. 160°C B. 320°C C. 175° D. 273°
Solution: $\frac{2x-32}{180}=\frac{x}{100}\Rightarrow x=160°$ **Answer: A**

235. The temperature of a body on Kelvin scale is found to be X_{ki} when it is measured by a Fahrenheit scale, it is found to be x°F. Then x is

A. 302.25 B. 574.25 C. 313 D. 40

Solution: $\frac{x-273}{100}=\frac{x-32}{180}\Rightarrow x=574.25$

Answer: B

236. On a hypothetical scales x, the ice point is 40° and the steam point is 120°. For another scale Y, the ice point and steam point are −30° and 130° respectively. If X reads 50°, then Y would read

A. −5° B. −8° C. −10° D. −12°

Solution: $\frac{Y-(-30)}{130-(-30)}=\frac{50-40}{120-40}\Rightarrow Y=-10$

Answer: C

237. 2 kg of ice at −20°C is mixed with 5 kg of water at 20°C. What mass of water remain in the container? (C = 1kcal/kg°C, C_{ice} = 0.5 kcal/kg°C, L_f = 80 kcal/kg)

A. 7kg B. 6kg C. 4kg D. 2kg

Solution: Heat lost by water in decreasing its temperature from 20°C to 0°C = (5)(1)(20) = 100 k cal

Heat gained by ice in increasing temperature from −20°C to 0°C = (2)(0.5)(20) = 20 kcal

The remainder 80 k Cal shall convert 1 kg of ice to water.

∴ Water in vessel = 6kg

Answer: B

238. 80 g of water at 30°C is poured on a large block of ice at 0°C. The mass of ice that melts is?

A. 30g B. 80g C. 150g D. 1600g

Solution: XL = 80(1) (30 − 0) ⇒ x= 30 g

Answer: A

239. 100g of ice is mixed with 100g of water at 100°C. The final temperature of the mixture is

A. 10°C B. 20°C C. 30°C D. 40°C

Solution: 100(80) + 100(1)(T) = 100(1)(100 − T) ⇒ T = 10°C

Answer: A

240. One gram of ice is mixed with one gram of steam. After thermal equilibrium, the temperature of the mixture is

A. 0°C B. 100°C C. 55°C D. 80°C

Solution: Heat required to melt 1g of water at 0°C into 1g of water at 0°C is 80 cal. Heat required to convert 1g of water at 0°C into 1g of water of 100°C = (1)(1)(100) = 100 cal

Heat required to condense 1g of steam = 540 cal clearly, whole of steam is not condensed. So, temperature of the mixture is 100°C.

Answer: B

241. If the temperature of the sun were to increase from T to 2T and its radius from R to 2R, then the ratio the radiant energy received on earth to the previous will be

A. 32 B. 16 C. 4 D. 64

Solution: Using Stefan's law

$P \sim AT_4$ and $A \sim R^2$

$P' \sim (2R)^2(2T)_4$

$$\frac{P'}{P} = 64$$

Answer: D

242. The area of the glass of a window of room is 10 m^2 and thickness is 2mm. The outer and inner temperatures are 40°C and 20°C respectively. If thermal conductivity of glass is 0.2, what is the heat flowing in the room per second?

A. 3×10^4J B. 2×10^4J C. 30J D. 45J

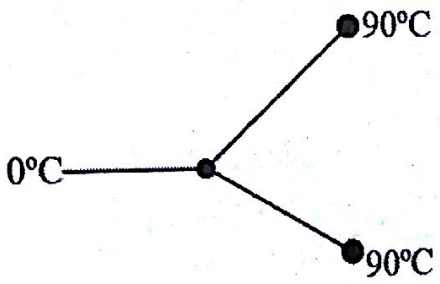
Solution: $Q = \frac{(\theta_1 - \theta_2)}{d} = \frac{0.2(10)(20)(1)}{2\times10^{-3}} = 2\times10^4\,J$ **Answer: B**

243. The emissivity and surface area of tungsten filament of an electric bulb are 0.35 and 0.25 × $10^{-4}m^2$ respectively. If $\sigma = 5.67 \times 10^8 w/m^2k^4$, the temperature is 3000k, then power of bulb is about

A. 40W B. 143W C. 3000W D.1050W

Solution: $\frac{Q}{t} = Ae\sigma T^4$ **Answer: A**

244. The rods made of same material and same cross section and length are given. If the left and right ends are kept at 0°C and 90°C respectively, the temperature of the junction of three rods will be



A. 45°C C. 30°C

B. 60°C D. 20°C

Solution: $\frac{kA(T-0)}{\ell} = \frac{kA(90-T)}{\ell} + \frac{kA(90-T)}{\ell}$

$T = 60°C$ **Answer: B**

245. The energy emitted per second by a black body at 27°C is 10J. If the temperature of the black body is increased to 327°C the energy emitted per second will be?

A. 20J B. 40J C. 80J D. 160J

Solution: $\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4$ **Answer: D**

246. A spherical black body with a radius of 12cm radiates 450W power at 500k. If the radius were halved and the temperature doubled, the power radiated would be:
A. 225W B. 450W C. 900W D. 1800W

Solution: $\frac{P_2}{P_1} = \frac{r_2^2}{r_1^2}\left(\frac{T_2^4}{T_1^4}\right)$ **Answer: D**

247. An ideal material for making cooking vessels must have
A. Small conductivity and large heat capacity.
B. Large heat capacity and large conductivity.
C. Small heat capacity and large conductivity.
D. Small heat capacity and small conductivity. **Answer: C**

248. Morning sun is not so hot as the mid-day sun because
A. Heat rays travel slowly in the morning.
B. Sun is cooler in the morning.
C. Earth is farther away from the sun in the morning.
D. The sun rays travel a long distance in the morning.
Hint: In the morning sun rays are not vertical. **Answer: D**

249. One a cold morning a metal surface will feel colder to touch than a wooden surface because
A. Metal has high specific heat.
B. Metal has high thermal conductivity.
C. Meal has low specific heat.
D. Metal has low thermal conductivity.
Explanation: Due to high thermal conductivity of the metal heat flows quickly from the body to the metal. **Answer: B**

250. Heat given to a gas at constant temperature is used up
A. In increasing internal energy.
B. In increasing temperature.
C. In doing external work.
D. In increasing temperature and doing external work.
Answer: C

251. Good absorber of heat are
A. Poor emitters C. Good emitter
B. Non – emitters D. Highly polished **Answer: C**

252. A piece of glass is heated to a high temperature and then allowed to cool; if it cracks, a probable reason for this is following property of glass

A. Low thermal conductivity
B. High thermal conductivity
C. High specific heat
D. High melting point

Answer: A

253. As compared to the person with white skin, the person with black skin will experience

A. Less heat and more cold
B. More heat and more cold
C. More heat and less cold
D. Less heat and less cold

Answer: B

Unit 8: Conceptual Questions

1. The purity of gold can be tested by weighing it in air and in water how? How can we identify a fake gold made by Gold plating some cheaper material?
2. A cargo ship travels from Atlantic Ocean to Lake Ontario via St. Lawrence River. Where will the ship sink more? Why?
3. A piece of iron is glued to the top of a block of wood. How can we make the block float on a bucket of water?
4. An ice cube floats in a glass of water. When the ice melts, will the water level in the glass change? Explain
5. Bernoulli's equation tells us that "where there is higher fluid speed, there is lower fluid pressure and vice versa" is this statement always true? Explain
6. Airports at high elevations have longer runways for take offs and landings than do airports at sea level. Why?
7. Steel is much denser than water, how then do ships made of steel float?
8. Hurricane will often lift the roof of a house. Use Bernoulli's equation to explain how this happen
9. When you drink a juice through a straw, you reduce the pressure in your mouth what let the juice in?
10. Pascal used a barometer with water, why is it impractical to use water for a typical barometer?
11. When is the buoyant force on a swimmer greater? After he exhale or inhale air? Explain
12. Explain why a sealed bottle partially filled with a liquid can float?